

Relational Frictions Along the Supply Chain: Evidence from Senegalese Traders*

Edward Wiles
Harvard University



Deivy Houeix
Harvard University

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Abstract

We study search and trust frictions in international sourcing, and whether the growth of “social commerce” in lower-income countries can alleviate them. We document that firms use social media as sourcing infrastructure in ways related to these frictions. Guided by a relational-contracting model, we run a field experiment with 1,862 Senegalese garment retailers, matching them to suppliers in Türkiye (search) and cross-randomizing information about types (adverse selection) and incentives (moral hazard). New matches expand access to varieties and quality, but only develop into relationships when trust frictions are alleviated. Structural estimates imply trust frictions are complements and substantially limit trade.

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1 Introduction

Search and trust frictions have historically made it hard for small firms to access foreign inputs. Consider a clothing wholesaler in Senegal who wants to start selling high quality European-made jeans. First, they must find a supplier and some way of seeing what this supplier sells—a classic search problem. Second, even if they find one and place an order, they typically pay before observing quality—exposing them to trust barriers in the form of both adverse selection (some suppliers may be more reliable than others) and moral hazard (suppliers may have incentives to shirk). These frictions may be particularly severe in lower-income settings, whose large informal sectors complicate information aggregation and lower state capacities make contracts hard to enforce. While there is a long theoretical tradition on search ([Stigler, 1961](#)) and trust ([Shapiro, 1983](#)), separate identification of these frictions and of how they interact remain rare because researchers typically do not observe variation that is both exogenous and maps to the theoretical primitives.

In recent years, the rapid growth in smartphone ownership in lower-income countries has changed how small firms buy and sell. Across Africa, online sales have expanded quickly—B2C e-commerce revenue doubled between 2019 and 2024 ([Statista, 2025](#))—and much of this activity runs through social media rather than traditional marketplaces. In a survey of MSMEs in six African countries, [GSMA \(2023\)](#) report that social media is the most widely used online channel, regardless of the size of the business. This growth in “social commerce” may reflect the fact that social media can alleviate both search and trust frictions in supply chains: suppliers can broadcast assortments at low cost (search), and buyers can share information and coordinate action to deter cheating (a modern analogue of [Greif \(1993\)](#)). However, despite its prevalence, there is little evidence on the extent to which social media can serve as a trade-enhancing infrastructure.

In this paper, we provide experimental evidence on the magnitude, mechanisms, and remediability of both search and trust frictions in access to foreign inputs, and evaluate to what extent widely used social media platforms can relax them. We run a field experiment in a large international import market that exploits common features of social media to lower search costs and manipulate trust primitives. Specifically, we randomly allocate 1,862 small garment firms in Senegal to treatment arms that connect them to new suppliers in Türkiye (search), and cross-randomize information about the types (adverse selection) and incentives (moral hazard) of these suppliers. We measure how these interventions affected their access to foreign goods, supplier relationships, and profits and sales, using data from a mystery shopping exercise, real-time transactions from Senegal’s largest mobile payments provider, and a follow-up survey.

We begin by providing, to our knowledge, the first systematic evidence on a growing but largely undocumented phenomenon: how small firms in lower-income countries use social media for B2B trade. We conducted a large-scale baseline survey of 1,862 garment retailers and wholesalers in Dakar, from which the picture that emerges is striking: firms have built a sophisticated sourcing infrastructure that leverages social media platforms in ways consistent with alleviating search

and trust frictions. On the search side, 86% of firms are members of WhatsApp groups in which suppliers (both domestic and international) continuously disseminate information about products and prices. The median firm is in four such groups. On the trust side, 23% of firms are in peer information-sharing WhatsApp groups, whose explicit purpose is for businesses to exchange information, and two-thirds of firms have recommended or warned against particular suppliers to others in the past year. Facebook, Instagram, and TikTok are also popular for sourcing, each used by around a third of firms. Meanwhile, formal B2B e-commerce platforms like Alibaba are almost entirely absent: only 6% of firms use them regularly, and usage is uncorrelated with firm size. In short, firms have organically repurposed social media into trade infrastructure. Motivated by these facts, we design an experiment that leverages the features of social commerce to create exogenous variation in these frictions in a large international import market.

To be precise about these frictions, we develop a dynamic model of relational contracting with sequential search for suppliers and both adverse selection and moral hazard. A firm can pay a search cost to match with a random supplier abroad, who can take a costly but unobservable action that increases the probability of high quality output. Bad-type suppliers cannot do this, while good-type suppliers will only do so if the future value of the relationship exceeds the current cost. We derive a wedge equation showing that the trust frictions distort quantity downwards on the optimal contract. As our experiment cross-randomizes interventions that target adverse selection and moral hazard, yet relatively little work characterizes how they interact in dynamic settings, we pay particular attention to this. We show that interventions that alleviate them are generally complements: if the firm's beliefs about a supplier's type exogenously improve, then they buy more today, which flattens the quantity schedule. This weakens the supplier's dynamic incentives and therefore increases the return to alleviating moral hazard. This interaction is important: since the return to alleviating one friction depends on the severity of the other, analyses that study each friction in isolation may understate or overstate the value of policies that address both.

The model highlights the types of variation that the ideal experiment aiming to isolate the three frictions would generate, as well as how social commerce may affect them. To isolate search frictions, treatments should either create matches or lower the cost of finding a random new supplier. To isolate adverse selection, treatments should either directly give information about a particular supplier or improve the ability to learn this information over time, but should not affect suppliers' incentives. To isolate moral hazard, treatments should strengthen the incentives of the supplier or firms' perceptions of these incentives, but should not provide any other information. More broadly, we view social commerce as making this type of remote contracting feasible (so search costs become finite), facilitating social learning (alleviating adverse selection), and facilitating joint punishment strategies (alleviating moral hazard).

The experiment comprises three treatments that aim to target the frictions while being of real-world interest in their own rights by mirroring common ways that firms use social commerce. The treatments involve connecting a random 80% of these 1,862 garment retailers and wholesalers

in Dakar with subsets of a pool of 30 suppliers that we recruited in Türkiye—the second-largest source of ready-to-wear garments in our setting—and then cross-randomizing the provision of additional trust-relevant information about the types (adverse selection) and incentives (moral hazard) of these suppliers. All treatments are randomized and implemented at the buyer level.

In the search treatment, we add treated firms to the supplier WhatsApp groups of three different suppliers in Türkiye. This creates three new matches for each firm and provides a channel for them to see what these suppliers sell. Among these Search-treated firms, half receive the Adverse Selection (AS) treatment, in which we add them to a fourth WhatsApp group that mirrors the real-world information-sharing discussion groups described above. This buyer-only WhatsApp group contains 20-30 firms in the same treatment arm, all of whom have been newly matched with the same three suppliers. We also seed these discussion groups with initial information: each firm receives a positive recommendation for one supplier, based on real mystery orders we commissioned prior to the study. This treatment thus directly provides information about types and facilitates social learning. Finally, a cross-randomized half of the Search-treated firms receive the Moral Hazard (MH) treatment, in which we inform and emphasize to them that we have given all of the study suppliers high-powered incentives. In particular, we explain to treated firms that we will ask the firms in the study to rate the suppliers, and that we will remove suppliers that consistently receive negative feedback, meaning they will lose access to a large number of potential clients. We make clear that the suppliers know this and therefore have strong incentives to exert effort. This treatment thus aims to shift firms’ perceptions about the suppliers’ incentives by invoking a joint punishment strategy with the experimenter acting as the coordination device.

Buyers are thus divided equally across five arms: Pure Control, Search Only, Search + AS, Search + MH, and Search + AS + MH. We interpret the experimental evidence in two layers. First, whether creating new matches is sufficient to generate relationships, or whether addressing trust frictions (broadly defined) is necessary. We do this by comparing Pure Control, Search Only, and the trust arms jointly, which does not depend on how AS and MH are separated. Second, the relative importance of the trust frictions and their interaction, for which we compare the three trust arms. Decomposing AS and MH in a real-world B2B setting is challenging, as providing information about types could change incentives, and vice versa. A key feature of our design that aims to mitigate this is that all randomization is at the buyer level, meaning that the actual suppliers and their incentives are held fixed. The suppliers were not made aware that some buyers would be in discussion groups or even that there were randomized treatments at all—from their perspective, they were simply provided with a large and undifferentiated pool of new contacts, with no obvious way to tell them apart even if they wanted to. The trust frictions are therefore identified by changing the information environment that buyers face: the AS treatment primarily provides buyers with information about supplier types, and the MH treatment primarily provides buyers with information about supplier incentives.

We measure outcomes using three independent data sources. The first is a mystery shopping exer-

cise in which trained surveyors act as real customers and attempt to buy high-quality goods from the study firms. We designed this to measure whether the treatments improve access to foreign goods along three pre-specified dimensions: horizontal (access to a wide set of differentiated varieties), vertical (access to high quality varieties), and price (cheaper access to otherwise available varieties). Pooling all treatments, treated firms are 7.1 percentage points ($p < 0.01$) more likely to have a close match to the shopper’s request (a 23.1% increase) and, conditional on having a close match, the purchased goods score 0.412 standard deviations ($p < 0.01$) on a vertical index that includes quality evaluations by hired experts. We find no significant change in prices, suggesting that gains stem primarily from access to a wider set of varieties and higher-quality varieties. We do not find evidence that the effects in the trust-treated arms are larger than Search Only.

These short-run mystery shopping results suggest that there are gains from trade available from connecting firms to suppliers abroad. However, to realise these gains in practice, firms must develop these connections into medium- or long-run relationships. We measure this using data from our two other sources: a follow-up survey that we conducted after 3 months, and real-time, transaction-level administrative data from Senegal’s largest mobile money provider, tracking a share of transactions with study suppliers for up to 22 months after the study started.

In the survey data, we find that (pooled) treatment increases the share of firms with a regular supplier in Türkiye by 3.7 percentage points ($p = 0.088$), a 22.2% increase. Disaggregating the treatments matters: the effect comes primarily from the trust arms, and, while the point estimate is larger for AS than MH, the bulk of the effect comes from the Search + AS + MH group at 7.5 percentage points ($p < 0.01$), consistent with the implication from the model that the trust treatments are complementary. We find no effect on the total number of suppliers, suggesting that firms have substituted away from a local wholesaler and towards importing directly. The mobile money data exhibits a very similar pattern (although the estimates are less precise): the trust groups, and in particular the Search + AS + MH group, order considerably more from study suppliers than Search Only. The fact that we find this pattern in two completely independent datasets leads us to conclude that the trust treatments played an instrumental role in converting the initial connections into relationships.

Finally, we analyze measures of monthly profit and sales from our survey, which display the same broad pattern. Increases in profits and sales are concentrated in the upper part of the distribution: limited effects below the median, significant gains in the order of 15% to 20% in the upper third, and larger but imprecise effects at the very top. This is consistent with the model, where the gains from alleviating trust frictions scale with firm size. The OLS average treatment effects are correspondingly large—43.8% and 40.2% increases in profits and sales—reflecting the pull of the upper tail; the magnitudes reduce by half (but remain significant) when we winsorize at the 1% level. As with the relationship outcomes, the point estimates on AS are larger than MH, but are mostly driven by Search + AS + MH. We thus conclude that firms are able to realise meaningful gains from accessing suppliers abroad by using social media to overcome search and trust frictions.

In the last part of the paper, we use the experimental treatment effects as moments to estimate the model. The estimates imply that firms behave as if three-quarters of suppliers in the market are bad types, and that moral hazard requires paying an incentive premium of around 9%. To our knowledge, these are the first experimental estimates of these structural parameters, which can be used to inform quantitative models in similar settings. Estimated search costs are large but imprecise. We use these estimates to simulate the gains from policies that alleviate the frictions in the entire marketplace (rather than only among a few suppliers) to various extents. We find that these gains are large, though, as in the experiment, the return to alleviating moral hazard is substantially higher when adverse selection is alleviated simultaneously.

Taken together, the evidence points to a dual constraint: reducing search frictions expands access to varieties and quality, but alleviating trust frictions—especially when done jointly—is necessary for these matches to develop into lasting and profitable relationships. We show how firms use social media infrastructure to find, learn about, and build relationships with suppliers.

This paper contributes to several strands of literature. First, we contribute to the literature on information and contracting frictions in firm-to-firm relationships. Recent experimental work finds that reducing search or matching frictions can substantially improve firm outcomes, while also suggesting that the transition from potential match to active trading relationship is not automatic (Bergquist et al., 2024; Rudder and Dillon, 2024; Cai et al., 2024). Observational and structural work, meanwhile, suggests that trust and contracting frictions are important determinants of the terms, scale, and durability of relationships (Antràs and Foley, 2015; Macchiavello and Morjaria, 2015, 2021; Monarch and Schmidt-Eisenlohr, 2023; Startz, 2025; Brugués, 2026). We bring these strands together by experimentally varying both matches and trust-relevant information in the same setting. This helps reconcile why matching interventions have large effects in some settings but limited persistence in others: whether a match develops into a relationship depends on the trust environment (which depends on market and institutional features such as whether quality is observable, payments occur before delivery, partners are socially or geographically distant, and past behavior is visible). We also decompose trust into adverse selection and moral hazard, two frictions often studied separately, and we show how their interaction matters.

Second, we contribute to a growing literature on the role of technology in reducing trade frictions. Prior work has documented how the internet (Freund and Weinhold, 2004; Fernandes et al., 2019; Akerman et al., 2022) and formal platforms like Alibaba (Lendle et al., 2016; Chen and Wu, 2021; Carballo et al., 2022; Bai et al., 2023) reduce the impact of distance and facilitate cross-border trade. However, most of this evidence comes from formal platforms in high- or middle-income settings. We show that, in a typical lower-income setting, the vast majority of firms operate outside these platforms, relying instead on informal, flexible, and highly relational tools like WhatsApp and Facebook—a form of “social commerce” rather than traditional e-commerce. As emphasized in Szeidl (2025)’s review article, we know little about how these new and widespread technologies shape trade frictions. We provide both descriptive and causal evidence: we document that firms

use social media as sourcing infrastructure and experimentally show that it can alleviate both search and trust frictions.

Finally, we contribute to a broader literature on community-based enforcement in the absence of formal institutions. A long tradition studies how groups sustain cooperation through informal monitoring, information sharing, and collective punishment, with examples spanning historical trade networks, where Maghribi traders used multilateral punishment systems to discipline overseas agents (Greif, 1993, 2006); the growth of microcredit, where joint liability that leveraged local information and social sanctions was seen as a central pillar (Morduch, 1999; Ghatak and Guinnane, 1999); informal and interfirm lending, where networks are used to sanction defaulting customers (McMillan and Woodruff, 1999; Karlan et al., 2009); and theoretical work identifying the conditions under which cooperation is feasible (Kandori, 1992; Ghosh and Ray, 1996).¹ A common constraint on these mechanisms is that they require information to flow and punishment to be coordinated, which is costly and limits their scale. Our study highlights how social media may relax this constraint: digital buyer networks can aggregate and transmit information about supplier quality at near-zero marginal cost, enabling the kind of multilateral enforcement that Greif (1993) describes at a scale and speed that was previously infeasible.

The rest of the paper is organised as follows. Section 2 presents the setting and descriptive evidence on how firms use social media in supply chains. Section 3 describes the model. Section 4 describes the experimental design. Section 5 describes the data and methods. Section 6 presents the results. Section 7 presents the model estimation. Section 8 concludes.

2 Setting

2.1 Ready-to-Wear Garments in Dakar

Our study focuses on the ready-to-wear garment industry in Dakar, the capital and economic hub of Senegal. This industry exhibits substantial horizontal and vertical differentiation, making it ideal for our study: horizontal differentiation (a wide range of varieties) is well suited for studying search frictions, and vertical differentiation (the presence of high and low qualities) is well suited for studying trust frictions. It is also an important industry in its own right: in a consumer survey that we conducted with 400 households, ready-to-wear garments represented an average of 48% of clothing expenditure and 6% of total household expenditure.

Within this industry, our study focuses on goods made in Türkiye. We chose Turkish-made goods for two reasons. First, Türkiye is the second largest source of ready-to-wear garments in Dakar (after China). Second, Turkish-made goods have a reputation for being higher quality than goods

¹There is also a wide literature outside of economics providing rich qualitative and historical accounts of how shared ties substitute for formal institutions. For example, Cohen (1969) studies the Hausa trade diaspora in Yoruba towns in 1960s Nigeria, Weidenbaum and Hughes (1996) studies the "bamboo network" of Chinese entrepreneurs across Southeast Asia in the latter half of the 20th century, and Chin et al. (1996) studies the role of Korean immigrants in Los Angeles in 1968-1977 in facilitating trade in the wig industry.

made in China, which is ideal for studying trust frictions.² In fact, in this setting, asking for a good “Made in Türkiye” is a common way to signal willingness to pay for quality.

2.2 Sample

Firms in Senegal The main subjects of the study are 1,862 small firms in the ready-to-wear garments industry in Dakar. These firms are typical of small, informal, owner-operated businesses in many large cities in lower- and middle-income countries. Our sample includes both firms that operate a physical store in a market (33%) and online-only firms that sell exclusively through platforms like WhatsApp and Facebook (67%). Firms with a physical store were recruited through a census in selected markets that sell both high and low quality goods while online-only firms were identified through a combination of advertisements on Facebook and snowball sampling.

Across African cities, the prevalence of firms that operate exclusively online has grown rapidly in recent years, yet they are typically not captured in existing datasets. Including both physical and online-only firms in our study thus allows us not only to close this representativeness gap, but also to offer an opportunity to document and understand the behavior of these firms. 93% of online firms and 83% of firms with a physical store agreed to participate when contacted. To ensure the relevance of the study, we conducted a short screening survey with some eligibility criteria (see Appendix F.1), which were met by 85% of online firms and 89% of physical-store firms.

At baseline, 91% of firms sold Turkish-made goods, which represented 44% of sales on average. Average monthly profit is 227 USD and 33% of firms sell wholesale. 31% of firms have at least one regular supplier based abroad (whom they access mainly through travel or the social commerce channels described in Section 2.3); most others buy imported goods from wholesalers in Dakar. Firms are wary of unknown suppliers abroad. 60% know more than one other firm that has had a bad experience purchasing without seeing the goods. When asked, if ordering from 10 unknown suppliers abroad, what share would arrive as contracted, the median firm answered only half.

There are many ways to source goods—traveling abroad to buy from suppliers in person, buying remotely from suppliers using relational contracts, buying from local wholesalers, etc.—that each have costs and benefits, and determining the optimal market structure depends on many factors beyond the scope of this paper. What is clear is that access to suppliers abroad was a valued margin for study firms: the primary benefit of participation was the possibility of being connected to such suppliers, and lack of interest was not a binding constraint in recruitment. The fact that many firms valued these connections but had not established them directly is itself suggestive of the search and trust frictions that we study.

²To quantify this, in our consumer survey we showed households images of garments, randomizing whether we said they were made in Türkiye or made in China, and asked for their willingness to pay. We plot the distribution in Figure 4. The Türkiye CDF is shifted to the right of the China CDF, with an average premium of 34% ($p < 0.01$).

Suppliers in Türkiye The study involves connecting firms in Senegal with suppliers in Türkiye. Our experiment involved 30 suppliers. We conducted a census of two large neighborhoods of Istanbul that are textile wholesale and export hubs to many parts of the world, including to several countries in West Africa. Among the 89 suppliers that met our inclusion criteria—including being based in Istanbul, regularly exporting ready-to-wear garments to West Africa, of Senegalese nationality, and using WhatsApp to export to Senegal (see Appendix F.2 for further detail)—we then conducted a mystery shopping exercise to screen out any suppliers that were unwilling in practice to engage with new customers or were otherwise unresponsive. We conducted this mystery shopping in randomly-ordered waves and invited suppliers to participate in the study on a rolling basis, with recruitment closing upon reaching our target of 30 suppliers. This exercise also provides baseline information, which we use in the Adverse Selection treatment. Since our selection criteria were primarily about being active—which would be visible to firms anyway—we expect the supplier pool to be a reasonable representation of the type distribution.

The suppliers in our study are not manufacturers—rather, they are agents of Senegalese nationality, based in Istanbul, that specialize in connecting the goods sold by Turkish manufacturers or wholesalers with African markets. Social scientists have extensively documented the importance of such intermediaries with shared ties in both present and historical trade (see Rauch (2001) and references therein).³ Indeed, among the 31% of firms in our study with a supplier abroad at baseline, 72% report that at least one such supplier is a Senegalese national. The supply chain thus has (at least) two layers of intermediation: foreign manufacturers and wholesalers sell to Senegalese agents abroad, who in turn sell to firms in Dakar—either directly, or through local wholesalers. Our experiment lowers the cost of the direct channel, which 22% of firms already use, rather than introducing a new organizational form.

2.3 Social Media in Supply Chains

In this section, we present new descriptive statistics on how small firms use social media to interact with and learn about suppliers. While the growth of firms in LMICs using social media as a way to buy and sell goods—often called social commerce—has attracted considerable media attention (The Economist, 2023; Le Monde, 2023; The New York Times, 2024), there are few statistics available to formally document this phenomenon. One exception, a survey of six African countries by GSMA (2023), finds that, among firms that have at least some online presence for selling to customers, 60% use social media as their *only* channel of doing so. We are not aware of any comparable statistics for B2B trade, and the evidence we present below aims to help fill this gap.

Suppliers, both domestic and international, use social media in a variety of ways that can comple-

³In our setting, shared nationality substantially reduces logistical barriers, such as language differences, destination-specific knowledge, and payment technologies. However, as the statistics described above show, it does not eliminate search and trust frictions. Indeed, our experiment finds that these frictions remain important even when buyers and suppliers share a culture and country of origin. The shared payment technology also provides a unique measurement advantage to the study: these agents accept payment through Senegal’s largest mobile money platform, so we can use the platform’s administrative data to track transactions directly.

ment or substitute for face-to-face trade, ranging from public-facing activities that expand reach, such as managing a Facebook business page or advertising products on TikTok, to more private activities, such as using WhatsApp groups to engage with regular customers. Indeed, social media companies are creating features to facilitate this. For example, WhatsApp Business, launched in 2018, allows a supplier to maintain a profile with product catalogs, run targeted advertisements on Facebook and Instagram that click through to this profile, and manage inquiries at scale (see Panel (a) of Figure 2 for an example of a profile). Firms can also use social media to exchange information about suppliers, and the fact that firms can broadcast negative experiences means that suppliers' reputations are at least somewhat public. Our experiment focuses on WhatsApp as suppliers and firms actively use WhatsApp groups in two specific ways that appear closely related to alleviating search and trust frictions. We can thus mirror these groups to provide a natural way to experimentally target different frictions (as opposed to, for example, experimentally creating Facebook business pages, which do many things at once). We first describe these two types of WhatsApp groups, and then present statistics on broader social media use.

Supplier Groups Many suppliers in our setting—both domestic and international—operate WhatsApp groups with their clients to advertise goods, post prices, and highlight new items in stock. We refer to these as “supplier groups” (see Figure 2 Panel (b) for an example). A typical group is managed by one supplier contains 50-100 of their clients. These groups are curated client lists rather than mass broadcast channels: suppliers add buyers by invitation and often remove members that have not made orders recently. These are not discussion groups; indeed, usually only the supplier has permission to post. Buyers interact with the supplier by sending private messages to negotiate prices or make inquiries. We view these groups as digital storefronts where clients can browse suppliers' offerings. This dissemination leverages canonical social media features such as social connectivity (group membership, contact lists, forwards), platform engagement (notifications, read receipts), and identity/reputation (supplier profiles visible to clients).

Supplier groups primarily reduce search frictions by allowing firms to observe varieties and prices from suppliers at distance, and then to negotiate and follow up at minimal cost. Moreover, these groups are designed to integrate with the large growth in B2C social commerce: firms can directly forward supplier images to their own customers to gauge interest or make targeted sales. To a lesser extent, supplier groups may also mitigate trust frictions as a large, active group is a valuable asset that takes time to build, raising the reputational cost of dishonest behavior.

Firm Discussion Groups Firms also use WhatsApp groups to share information—for example, about suppliers—with other firms, which we refer to as “discussion groups”. These primarily address trust frictions as they allow firms to exchange experiences and warnings. In our sample, 23% of firms are members of at least one group whose explicit purpose is information sharing between businesses. An important feature of these groups is that they also serve as directories through which firms identify other buyers they can contact privately. This is consistent with the

broader patterns we document: 67% of firms have shared recommendations or warnings about suppliers with other firms in the past year—a form of informal collective punishment in which firms broadcast negative experiences to deter future misbehavior (see Figure A1 for examples). While these WhatsApp discussion groups are a novel modern phenomenon, they are part of a much older social mechanism of informal, community-based systems of information sharing and enforcement, that are often sustained by social networks or social norms. Indeed, our setting is characterized by strong social norms of reciprocity and mutual aid.

Statistics on Social Media Usage In Figure 3, we present statistics from the 1,862 firms in our baseline survey. We present sample-wide averages here, but we also show in Appendix Figure A3 that the results are identical among firms with and without a physical store, reflecting the widespread nature of this phenomenon. In Panel (a), we plot the share of firms that reported using different types of social media to obtain information about suppliers. WhatsApp is ubiquitous: 92% of firms use WhatsApp Status (a feature, used less often in the United States, that broadcasts content to all contacts for 24 hours), and 86% are in at least one supplier WhatsApp group. TikTok, Instagram, and Facebook are also popular for sourcing, each used by about a third of firms. Panel (b) shows the distribution of the number of supplier WhatsApp groups that firms belong to. Firms belong to many supplier groups—almost half of firms are in 5 or more—allowing them to continuously observe information about different varieties and prices. Moreover, firms are not simply passive observers: the distribution of the number of groups that they have bought goods from in the past year is almost identical.

Panel (c) shows why firms use these groups: most report that the groups help them follow new product releases, expand the set of varieties that they can offer, and compare prices across suppliers. Finally, in Panel (d), we show the location of the suppliers running these groups. The majority of firms (81%) are in a group with a supplier located in Senegal, showing that social commerce is used actively for domestic trade. A large minority (27%) are in at least one group with a supplier located abroad, including 21% in Türkiye, 12% in China, and 6% in Dubai.

Overall, firms are very familiar with supplier WhatsApp groups, but, since only 21% are in a group with a supplier in Türkiye, our experiment is still able to generate meaningful variation.

2.4 e-Commerce in Supply Chains

While social commerce appears to have grown rapidly, formal B2B e-commerce platforms like Alibaba have had limited success in this market. They are seldom used by the firms in our survey: 85% have never bought from these platforms, and, of those who have, half do so less than once a year, despite 88% having heard of them. These rates of non-usage are uncorrelated with firm size (see Appendix Figure A2). When asked why they do not use such platforms, the most common answers were that firms find them too complicated (40%), often due to language barriers and bank access requirements, and that firms do not trust them (33%). This is consistent with a broader trend

of formal e-commerce platforms having limited success at penetrating African markets.⁴

The picture that emerges is one in which firms have organically built sourcing infrastructure that leverages social media in ways that appear closely related to alleviating search and trust frictions, while formal platforms remain peripheral. Yet, social commerce likely does not fully eliminate search and trust frictions, and our experiment tests the extent to which the specific features of social commerce can further alleviate these frictions.

3 Theory

In this section, we describe a dynamic model of relational contracting with adverse selection and moral hazard, embedded within a sequential search framework. We use this to discuss how social commerce may affect the frictions, analyze how the frictions interact with each other, and highlight what types of variation would separate them. In Section 7, we use the experimental treatment effects as moments to estimate the parameters that govern the frictions. While the forces in the model are all canonical, we are not aware of existing work that combines them in this way.⁵

3.1 Setup

Time is discrete and infinite horizon. All players are risk neutral, with common discount factor δ . Every period, a firm with productivity z buys high quality inputs q_t^j from one or more suppliers $j = 1, 2, \dots$ and sells them downstream according to a downward-sloping demand curve that generates revenue $zr(q_t^1, q_t^2, \dots)$, where r satisfies gross substitutes and is strictly increasing, concave, and with $\lim_{q \rightarrow \infty} \partial r / \partial q = 0$ in each argument. Inputs may also be of low quality, and for simplicity we set the productive value of low quality inputs to zero. The firm can always buy high quality inputs without frictions from local suppliers at constant price c_l , which can be thought of as visiting the wholesale market in Dakar and directly inspecting quality.

3.2 Three Frictions

Sequential Search The firm can also engage in international trade. To do so, the firm must pay a search cost s to match with a random supplier abroad. Upon matching, both parties observe a draw of a fixed match-specific productivity shifter, $\psi \sim G$, that transforms inputs q_t into effective inputs ψq_t . The rationale for engaging in international trade may be horizontal (if ψ is interpreted as representing newer varieties or if the revenue function r features love-of-variety), vertical (if ψ

⁴B2C marketplaces have expanded but remain small. Jumia—the largest pan-African platform—served 6 million consumers across nine countries in 2024, still modest relative to national retail sales. Local B2B distribution apps (such as Twiga in Kenya or TradeDepot in Nigera) are also emerging, but these primarily aim to link domestic FMCG suppliers to micro-retailers rather than facilitate general B2B trade, and adoption among our surveyed firms is negligible.

⁵The dynamic moral hazard part is closest to Ray (2002); the adverse selection part is closest to Yang (2013). In the dynamic contracting models closest to ours, the principal’s outside option is usually taken as primitive—here, it is endogenous via (standard) sequential search. While existing models often include both moral hazard and adverse selection, relatively little work characterizes how interventions that separately target them interact. We also abstract from behavioral frictions—e.g., overly pessimistic priors—that can depress search; see Szeidl (2025) for a review.

is interpreted as a quality shifter), or to obtain lower prices (if ψ is interpreted as a price shifter).⁶ We provide a formal proof in Appendix B, but standard arguments imply that, if the firm ever searches, they will do so until they obtain a draw that exceeds an endogenous cutoff, and we write the expected utility of doing this as $\bar{U}(z)$. As the search cost is fixed, there exists a cutoff $\bar{z}$ such that only firms with $z > \bar{z}$ ever search, an importing equivalent of the result in Melitz (2003).

Moral Hazard If the firm matches with a supplier abroad, they play a repeated game. The stage game has three sub-periods. First, the firm orders a quantity q_t at price p_t and pays $p_t q_t$ upfront. Second, the supplier chooses an unobservable action $a_t \in \{0, 1\}$. If they choose $a_t = 1$, they pay cost $c q_t$ and the goods are high quality with probability 1, where $c > 0$ is a constant marginal cost. If they choose $a_t = 0$, they instead pay a lower cost $(1 - \xi)c q_t$ with $\xi \in (0, 1)$ but the goods are high quality only with probability λ . Avoiding the action and cutting a share ξ of the cost may, for example, be interpreted as purchasing the goods from a cheaper but riskier manufacturer that only delivers with probability λ . Third, the good arrives and the firm observes the quality. We focus on grim trigger strategies, in which both parties take their outside option if a low quality realization occurs, because these strategies provide maximal incentives.

While the action a_t is unobservable, the firm can incentivize the supplier to choose $a_t = 1$ by promising future rewards tied to repeated high quality realisations. In particular, any equilibrium in grim trigger strategies that sustains $a_t = 1$ must satisfy the following Dynamic Incentive Compatibility Constraint (DICC),

$$(1 - \lambda)\delta V_{t+1} \geq \xi c q_t \quad (\text{DICC})$$

where $V_{t+1} \equiv \sum_{n=1}^{\infty} \delta^n (p_{t+n} - c) q_{t+n}$ is the supplier's (endogenous) discounted sum of future profits. Intuitively, the supplier has already received the payment $p_t q_t$, so they will only choose $a_t = 1$ if the increased probability $(1 - \lambda)$ of obtaining the future value of the relationship, δV_{t+1} , exceeds the additional cost $\xi c q_t$. We can see that the constraint holds trivially for $\xi = 0$, and that larger ξ implies a more binding moral hazard problem. This constraint highlights two important features of any equilibrium. First, the supplier needs to be paid above marginal cost at some point to provide incentives. Second, the value of the relationship provides a (time-varying) ceiling on contractible quantity, as going beyond this would make the supplier too tempted to cheat.

Adverse Selection There are two types of suppliers abroad: a share $1 - \mu$ are good and the remaining μ are bad. A supplier's type is fixed over time, known to the supplier, and unobservable to firms. Good type suppliers are as described above. Bad types, in contrast, are unable to choose $a_t = 1$ (for example, because they are unskilled). We assume that λ is sufficiently low that, if the

⁶We do not model multi-variety firms as this would introduce considerable complexity that our data are not well suited to test. Our mystery shopping exercise helps understand the mechanisms for potential gains from trade, but we would need strong assumptions to extrapolate from this to a welfare-relevant horizontal demand parameter without detailed data on consumer demand for varieties. We thus interpret ψ broadly as a shifter that can encompass horizontal, vertical, or price-based gains as the rationale for forming relationships.

firm knew that the supplier was a bad type, they would prefer to exit the relationship and take their outside option. We denote the firm's beliefs that a given supplier is a bad type after observing t high quality realisations as μ_t , with $\mu_0 = \mu$. The firm updates this belief each period using Bayes' Rule. To prevent unrealistic equilibria in which the firm can learn the supplier's type without taking any risk by repeatedly ordering $\epsilon > 0$, we also impose that, for order sizes below some $\underline{q}$, learning is an increasing function of quantity, $\mu_{t+1}(q_t)$, with $\mu_{t+1}(0) = \mu_t$ and $\mu_{t+1}(\underline{q}) = \mu_{t+1}^{\text{Bayes}}$.

3.3 Relationship with Social Commerce

We interpret the growth of social commerce as affecting these three information frictions as follows. For moral hazard, it facilitates a shift towards a joint punishment equilibrium. For simplicity, suppose there are N symmetric firms for whom the technology allows them to coordinate behavior through social networks. Then, the supplier risks losing not just V_{t+1} but NV_{t+1} if they cheat. This strengthens suppliers' incentives, and, from inspecting the DICC, is equivalent to a reduction in the contracting friction parameter from ξ to $\xi' = \xi/N$.

For adverse selection, we interpret social commerce as facilitating social learning. For simplicity, suppose that N firms that match with the same supplier agree to share quality realisations, which they may choose to do as they are jointly better off coordinating if competition is not too strong (e.g., if they operate in geographically dispersed markets), or because they share social ties or are subject to social norms that yield non-monetary payoffs.⁷ Then, the firm learns faster: for example, if the supplier is a good type, the firm's posterior after one period updates to μ_N rather than μ_1 .

Finally, for search, social commerce both lowers the search cost s —for example, by allowing suppliers to create business profiles that appear on searches on Facebook or Instagram—and, at more fundamental level, helps make this type of remote contracting feasible by significantly lowering the cost for firms to interact and exchange media bilaterally with suppliers abroad.

3.4 Further Contracting Details

Outside Options: The firm's outside option, denoted $\bar{U}$, is given by the maximum of the expected value of searching again or of buying locally forever. Suppliers have outside option 0, reflecting the idea that relationships are separable for the supplier.

Contracts and Enforcement Constraints: In period 0 of the relationship, the firm makes a take-it-or-leave-it offer of a long-term contract to the supplier, which is a sequence $\{q_t, p_t\}_0^\infty$. The contract is relational: both parties can agree on a long-term plan, but are unable to commit to it. Both parties therefore face a Dynamic Enforcement Constraint (DEC) that requires that in every period

⁷ A more formal and general treatment of firms' incentives to share information in a dynamic setting can be found in Golosov et al. (2014). If deviations from truth-telling are at least occasionally discoverable, then the group can exclude any firm that deviates from accessing future information revealed in the group. Hence, for sufficiently patient firms, truthful, state-by-state disclosure is optimal, so full revelation can be sustained in equilibrium with repeated cheap talk, including under competition or when disclosure carries stigma (e.g., revealing having been cheated).

t it must be optimal for them to not walk away: $U_t \geq \bar{U}$ and $V_t \geq 0$ for all t . The DEC prevents the firm from (for example) promising large bonus payments to the supplier in the distant future, because when the distant future arrives the firm would prefer to renege.

Limited Liability: The contract must satisfy $p_t q_t \geq c q_t$ for all t ; that is, the supplier must make weakly positive profits period-by-period. We impose this restriction to prevent large period-0 rent extractions that we do not see in our empirical setting. Instead, in our discussions with suppliers, we often heard of selling at cost early in the relationship to establish a reputation, which this restriction permits (and which will occur in equilibrium).

Equilibrium: We focus on Perfect Public Equilibria in which the good type supplier chooses $a_t = 1$. We also restrict attention to pooling equilibria, which we show in Proposition 1 in Appendix B is without loss. Intuitively, the firm cannot commit to not immediately terminating the contract if it learns that the supplier is a bad type, which means that bad types would always earn 0 upon revealing their type, but can earn a positive payoff by mimicking the good type.

3.5 Results

The optimal contract is the solution to the dynamic program formed by the firm maximising its discounted lifetime profits subject to the DICC, the DEC, and limited liability. We state this program explicitly and characterize the solution in Appendix B.⁸ Here, we highlight three aspects that are relevant to the interpretation of the experiment.

Result 1 (Intensive Margin). *Trust frictions (adverse selection and moral hazard) distort quantity and value purchased from the supplier abroad downwards.*

This result can be seen most intuitively in the following wedge equation relating marginal revenue and marginal cost, derived from the FOCs of the dynamic program:

$$r'(q_t^*) = \underbrace{\frac{1}{1 - \mu_t(1 - \lambda)}}_{\text{Adverse Selection}} \underbrace{(1 + \xi \rho_t^*)}_{\text{Moral Hazard}} c, \quad (1)$$

where ρ_t^* is a term proportional to the Lagrange multiplier on the DICC. Both wedge terms are weakly greater than 1, and so distort q_t^* downwards. The adverse selection term captures the firm's incentive to lower quantity because with probability $\mu_t(1 - \lambda)$ the supplier is a bad type and delivers low quality. The moral hazard term captures the firm's incentive to lower quantity (and increase the price) to ensure the supplier is not tempted to cheat by avoiding a share ξ of the

⁸In brief, the optimal contract is backloaded, with both quantities and markups rising over time. In an initial phase, quantity builds up ($q_{t+1} > q_t$) and the supplier invests in the relationship by selling at cost ($p_t = c$). Second, in a mature phase, quantity converges to a long-run q^{LR} and the supplier charges positive markups to reap the rewards of their reputational investment, i.e., $p^{LR} > c$. The long-run price premium makes the supplier the residual claimant: p^{LR} is as large as possible to maximise dynamic incentives while ensuring that the buyer still prefers to participate. Intuitively, the contract is the dynamic equivalent of selling the firm to the agent, where the buildup phase is a second-best way for the firm to extract surplus when charging the full rent upfront is prohibited by limited liability. This type of contract is a very general prediction of models of dynamic moral hazard without commitment (Ray, 2002; Martimort et al., 2017).

cost, which matters in proportion to how binding the DICC is (captured by ρ_t^*). Treatments that alleviate these frictions should therefore increase quantity on the intensive margin.

Result 2 (Extensive Margin). *The buyer's period-0 relationship value is decreasing in μ and ξ . Moreover, the productivity cutoff for importing, $\bar{z}$, is increasing in μ and ξ .*

This result highlights that adverse selection and moral hazard limit the gains from trade, which prevents some otherwise productive matches from developing relationships and causes some firms to opt out of searching for an international supplier altogether. Treatments that alleviate these frictions should increase the share of firms that import directly.

Result 3 (Relaxing Trust Frictions is Complementary). *Under a mild technical condition (stated in Appendix B), the buyer's period-0 relationship value satisfies $\partial^2 U_0(\mu, \xi) / \partial \mu \partial \xi > 0$.*

This result shows that interventions that lower the trust frictions are complements.⁹ The intuition is as follows. Suppose that the adverse selection problem is exogenously alleviated (μ decreases). Equation (1) then implies that, all else equal, the buyer's desired q_t increases. But then the DICC becomes even more binding (ρ_t^* increases), so the return to alleviating moral hazard (lowering ξ) is higher. That is, alleviating one friction may have little impact if the firm does not want to buy much anyway because of the other friction. This is formalized in the positive cross-partial above.

3.6 Implications for the Experiment

The model makes precise the three frictions that we will study in the experiment: search, adverse selection, and moral hazard. Search frictions limit the share of firms that import from suppliers abroad and Results 1 and 2 show that adverse selection and moral hazard not only reduce the share of firms that import but also lower the quantity and value ordered among firms that do import. To relax the search friction, we need a treatment that either lowers the cost of matching with suppliers abroad or improves the likelihood of finding a suitable match. To relax adverse selection, we need a treatment that improves beliefs (or improves the ability to learn) about a supplier or suppliers in general. To relax moral hazard, we need a treatment that either improves suppliers' incentives or changes firms' perceptions of suppliers' incentives. Finally, since our experiment will cross-randomize treatments for both adverse selection and moral hazard, Result 3 highlights that these treatments are complements in the model.

In addition to these predictions, the model also implies a variety of dynamic patterns, such as value traded building up over time. While we do find descriptive and anecdotal support for these patterns, which we discuss and relate to the literature in Appendix G, we do not emphasize formally testing them because statistical power would depend on the number of relationships, not the number of firms, in the study. In contrast, we designed the experiment to generate exogenous

⁹The condition rules out an extreme case where the revenue function is so elastic and λ is so large that improving beliefs causes the buyer to decrease early quantities (distorting them even further) in exchange for higher rents later. The full (necessary and sufficient) condition is technical and is thus stated in Appendix B. A simpler sufficient condition is $dV_0/d\mu < 0$, i.e., the supplier would prefer the buyer to believe they are a good type.

variation to test outcomes concerning access to foreign goods and relationship *formation*, which are powered at the firm level.

4 Experimental Design

4.1 Treatment Conditions

The experiment comprises three treatments that aim to target the three frictions—Search, Adverse Selection (AS), and Moral Hazard (MH)—while being of real-world interest in their own rights by mirroring common ways that firms use social commerce. The treatments involve connecting a random 80% of the 1,862 garment retailers and wholesalers in Dakar—whom we refer to interchangeably as firms or buyers—with the 30 suppliers in Türkiye, and then cross-randomizing the provision of additional trust-relevant information about the types (AS) and incentives (MH) of these suppliers. All treatments are randomized and implemented at the buyer level, meaning that the actual suppliers, their incentives, and the matching procedure are held fixed. AS and MH (which we refer to jointly as trust frictions) are therefore identified by comparing how buyers respond to the additional information tailored to target the different theoretical channels. We present a visualization of the design in Figure 1.

We divided the 30 suppliers into 10 buckets of 3 suppliers each. All suppliers in the same bucket specialized in the same sub-sector (men’s clothing, women’s clothing, or shoes and accessories). We asked each supplier to create a new supplier WhatsApp group for the study following standard conventions of this ubiquitous phenomenon (as described in detail in Section 2.3), including restricting posting rights to the supplier’s phone number. Suppliers knew that they would be connected to potential buyers through these WhatsApp groups, but were not told that buyers would be randomized into different treatments; from their perspective, their role was simply to operate the group and supply customers as they normally would. As each supplier operated one study supplier WhatsApp group, their posting behavior is constant across trust treatments.

Search 80% of firms receive the Search treatment. The model highlights that search costs can be identified by lowering the cost of matching with a supplier in Türkiye, so this treatment directly creates matches. We add Search-treated firms to the supplier WhatsApp groups of 3 different study suppliers, thereby connecting them and allowing the firm to see what the supplier sells, which we interpret as paying the search cost s three times on behalf of the firm. To minimize any inference of endorsement by the study team, we emphasize that we are not guaranteeing orders and we deliberately do not provide any further information beyond stating that the suppliers are exporters based in Türkiye and were recruited in a similar manner to how the firm itself was

recruited.¹⁰ We communicate to the control group that unfortunately we cannot add them to any supplier groups at this time, but that we might do so at the conclusion of the study.

Adverse Selection (AS) 50% of firms that receive the Search treatment also receive the AS treatment. The model highlights that identifying adverse selection requires either improving the ability to learn about a supplier’s type or providing information directly. This treatment does both. We add treated firms to a fourth WhatsApp group. This group is modeled after the information-sharing discussion groups described in Section 2.3: it does not contain any suppliers, but instead contains 20-30 other buyers in the AS treatment that were (randomly) allocated to the same bucket of three suppliers as part of the Search treatment. We explain that all members of this discussion WhatsApp group have been matched with the same three suppliers and that the purpose is for buyers to share information about the suppliers (either directly in the group or privately). A member of the study team moderates discussion and periodically encourages firms to share information.¹¹ While many firms participate in similar discussion groups at baseline, those groups are not organized around the specific study suppliers. The experiment is able to generate variation because we matched many firms to the same new suppliers, so it is possible for us to create a targeted information network that study firms could not have assembled on their own. We discuss how these groups influenced buyer perceptions and behavior in Section 6.2.1.

Since no firms have experience with the suppliers at this point, we seed the groups with initial information. Treated firms receive a phone call 2-3 days after recruitment from a recommender.¹² The recommender is part of a team of firms—who are not subjects in the study—that we hired prior to the study to make multiple mystery orders from all of the suppliers. On the call, the recommender (truthfully) describes a positive experience ordering from one of the suppliers that the firm was matched with and sends a photo of the item that they ordered (we designed the supplier buckets in a way that ensured a positive recommendation was possible in all buckets). The recommender is also in the discussion WhatsApp group, and posts a similar message there. To ensure firms were not surprised to receive this call, we asked all firms at the end of the baseline survey if they would be willing to call a few other firms to discuss their experiences working with the study suppliers. The information sharing in the group could in principle lead to more or less trade, depending on what information is shared. But given the fact that the seed always provided a positive and specific signal, we expect this arm to increase trade. Moreover, we deliberately

¹⁰We do not think that implicit endorsements are an important concern for our results, for at least two reasons. First, we conducted a brief phone survey with all firms one week after recruitment and asked how much, on a scale of 1 (very high trust) to 5 (no trust at all), they would trust one of the matched suppliers if they were to make an order. Only 32% of firms reported a trust level of 1 or 2. Second, endorsement effects would lead us to underestimate the effect of the trust treatments, which would strengthen our primary finding that the trust frictions are large.

¹¹The Senegalese ready-to-wear garment market is very large and most firms in the study are unlikely to be in meaningful competition with each other. Moreover, as discussed in Section 3.3, even under competition information sharing can be sustained in a repeated game if the gains to coordination are high, social ties allow for non-monetary payoffs, or social norms are strong. This is a setting where social norms strongly emphasize reciprocity and mutual aid.

¹²All firms not in this treatment condition instead receive a “placebo” phone call from a surveyor, asking them for their opinion about supplier WhatsApp groups in general.

implemented this treatment by having firms share information among each other—rather than the experimenter providing information directly—to capture the spirit of social learning.

Moral Hazard (MH) 50% of firms that receive the Search treatment also receive the Moral Hazard treatment, cross-randomised with the Adverse Selection treatment. The model highlights that identifying moral hazard requires either shifting suppliers’ incentives or shifting firms’ perceptions of suppliers’ incentives. Since all treatments are implemented at the buyer-level (the suppliers and their incentives are fixed across the treatment groups), this treatment does the latter. The MH treatment takes place at the end of the baseline survey. The surveyor (truthfully) informs and emphasizes to treated firms that the study has given all of the suppliers strong incentives to encourage reliable fulfillment, deliberately avoiding any language suggesting pre-screening. The full statement is printed in Appendix F.3. In brief, we explain that we will ask the firms to rate the study suppliers, and we will punish suppliers about whom we hear bad feedback. The punishment consists of removing the supplier from the study, meaning they will lose access to the large number of potential customers we have matched them with.¹³ Importantly, we stress that the suppliers are very aware of this and therefore have strong incentives. We deliberately used the loss of future business from study firms as the incentive effect—rather than, say, a financial bonus for good behavior—to capture the spirit of a joint punishment strategy, with the experimenter serving as the coordination device.

After delivering this message, the surveyor provides a *business card* to the firm. The business card has a phone number to call, and prominently highlights that this number should be used to rate the suppliers and/or to signal any problems. Untreated firms receive a similar card, but without any mention of ratings. Both cards can be seen in Appendix Figure A4.

As described above, all suppliers face the same high-powered incentives and we randomize whether this is made salient to firms.¹⁴ This implies that we should not expect the MH treatment to affect supplier defaults (conditional on a relationship forming). As in the other treatment arms, we make clear that we are not endorsing the suppliers or guaranteeing their behavior. The additional information in the MH treatment arm is simply that we have provided them with incentives. To the control firms, we explain that our only involvement is the initial connection. Overall, the primary function of this treatment is to shift firms’ views about the suppliers’ incentives while providing minimal information about the suppliers’ types.

¹³This punishment is credible as the supplier WhatsApp groups were newly created for the study, meaning that we are administrators and thus can replace the supplier with another from the wait list if necessary. We required that suppliers confirm that they understood and consented to this in order to participate in the study.

¹⁴While the model does not have a formal notion of incorrect beliefs, an untreated firm would not propose or agree to a contract that they believe violates the Dynamic Incentive Compatibility Constraint. This also means that this perception can persist as the firm never observes how the supplier would have behaved under a weaker contract.

4.2 Interpreting the AS and MH Variation

Two features of the design are crucial to interpreting how the treatments aim to separate AS and MH: (i) all randomization was at the buyer level, and (ii) suppliers knew only that we would match them to buyers and would collect ratings from these buyers. Each supplier was connected to around 150 buyers spanning all combinations of the trust treatments (S Only, S+AS, S+MH, S+AS+MH), but, from the supplier's perspective, this is simply a large and undifferentiated pool of new contacts for them to serve as they normally would. They were not told that buyers had been randomized into different treatment arms, and, even if they were, there would be no obvious way to distinguish which of these many new buyers are in which arm. The identification of the trust frictions thus comes from changing the information environment that buyers face regarding the study suppliers, which is feasible precisely because our experiment matches many buyers with these new suppliers.

One potential concern is that the AS treatment could also alleviate MH, which would happen if firms in the buyer-only WhatsApp discussion groups infer from this that suppliers have stronger incentives. To mitigate this, we explicitly (and truthfully) tell firms that suppliers are unaware of these discussion groups, so if firms used the group to coordinate behavior then that does not affect MH, but, rather, is precisely the social learning mechanism of AS. In principle, firms could try to convince the suppliers that these groups exist to attempt to influence their behavior. However, doing so credibly would be difficult as such a claim is essentially cheap talk, especially coming from a would-be buyer whom the supplier does not know. Moreover, suppliers generally know that buyers sometimes use WhatsApp groups to share information (as described in Section 2.3), so a generic claim to that effect would provide little new information—the buyer would have to be highly specific and detailed. Ultimately, we did not hear of buyers attempting to do this, and the study team did not receive any questions from suppliers about the presence of buyer-only discussion groups.

Another potential concern is that the MH treatment could also alleviate AS, which would happen if firms interpret the monitoring language as implying that suppliers have been pre-screened. To this end, we designed the MH script to emphasize incentives going forward (the script repeats three times that the suppliers have incentives to do a good job) and says nothing about supplier selection or past quality. Although firms could, in principle, wait for the other firms in the study to report problematic suppliers, in practice they typically decided quickly whether or not to initiate a relationship (in our mobile money data, 70% of firms that ever ordered from a study supplier made their first order within the first month, with no significant differences across treatment arms). Moreover, the script does not imply that any enforcement processes would be particularly fast.

Ultimately, we are interested in two sets of economically important comparisons. First, we want to measure whether alleviating search frictions by creating matches is sufficient to generate new relationships, or whether it is also necessary to address trust frictions broadly defined. We can do so by comparing Search Only to Pure Control, and then the trust arms jointly to Search Only,

which is valid regardless of whether our design achieves an idealized separation of AS and MH. Second, we want to understand the relative importance of the trust frictions and how they interact, which we can do by comparing the various combinations of the trust arms to Search Only. While we cannot guarantee that there were no cases where a buyer made the additional inferential steps described above for cross-contamination to occur, the core identifying content of our design is that the AS treatment primarily provides buyers with information about supplier types, while the MH treatment primarily provides buyers with information about supplier incentives. The cross-randomized design implements this separation at scale in a real-world firm-to-firm setting by targeting these different buyer-side information channels, while holding fixed the actual suppliers and their incentives and ensuring that, from the supplier side, buyers appeared simply as a large pool of new potential customers rather than members of different experimental arms.¹⁵

4.3 Randomisation and Balance Check

We randomly assigned firms to one of the five groups shown in Figure 1, stratifying on sub-sector (men’s clothing, women’s clothing, shoes & bags), whether the firm has a physical store, and whether the firm had prior direct importing experience. To check that the randomization operated as expected, we report a balance check in Appendix Table A1. The differences in means across groups are small and insignificant, and a joint test has p -value 0.958.

4.4 Sub-Treatments

We also cross-randomised two additional sub-treatments within the pure control group. The first aims to test whether the binding constraint behind the limited use of formal B2B e-commerce platforms is that firms do not know how to use them. We thus provide a short training on Alibaba that covers how to install the app, search for products, contact suppliers, and make purchases and arrange delivery. The second is a placebo to ensure that any effects of the adverse selection treatment are not driven by connecting firms *per se* (as in Cai and Szeidl (2018)). We thus create similar WhatsApp discussion groups here, where no firms have been connected with suppliers.¹⁶

¹⁵We intentionally did not use a Karlan and Zinman (2009)–style two-stage design. That approach requires the experimenter to set and randomize *post-match* contract terms at the relationship level to separate selection from behavior. In our B2B context, transaction terms are multidimensional (price, delivery, payment type, quality), privately negotiated, and often renegotiated, so it is not feasible or realistic to re-set terms for a subset of relationships. We thus instead varied the information provided to buyers about types (AS) and incentives (MH). Moreover, by building on commonly used social commerce infrastructure, our design mirrors the buyer-driven approaches that actually occur in this market.

¹⁶We find no effects of the placebo groups (see Appendix Table A9), which is not surprising as the intervention in Cai and Szeidl (2018) is significantly more intensive, with firms meeting in person and at length every month for a year.

5 Outcomes, Data, and Empirical Methodology

5.1 Data and Outcomes

Consumer Survey In March 2024, we conducted a 15-minute survey with 400 households in several areas of Dakar. Enumerators selected respondents by starting from a random point and then visiting every fifth household along the route. We use this to measure the relationship between consumer willingness to pay and our measures of garment quality (described below), and to calculate the statistics on household clothing expenditure presented in Section 2.1.

Baseline and One-Week Surveys Upon recruiting a firm, we conducted a 30 minute baseline survey covering their supplier relationships, social media use, e-commerce use, and profits, and is the source of most of the statistics presented in Section 2. In a short additional follow-up survey one week later, we asked for their views about their access to Turkish-made goods, and, if applicable, their beliefs about the trustworthiness of one of the suppliers we matched them with.¹⁷

Access to Foreign Goods We designed and implemented a mystery shopping exercise to measure—using real behavior—whether the treatments expand access to high-quality foreign-made goods, as well as to separate between different theoretical mechanisms for potential gains: horizontal (access to a wider set of differentiated varieties), vertical (access to higher quality varieties), and price (cheaper access to otherwise available varieties). We describe the full protocol in Appendix F.4, and here summarize the key details. Two weeks after a firm completes the baseline survey, a mystery shopper—a trained surveyor that acts as a real customer—contacts the firm over WhatsApp. The firm does not know that the mystery shopper is part of the study team, but is expecting the contact as we explain during the baseline survey that we will connect firms with customers. The mystery shopper describes that they are looking to buy a certain product for an event. We created a bank of 28 products, each defined by five criteria unrelated to quality, such as color and sleeve style (see Appendix Figure A5 for two examples).¹⁸ The primary outcome for this horizontal component, pre-specified in our PAP, is an indicator for whether the firm had a product that matched at least three of the five criteria.¹⁹

If the firm has the product, the mystery shopper buys it (with some exceptions described in Appendix F.4), and, once it arrives, two expert tailors assess its quality according to a 50-point score-

¹⁷As an early, self-reported measure of how the search and trust treatments shift perceptions, Appendix Table A2 reports one-week treatment effects on access and trust outcomes. All treatments significantly raise self-reported access to Turkish-made goods. The trust treatments jointly increase self-reported supplier trustworthiness, although it is individually significant only for the Search+AS+MH group. As these measures are taken after just one week, changes in trust primarily reflect the researcher-controlled components of the intervention rather than new relationship formation.

¹⁸We chose products that were fairly general—such as a red polo shirt or a black knee-length dress—and, for each product, we monitored whether each supplier had advertised the specific image in their group, and, if so, excluded it from firms that were or would have been matched with that supplier.

¹⁹Our definition of having a good was that the firm said that they could deliver it within two weeks, as this matches the natural cadence of retail trade for the firms without physical stores (and is common for firms with physical stores).

card that we developed (printed in Appendix F.5). To validate the quality measure, we presented a subset of these purchased goods to respondents in the consumer survey and elicited willingness to pay (WTP). We construct an indicator for high quality that is 1 if the quality score is above the median of its product type, and plot the distribution of WTP separately for high and low quality goods in Figure 5. The high quality distribution is a clear rightward shift relative to the low quality distribution, with an average premium of 35% ($p < 0.01$). In Panel (b), we show that WTP increases with the raw quality score, though it becomes flat towards the left tail as, below a certain point, consumers simply view goods as “low quality”. The outcomes are these two measures: the indicator for high quality, and the raw score.

We also measure whether the good was made in Türkiye. As we showed in Figure 4, it is common for consumers to ask if a good was made in Türkiye as this is strongly associated with higher quality. This suggests that quality is difficult for consumers to observe and so product origin plays an important role in WTP.²⁰ For most goods, we record this information from the label, and the outcome is 1 if the label says “Made in Türkiye” and 0 if it says it was made elsewhere.²¹

The mystery shopping also had a secondary goal of providing treated firms with an opportunity to experiment with the study suppliers. Nothing in the procedure makes this explicit, the goods are not specific to the study suppliers, and the mystery shopper does not know the treatment status of the firm, but if the firm was considering making an order, then the mystery shopper reduces the risk that they will be unable to find a buyer. We mystery shop at *all* firms (including Pure Control), so this secondary goal does not confound our analysis.

Followup Survey We conducted a 30-minute followup survey with similar questions to the baseline survey around 3 months after a firm is recruited to the study. We successfully surveyed 90% of the sample (1671 firms). The followup rate is very similar and not significantly different across the four treated groups, but is 5 percentage points higher and significantly different in pure control.²² The main outcomes relate to supplier relationships, profits and sales, and e-commerce use.

Mobile Money To complement the survey data, we also use real-time, transaction-level data from the largest mobile money provider in Senegal, Wave Mobile Money, made available for this study. This data contains the value, timestamp, sender, and receiver for the universe of transfers between the phone numbers of study firms and those of study suppliers. This data complements survey-based measures and has several advantages: (1) we can see transaction profiles over time, (2) it continues after the followup survey, (3) it captures real transactions (rather than self-reports).

²⁰We pre-specified this outcome, but did not attach it to either the horizontal or vertical dimensions. Since the consumer preference for Turkish-made goods reflects a preference for quality, it seems more fitting to include it under the vertical dimension.

²¹For the few goods where this was not possible, the hired experts evaluated origin based on design and construction cues. We show in Table A4 that the results are not sensitive to how we treat these goods.

²²We present Lee (2009) bounds for the main survey outcomes in Appendix Table A8. The results are similar, in line with the differential attrition being relatively small and a higher followup rate in control than treatment (i.e., the lower bounds trim the left tail of the control distribution, which is bounded for most outcomes).

While this data cannot capture the full universe of transactions, we expect that a non-trivial share takes place through this provider, for two reasons. First, as described in Section 4.1, we hired a set of non-study firms to make mystery orders from the suppliers prior to the intervention, and every supplier accepted payment through this provider. Second, in the baseline survey, 86% of firms reported regularly using this provider to pay suppliers for remote transactions, with 35% exclusively so.²³ However, larger wholesale orders are more often conducted through formal channels such as bank transfers or international remittance services (e.g., Western Union). This distinction is reflected in the baseline survey data: wholesalers are significantly more likely than retailers to use such methods—12% versus 4% of them, respectively.

5.2 Empirical Methodology

Our primary empirical method, pre-specified in our PAP, is the following OLS specification

$$y_i = \alpha + \sum_{j=1}^4 \beta_j T_{ji} + \delta y_{0i} + \gamma_s + \rho' X_i + \varepsilon_i, \quad (2)$$

where y_i is the outcome for firm i and T_{ji} for $j = \{1, 2, 3, 4\}$ are indicators for treatment arms Search Only, Search + Adverse Selection, Search + Moral Hazard, and Search + Adverse Selection + Moral Hazard. y_{0i} is the outcome measured at baseline, if available. γ_s are stratum fixed effects. X_i are firm-level covariates selected by Double Lasso (Belloni et al. (2014)).²⁴ We also report a similar regression that pools the four treated groups.

Inference Our primary method of inference is randomisation inference, as recommended by Athey and Imbens (2017) and Young (2019). We compute two-sided p -values for the sharp null of zero treatment effect using 2,000 permutations of the t -statistic. We report conventional heteroskedasticity-robust standard errors in parentheses, but we do not use these for inference directly. As each regression estimates four coefficients, we also calculate Romano-Wolf multiple-testing adjusted p -values in square brackets (Romano and Wolf, 2005). Since there are three combinations of trust treatments, we report the p -value of a joint test that all four coefficients are equal (computed by permuting the F -statistic) in order to test the null that none of the trust treatments have any effect.

Indexes To account for multiple hypothesis testing across outcomes, in tables that present more than one outcome measuring the same broad construct, we also present an index that aggregates the outcomes using the standardised inverse-variance weighted method of Anderson (2008). Since the disaggregated regressions may include different covariates y_{0i} and X_i , before indexing we first residualise each outcome using the covariates that were included in their respective regressions.

²³For comparison, 37% reported using the second-most-common provider, with 2% of them exclusively. Firms using this mobile money provider are more likely to be female-owned, sell online, be relatively new, and earn lower profits.

²⁴This means that, prior to each regression, we run lasso to predict y_i and each T_i and include the union of selected covariates. In practice, this selects few covariates and thus makes little difference. Nonetheless, in Appendix D, we reproduce the main tables using only treatment indicators. The results are very similar.

Quantile Regression Prior evidence suggests that some of our outcomes, notably profit and sales, may have thick tails and have important distributional treatment effects (Meager, 2022). We therefore included in our PAP that we may use quantile regression to examine distributional treatment effects. These follow the same specification, except that we omit the stratum fixed effects (γ_s) and the Double-Lasso-selected covariates (X_i) as quantile regressions are much more demanding and the Belloni et al. (2014) covariate selection method is designed for linear models.

6 Results

In Section 6.1, we present results on access to foreign goods, as measured by our mystery shopping activity. In Section 6.2, we present results on supplier relationships. In Section 6.3, we present results on profit and sales. In most tables, we show the pooled regression in Panel A, and then disaggregate across the treated groups in Panel B, with standard errors in parentheses, Romano-Wolf multiple-testing adjusted p -values in square brackets, and the p -value of a joint test that all coefficients are equal at the bottom.

6.1 Access to Foreign Goods

As described in Section 5, we designed a mystery shopping exercise to obtain a revealed preference of access to foreign goods, and to separate potential gains across horizontal, vertical and price-based mechanisms. Table 1 reports the outcomes of this exercise.

Horizontal In Column 1, we report the main horizontal outcome, which is an indicator for whether the firm had a product with at least three horizontal criteria.²⁵ Pooling the treatments together, treated firms are 7.1 percentage points more likely to find a suitable good ($p = 0.010$). This is a 23.1% increase from the control mean of 30.7%. In Panel B, we see that the effect is broadly similar in all four treatment groups and we do not reject the joint null that they are all equal, which implies that the effect is not different for the groups with the trust treatments. The results thus show that relaxing the Search friction by connecting firms with suppliers abroad via social media led to a sizable increase in the set of varieties that they can provide to real customers.

²⁵Our PAP specified this indicator as the main outcome. For robustness, we use the raw number of criteria as an outcome in Appendix Table A3. The pattern of the results is the same. We also disaggregate this outcome into an extensive margin effect (agreeing to sell a product at all) versus an intensive margin effect (how many criteria the product matched), and find that the effect comes mostly through the intensive margin.

Vertical The outcome in Column 2 is an indicator for whether the product is “High Quality”, defined as whether the product’s quality score is greater than its product-group median.²⁶ Pooling the treatments together, treated firms are 13.1 percentage points more likely to be high quality ($p = 0.058$), a 30.4% increase from the control mean of 43.1%. In Panel B, we see that the coefficient is positive and similar in all four treatment groups.

The outcome in Column 3 is the raw quality score out of 50. Here, the effect is both small and insignificant. As we noted in Section 5, this outcome is vulnerable to a long left tail having an outside influence. This is indeed what happens: in Appendix Figure A6 we plot the CDF of quality score by treatment status, and we see that, while the treatment CDF is to the right of the control CDF from the 30th percentile onwards (as reflected in the binary outcome), it also contains a handful of very low scoring goods that skew the mean.

The outcome in Column 4 is an indicator for whether the product was made in Türkiye. Pooling the treatments together, treated firms are 16.2 percentage points more likely to supply a good made in Türkiye, a 34.0% increase from the control mean of 47.7% ($p = 0.017$). In Panel B, we see that the effect is positive and similar for all four treatments.

Finally, to account for multiple hypothesis testing across outcomes, we aggregate these three outcomes into a vertical index using the standardised inverse-variance weighting method in [Anderson \(2008\)](#). The pooled coefficient is 0.412 standard deviations ($p = 0.005$), and similar across the four treatment arms. The results thus show that alleviating the search friction by connecting firms with suppliers abroad via social media increased their access to higher quality goods.

Price The outcome in Column 6 is the price charged to the mystery shopper. The coefficient is positive, small, and statistically insignificant, suggesting no meaningful price effects in either direction. While this could reflect two offsetting effects if the observed increase in quality causes prices to rise by an amount that offsets a decrease through a price-based mechanism, it at least suggests that a price-based mechanism is not so large as to overcome this.

Summary: Relaxing Search Frictions Expands Access to Foreign Varieties Putting together these results, we find that all of the treated groups saw large and significant increases in access to foreign goods. Treated firms are able to sell a wider set of varieties and higher quality varieties. The fact that we find no detectable effects on the price suggests that gains come more from improving access to otherwise unavailable varieties than from lowering prices for current varieties.

²⁶The sample size is lower for the vertical and price outcomes because the mystery shopper only proceeded to the negotiation phase (to measure price) if the firm had a suitable good, and then only purchased the good (to measure quality) under some additional conditions (see Appendix F.4). Two analyses in Appendix Table A5 investigate whether this introduces selection bias. First, we define versions of the vertical outcomes that are unaffected by this issue: one that is 1 if the firm had a suitable and high quality good (and 0 otherwise) and another that is 1 if the firm had a suitable good from Turkey (and 0 otherwise). We find positive and significant effects on both outcomes. Second, to ensure differences do not come from comparing different products, we replicate Columns 2-6 of Table 1 with fixed effects for the 28 products. The qualitative takeaways are the same. Finally, since we use PDS Lasso in all regressions, any observables strongly correlated with either the outcome or the treatments would likely already be controlled for.

Across all of these outcomes, the pattern is consistent: the results are driven by relaxing the search friction. We thus conclude that finding a supplier abroad is costly, and that social media can alleviate this friction. This does not necessarily mean that there are no trust frictions: these are small orders that have a customer already in place, so for many firms the risk may be low. However, if firms are to realise the gains from trade that these short-run results imply, they will need to develop these new connections into medium- or long-run relationships, which we look at next.

6.2 Building Supplier Relationships

6.2.1 Survey Data

In our follow-up survey, conducted after 3 months, we asked firms how many regular suppliers they had, and where those regular suppliers were based. We defined a regular supplier as any supplier from whom the firm had ordered at least twice in the past three months, and intended to continue the relationship. We analyse these outcomes in Table 2. In Column 1, the outcome is an indicator for whether the firm has a regular supplier in Türkiye, and in Column 2 it is the number of regular suppliers in Türkiye. In both cases, pooling all four treated groups shows that treatment caused firms to develop new relationships with suppliers in Türkiye. The pooled coefficients are 3.7 percentage points (an increase of 22.2% relative to control mean of 16.7%) with $p = 0.088$, and 0.083 suppliers (an increase of 37.4% relative to the control mean of 0.222) with $p = 0.023$.

Disaggregating by treatment arm, for both outcomes, the coefficient for Search + Adverse Selection + Moral Hazard is substantially larger: this group sees a 7.5 percentage point increase in the likelihood of having a supplier in Türkiye, and a 0.188 increase in the number of suppliers in Türkiye, both of which are highly significant (including after adjusting for multiple hypothesis testing). The effect is also larger for the Search + Adverse Selection group. Formally, we can reject the null that all four coefficients are equal. This suggests that relaxing trust frictions, and perhaps in particular relaxing them jointly (consistent with the complementarity in the theory), increased the likelihood that these new connections developed into relationships.

In the second part of Table 2, we look at whether these new relationships complement or substitute for existing relationships. In Column 3, the outcome is the total number of suppliers; in Column 4, the outcome is the number of suppliers in Senegal. The general direction looks closer to a world of substitutes: the coefficients on the total number of suppliers are close to zero, and the coefficients on the number of suppliers in Senegal, while noisy, are of similar magnitude (but opposite sign) to the coefficients on the number of suppliers in Türkiye. Finally, Column 5 shows an indicator for whether the firm said that they have ended a relationship with a regular supplier in the past 3 months. The coefficients are generally positive, which is also suggestive of substitutes.

To assess how the buyer-only discussion WhatsApp group in the adverse selection treatment influenced buyer perceptions and behavior, we included targeted questions in the follow-up survey. A majority (53%) reported that this group provided them with meaningful value. Specifically, 24%

stated that the group increased their trust in the suppliers or helped them obtain useful information about them. Another 12% indicated that the group facilitated new connections, and an equal share (12%) reported that they had privately contacted other firms from the group to learn more about the study suppliers—a figure that likely understates the true extent of private communication, since it captures only what firms chose to report in the survey. The implication is that observable group-level activity is a lower bound on the information that flows through these labeled firm discussion groups. In addition, 24% of firms reported knowing other firms who had placed orders with the study suppliers (compared to 13% in the absence of this treatment)—evidence pointing to the group’s role in disseminating information about peer purchasing behavior.

6.2.2 Mobile Money Data

As discussed in Section 5, we use data from the largest mobile money provider in Senegal to directly observe transactions between study firms and study suppliers. Before turning to formal regression results, we first present graphical evidence from the raw data. In Figure 6, we plot cumulative order value from study suppliers as a function of time in the study aggregated across the four treatment groups. In the initial month, the total value ordered is very similar across the four treatment groups. This is the period when the mystery shopping is taking place, so limited differences appear in line with the results in Section 6.1.²⁷ Outside of this period, differences start to open up, with all three trust groups ordering more than Search Only, especially the Search + Adverse Selection + Moral Hazard group, which is in line with the survey results.

We formally test these patterns in Table 3, where the omitted category is Search Only. The outcome in Column 1 is the total value transacted with study suppliers. The coefficients are positive and large—ranging from a 30% increase under AS to a 90% increase under AS+MH—but are imprecisely estimated and not significant. We then decompose this total value into the number of orders (defined as the number of weeks with a positive transaction value) and the average order value (defined as the value transacted in weeks with positive transaction values). We find no effect on the number of orders, and positive, large, and significant effects on the average order value. The mobile money data therefore suggests that the new matches translated into more meaningful relationships in the trust groups. This looks closer to an effect on the intensive rather than the extensive margin, though it is worth bearing in mind that this dataset does not capture all trade. In interpreting the magnitudes, recall that these are unconditional averages, that relationships are

²⁷We do not know if a given transaction corresponds to fulfilling a mystery shopper’s order, but we can do a back-of-the-envelope calculation of aggregate values. There are 1,500 treated firms, the mean value of a mystery shopping order was USD 21, the share of firms that we purchased from was 0.8×0.37 , and the mean markup from the baseline survey was 0.5. Thus, if every treated firm that sold a mystery shopping product sourced it from a study supplier (a very strong assumption) and paid with this method (another strong assumption), we would expect a total value of around USD 4,700, as compared to around USD 7,500 in the first month of the mobile money data. The mystery shopping therefore likely accounts for an important share of purchases in the first month, which is not unexpected since an explicit secondary goal of the mystery shopping was to subsidize experimentation. It need not be surprising that some firms ordered from study suppliers for the mystery shopping but not again; the model explicitly assumes that small orders provide limited learning about supplier type.

concentrated among a subset of firms, and that larger transactions tend to occur off-platform.

The next three columns repeat the same exercise, but exclude the first four weeks to remove entirely the mystery shopping period. The general pattern is similar.

6.2.3 Summary: Trust Treatments Convert Matches into Supplier Relationships

Putting together the results from Sections 6.2.1 and 6.2.2, we find that trust treatments made it more likely for these new connections to develop into meaningful relationships. In both datasets, the largest effects come from the Search + Adverse Selection + Moral Hazard group. This is consistent with the implication from the model that the trust treatments are complements, although we are not powered to formally test for a complementarity.

6.3 Profit and Sales

6.3.1 Average Treatment Effects

To see whether improved access to foreign goods and new relationships flow through to producer surplus, we report the results on profits and sales in Table 4. We measure profit using the summary survey question in De Mel et al. (2009) for the last 30 days, with a similar question for sales.

Columns 1-2 show the results on raw profit and sales. We find large and significant effects. For profit, the pooled coefficient is 82.4 USD ($p = 0.025$), or a 43.8% increase from the control mean. For sales, the pooled coefficient is 245.2 USD ($p = 0.051$), or a 40.2% increase from the control mean. When we disaggregate the four treated groups in Panel B, we see that (for both outcomes), while the effect is positive in all four groups, it is much larger and only significant in the Search + Adverse Selection + Moral Hazard group (including after adjusting for multiple hypothesis testing). Formally, we can reject the null that the four coefficients are equal. The same pattern holds when we index the two outcomes. This aligns with Table 2, which shows this group has significantly more suppliers in Türkiye.

To limit the influence of outliers, in Columns 4-5 we report the results when we winsorize the outcomes at the top 1%. The coefficients decrease in magnitude by around half, but the same pattern remains: the coefficient on the Search + Adverse Selection + Moral Hazard group is large and significant, including after adjusting for multiple hypothesis testing. We also report Poisson regressions, which are less sensitive to skewness, in Appendix Table A6. The pattern is the same.

6.3.2 Distributional Treatment Effects

It is now well established in the literature studying small firms in lower-income countries that profit and sales tend to be thick-tailed, and that these tails can have large effects on the coefficients

in OLS regressions (Meager, 2022).²⁸ Thus, as discussed in Section 5.2 and specified in our PAP, we use quantile regression to examine distributional effects.

Quantile Treatment Effects In Figure 7 Panel (a), we show the quantile treatment effects for profit for percentiles 5-95. Across all four groups, the coefficients are small and generally insignificant up to roughly the middle of the distribution. However, starting from around the upper third, the Search + Adverse Selection + Moral Hazard group coefficient becomes large and significant. Expressed relative to the control quantiles, these effects from roughly the 65th to 95th percentiles are close to constant, on the order of 15–20%. The coefficients for Search Only and Search + Adverse Selection are also large at the 95th percentile, but are very noisy. The results for sales in Panel (b) are similar.²⁹ We relate these distributional findings to the OLS estimates and the model in Section 6.3.3.

Threshold Regression As another way to analyse distributional treatment effects, we also run OLS regressions on indicators that are 1 if the outcome is greater than a threshold t , for a range of t . This is less affected by high variance in the tails as all that matters is whether the outcome is above the threshold. The results, reported in Appendix Figure A7, are similar to the quantile regressions: large and positive effects at the upper end of the distribution for the Search + Adverse Selection + Moral Hazard group, and some suggestive evidence of positive effects for the other groups at the very top.

6.3.3 Interpreting Profit Effects

The pattern of the profit results is similar to the pattern in Section 6.2. However, a few points warrant further discussion. First, several features of the data indicate that the conclusions are not driven by a handful of outliers that happen to be in the treatment group: (1) the quantile regressions show gains starting at around the 65th or 75th percentile, beyond what would be expected from a few extreme cases, (2) the threshold regressions are by construction robust to a small number of observations having outsize influence, (3) the p -values in Table 4 highlight that the patterns we observe are very unlikely under the sharp null, and (4) a placebo using baseline profit (Appendix Figure A9) shows no such pattern.

Second, the pattern of gains across the distribution is broadly consistent with the model and suggests a natural way to interpret the magnitudes in light of the large OLS estimates. In the model,

²⁸To verify that treatment effects are not driven by measurement error in the tails, we called back all firms whose profit was more than 5 times that at baseline and exceeded a threshold at endline, and we asked them to confirm their previous response. Out of 13 such firms, 12 confirmed that their previous response was correct.

²⁹While QTEs are only interpretable as the effect for firms at a given quantile under rank preservation, we think it is likely that these treatment effects do correspond to firms that are larger at baseline. In Appendix Figure A8 we show treatment effects on the transition matrix between baseline and endline profit: the largest coefficients are in the northeast region, suggesting that treatment helps large firms remain large. Firms in the 95th percentile of baseline profit have, on average, one employee; 30% report traveling abroad in the past five years; they are also more likely to be male-owned, to operate physical stores, and to be wholesalers (see Section 6.5 for heterogeneity analysis).

there is a productivity cutoff $\bar{z}$ above which firms engage in search for a supplier abroad. For firms with $z > \bar{z}$, the combined treatment generates (i) a level effect from a new match created by the search treatment, (ii) an effect that scales with z from the trust treatments, so that treatment effects rise with size and converge to a roughly constant percentage gain.³⁰ For firms with $z < \bar{z}$, revealed preference implies that the value of a new match is less than the search cost s , and our assumption that learning is limited at small order sizes further dampens the gains, so effects can be small (or even zero) and difficult to detect.

The empirical distributional effects line up with this logic. Assuming rank preservation for simplicity, firms in roughly the bottom half of the profit distribution show little to no detectable gains. Firms in the 65th-95th percentile show increases in profit on the order of 15% to 20% of the corresponding control quantile. These are economically meaningful but smaller than the OLS point estimate (though within the 95% confidence interval). Finally, firms in the extreme right tail (around the 95th-99th percentiles) exhibit very large but noisy gains that have a large influence on the OLS point estimate; as we show in Section 7, the model cannot fully quantitatively match the size of these tail effects. Taken together, the most prudent interpretation is that the intervention generates (i) little detectable profit impacts in the bottom half of the distribution, (ii) economically important profit increases in the upper third of the distribution, (iii) possibly very large, but rare and imprecisely estimated, effects at the very top.

Third, the profit effects appear before the mobile money data in Figure 6 diverge. We view this timing pattern as plausible in light of how firms transact. Larger orders with new suppliers in Türkiye that would cause changes in profits tend to take place off-platform, especially among wholesalers (who are more likely to use other payment methods like bank transfers, and, as we show in Section 6.5 drive profit effects). Ultimately, Table 2 shows that the Search + AS + MH group had significantly more regular suppliers in Türkiye after 3 months, and the profit results suggest that these new relationships were profitable. The subsequent divergence in the mobile money data, which is completely independent of our survey-based profit measure, therefore provides complementary evidence that the treated firms built and then exploited new trading relationships.³¹

6.3.4 Summary: Profit Gains from Reducing Trust Frictions

In Table 4, we saw large average effects on profit and sales. As in Section 6.2, the effects are concentrated in the trust groups, and in particular in the group with both trust treatments. This suggests that trust frictions matter for firm profits, and that alleviating them using social media

³⁰The search effect is constant in levels because the expected gain from a new match must equal s in equilibrium. This is true regardless of the quality of the current match, because, as we discuss in Section 7, we interpret the baseline equilibrium as the long-run steady state, in which the firm has already extracted its surplus and thus earns just enough to make sure it does not want to leave. The trust effect scales with z as the dynamic program is homothetic in z : there are no scale effects, so the gains to improving trust are constant in percentage terms.

³¹The firms that drive the early profit gains need not be the same firms that later transact more heavily via mobile money, and the frequency of larger restocking orders may differ from that of mobile money payments. We therefore do not expect a one-to-one mapping between the timing of profit and mobile money effects.

can unlock meaningful gains.

6.4 Limited Understanding of Formal Platforms is not the Binding Constraint

Our final set of outcomes examines a light-touch Alibaba training sub-treatment intended to probe one specific mechanism: whether very limited use of formal B2B platforms reflects platform complexity/know-how. As described above, the brief training was offered only once and only to firms that did not receive any of the main treatments, so our main analysis excludes firms in the main treatments (though results are similar if we include them).

We report the results in Table 5. The training has a first-stage: treated firms are 6.5 percentage points more likely to have heard of Alibaba, 11.3 percentage points more likely to have searched for goods on the platform, and 8.7 percentage points more likely to have compared prices on Alibaba with prices from their regular supplier. However, we detect no increase in actual purchases: the coefficient is 1.4 percentage points, with standard errors tight enough to rule out modest to large effects. Taken together, these estimates suggest that a lack of basic familiarity or simple usability is unlikely to be the primary binding constraint. At the same time, this design was not intended to—and cannot—disentangle alternative mechanisms such as language barriers, broader technological constraints (payments, logistics, dispute resolution), or social/relational factors (direct messaging, negotiation, supplier access). This points toward deeper frictions that social media may relax but current B2B platforms like Alibaba do not.

6.5 Heterogeneous Treatment Effects: Higher Profit Gains for Wholesalers

We pre-specified a set of heterogeneity analyses to explore how treatment effects vary with firm characteristics, and we report detailed tables in Appendix C. Several patterns emerge.

First, we compare retailers and wholesalers. We define wholesalers as the 33% of the sample for whom some (or all) of their sales are wholesale to other firms. While we find no consistent heterogeneity across mystery shopping or relationship outcomes, the profit gains appear concentrated among wholesalers. This aligns with the distributional treatment effects in Section 6.3. One possible explanation is that wholesalers, who typically buy and sell in bulk, stand to gain more immediately from access to new suppliers. Alternatively, the effects may reflect reduced travel costs for sourcing, given that wholesalers are more likely to travel abroad for procurement. While we cannot definitively distinguish between these mechanisms, Appendix Table A7 provides suggestive evidence that wholesalers in the treatment group are less likely to have traveled internationally for business in the three months between the intervention and the follow-up survey. The estimated effects are large and mostly negative, with a few significant coefficients, though wide confidence intervals suggest they should be interpreted with caution.

Second, we examine whether treatment effects differ between online-only firms and those with a physical store. The horizontal mystery shopping effects (e.g., custom order responsiveness) are

more pronounced among online-only firms, likely reflecting their greater flexibility in adapting to buyer requests. However, we find no systematic differences in vertical mystery shopping or relationship outcomes. Interestingly, the profit effects are instead concentrated among firms with physical stores, many (45%) of which are wholesalers. This suggests that while online-only firms may be more responsive to initial contact, firms with physical stores may be better positioned to convert new relationships into sustained profit gains. Since online-only firms are disproportionately women-operated (63% against 25% for physical stores), we observe a similar pattern when disaggregating by the gender of the firm owner.

Finally, we find no meaningful heterogeneity in treatment effects based on prior importing experience, initial exposure to Turkish-made goods, or existing membership in a WhatsApp group with a supplier in Türkiye. This suggests that the intervention may be useful for firms regardless of their baseline access to international sourcing networks.

7 Model Estimation

In this section, we use the results of the experiment to estimate the model from Section 3. We have two main goals. First, we want to estimate the parameters governing the frictions, as these are important inputs to quantitative models but few estimates are available in the literature. Second, since our treatments are only one particular way of alleviating the frictions, we want to evaluate the gains from trade available if the frictions were to be counterfactually alleviated in different combinations, to different magnitudes, or through different interventions.

7.1 Estimation Details

Modifications to the Model We modify the model in a few ways to make it estimable. First, we let the revenue function r be such that the firm faces and internalises a constant elasticity residual demand curve for goods, $zQ^{-1/\sigma} = P$, with $\sigma > 1$.³² The aggregate good Q is a linear combination of inputs from either a new supplier abroad that they are matched with, q_f , or their existing suppliers (who may also be abroad), q_e . For simplicity, we let firms' existing suppliers be summarized by constant price p_e . Timing is such that the firm chooses (q_f, q_e) , observes whether the order q_f is high quality and therefore whether $Q = q_f + q_e$ or $Q = q_e$, and then sells to their downstream buyers. The firm's stage game payoff is therefore the following,

$$(1 - \mu_t(1 - \lambda))z(\psi q_{ft} + q_{et})^{\frac{\sigma-1}{\sigma}} + \mu_t(1 - \lambda)zq_{et}^{\frac{\sigma-1}{\sigma}} - p_{ft}q_{ft} - p_e q_{et}, \quad (3)$$

where $z > 0$ is firm-specific productivity and ψ is match-specific productivity. We can see from inspection that the firm will always choose $q_{et} > 0$, but may choose $q_{ft} = 0$ if either beliefs μ_t , or

³²Since our experiment was not large enough to generate general equilibrium effects, we do not explicitly model competition between firms. Our counterfactuals remain valid so long as the policy does not affect very large numbers of firms. We do not expect that introducing general equilibrium effects would meaningfully affect the relative gains to alleviating the frictions (e.g., AS vs MH), but it would likely lower the absolute returns.

the prices $\{p_{ft}\}_t^\infty$ required to incentivise the supplier, are large.³³

Second, we need distributions for the heterogeneity. For the match-specific productivity, ψ , we normalize the supplier's base cost, c , to 1 and then let ψ be distributed lognormal with parameters (ψ_μ, ψ_σ) , which we will estimate. We draw the firm-specific productivity, z , from the empirical distribution that rationalises the control group's profits (described in more detail below).

Third, we need to take a stand on the baseline equilibrium. The sequential search process implies that there exists a cutoff value $\bar{z}$ such that firms with $z > \bar{z}$ will search until they find a supplier abroad with sufficiently high match-specific productivity, where $\bar{z}$ is a function of parameters to be estimated. We interpret the baseline equilibrium as the long-run steady state of the model, meaning that all firms with $z > \bar{z}$ have found a satisfactory supplier abroad and are in the mature phase of the relationship, which means they earn $(1 - \delta)\bar{U}(z)$ every period. For such firms, we then set p_e to be the constant price that delivers exactly $(1 - \delta)\bar{U}(z)$. For all other firms, p_e represents the cost of buying from a local supplier in Dakar, which we calibrate using the baseline survey. A firm's outside option, $\bar{U}$, is then defined by the value of purchasing at p_e forever,

$$\bar{U}(z) = \left(\max_{q_e} z q_e^{\frac{\sigma-1}{\sigma}} - p_e(z) q_e \right) / (1 - \delta).$$

Fourth, while taking μ to its endpoint of 1 implies the adverse selection problem is severe enough that trade is fully deterred, taking ξ to 1 does not have the same effect. To make this more symmetric, we introduce an additional term, $\kappa \in (0, 1)$, on the LHS of the DICC, and we allow good-type suppliers to have a different $\lambda_G \geq \lambda$ probability of success under low effort. The modified DICC is then given by $\kappa(1 - \lambda_G)\delta V_{t+1} \geq \xi c q_t$. The κ term could be interpreted literally as supplier impatience, but other microfoundations are available, such as agency problems internal to the supplier (e.g., if an agent gains the full value of shirking but only a share κ of the surplus). Since these parameters are not separately identified from ξ , we simply define $m \equiv \xi/(\kappa(1 - \lambda_G))$ and re-scale it to define a composite moral hazard parameter $\tilde{\xi} \equiv m/(1 + m) \in (0, 1)$ that has the property that $\tilde{\xi} \rightarrow 1$ fully deters trade.³⁴

Parameters Our goal is to estimate the parameters governing the three frictions: the search cost, s , the share of bad types, μ , and the composite moral hazard parameter, $\tilde{\xi}$. We also include the probability of high quality for the bad types, λ_B , which affects the speed of learning. Finally, we need to estimate the parameters governing the lognormal distribution of match-specific productivity, (ψ_μ, ψ_σ) , which are related to the search friction as they govern the expected returns to

³³As discussed in Section 3, to prevent a discontinuity in learning when going from $q_{ft} = 0$ to any positive value, we assume that for q_{ft} below a small $\underline{q}$, posteriors are given by $\mu_{t+1} = \mu_{t+1}^{\text{Bayes}} + (\mu_t - \mu_{t+1}^{\text{Bayes}}) \cdot (1 - q_{ft}/\underline{q})^p$, with $p \in (1, 2)$. Making learning a continuous function of q_{ft} ensures the problem satisfies appropriate regularity conditions, and generates similar behavior to a minimum order size without introducing non-convexities into the principal's choice set.

³⁴The resulting DICC is therefore $\delta V_{t+1} \geq \frac{\tilde{\xi}}{1 - \tilde{\xi}} c q_t$.

searching. We thus estimate these six parameters, and calibrate the remaining three, (σ, δ, p_l) .³⁵

Moments To leverage the exogenous variation in the frictions generated by the experiment, we estimate the model by simulating theoretical treatment effects in the model and then matching these onto the empirical treatment effects. We use the four treatment effects on likelihood of having a supplier in Türkiye as these provide the cleanest empirical expression of the extensive margin. We use the treatment effects on total value ordered after mystery shopping in the mobile money data to capture the intensive margin, expressed as percentage changes since this dataset does not cover all trade. Finally, we include profit treatment effects as these are a direct reduced form measure of producer surplus. Given that the treatment effect is null below the 65th percentile, we target the upper quantile treatment effects (specifically, quantiles 65, 75, 85, 95).

We simulate the three treatments in the model as follows. We implement search as the firm being matched at zero cost to a supplier abroad with match-specific productivity equal to the maximum of three draws from the distribution (to capture the idea that the firm is matched to three suppliers in the experiment). We implement adverse selection as the firm receiving one high quality signal realisation without having to purchase anything, meaning that they update as a Bayesian and thus begin the relationship with belief that the supplier is a bad-type of $\mu_1 = \mu_0 \lambda / (1 - \mu_0(1 - \lambda)) < \mu_0$. This intends to capture the recommendation call from the adverse selection treatment that explicitly told the firm about one positive order experience.³⁶ We implement moral hazard as treated firms solving for the optimal dynamic path under a joint-punishment strategy among study firms, as opposed to an individual grim-trigger strategy. That is, treated firms know that the suppliers stands to lose δNV_{t+1} instead of δV_{t+1} if low-quality is realised, where we calibrate $N = 5$ as the empirical ratio of the number of firms that made orders from study suppliers to the number of study suppliers.³⁷

Solving and Estimating the Model We discuss how we solve and estimate the model in detail in Appendix E. In brief, dynamic models of moral hazard are non-stationary in general and solving them numerically is non-trivial. We use the method of [Marcet and Marimon \(2019\)](#), which involves rewriting the problem as an as-if planner’s problem with Pareto weights that evolve endogenously to summarize past utility promises. For estimation, we use Simulated Method of Moments (SMM) with a weighting matrix equal to the inverse of the variance-covariance matrix of the empirical moments. For each guess of the parameters, we first calculate the value of ψ_μ and the empirical distribution of z that rationalise the control group’s share of firms with a supplier in

³⁵We set $\sigma = 3.02$ to match the average markup from the baseline survey, $\delta = 0.96$ to match the empirical firm survival rate in the 3 month gap between baseline and followup surveys (we interpret one period as one month), and $p_l = 12.29$ as the average input price from the baseline survey among firms without a foreign supplier.

³⁶We focus on the direct information provision, as opposed to any social learning in the discussion WhatsApp group, as we believe this approach is more transparent. Modeling the WhatsApp group would introduce several additional assumptions (and parameters), such as those governing firms’ incentives to share information and firms’ beliefs about when other firms will make orders.

³⁷This implicitly assumes that firms treat other firms as identical to themselves (i.e., V_{t+1} is the same). This simplifies the computation significantly, as otherwise we would need to repeatedly solve for a fixed point in V_{t+1} (for all t).

Türkiye and profit distribution, respectively.³⁸ Then, we use these to draw from the (z, ψ) distribution and calculate the moments. We compute standard errors by bootstrapping the estimation procedure 200 times and calculating the standard deviation of the estimates.

7.2 Parameter Estimates

We present the estimated parameters in Table 6.³⁹ The value of $\hat{\mu} = 0.76$ suggests a large presence of bad types. The value of the composite moral hazard parameter, $\hat{\xi} = 0.81$, is also large, although how much moral hazard actually constrains the firm depends on how severely the DICC binds. The search cost of $s = 73$ USD is somewhat large, although the standard error is relatively high.⁴⁰

While there is a large literature linking institutional quality and trade outcomes, there are relatively few papers that explicitly estimate micro-level structural parameters that can be readily plugged in to quantitative models, and fewer still that exploit experimental or quasi-experimental variation to do so. The closest estimate to our μ is provided by [Monarch and Schmidt-Eisenlohr \(2016\)](#), who use the dynamic patterns of US imports to back out the implied values of similar parameters for a range of countries. In their estimates, μ varies from 0.56 to 0.91, placing our estimate around the middle of this range. Since the role of ξ depends on its endogenous implications, a natural comparison is to [Startz \(2025\)](#), whose estimates for Nigerian traders imply that ordering from abroad requires paying a price premium to incentivize suppliers of around 15% on average. At the estimated parameter values in our model, calculating a similar object—discounted total incentive premium, i.e., $\sum_t \delta^t (p_t - c) q_t / \sum_t \delta^t c q_t$ —yields values of around 9%.

7.3 Counterfactual Reductions in Trust Frictions Market-Wide

We use the estimated parameters to consider the gains available from policies that reduce the trust frictions market-wide. This is different to our experiment, as in the experiment we only reduced the frictions among a few specific suppliers and thus firms' equilibrium search behavior was unaffected. There are many possible policies that could reduce the frictions, ranging from government-run trade promotion programs that facilitate matching with known and trusted suppliers, to business associations maintaining lists of verified suppliers abroad, to social media com-

³⁸Profits of firm i are given by $\pi_i = z_i^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p_e^{1-\sigma}$. Our definition of baseline implies a (parameter-dependent) mapping from z to p_e , so we can define a bijective function $\pi(z) = z^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p_e(z)^{1-\sigma}$. Then, we can construct a distribution of z by drawing π_i from the control group's empirical distribution and inverting this function to obtain z_i .

³⁹We plot the targeted moments in Panel A of Appendix Figure A10. The model is able to match the treatment effects reasonably well. It matches the survey relationship moments well, including the complementarity, though does not match the full scale of the profit moments. At more extreme values of the parameters it is able to match the profit moments (including the complementarity), but then tends to overshoot the relationship moments.

⁴⁰To understand how the estimates relate to the various moments, in Panel B of Appendix Figure A10 we present the [Andrews et al. \(2017\)](#) measure of local sensitivity, which is a local approximation to the mapping from moments to parameters. While the overall matrix is complex, as the frictions interact and some moments are highly correlated, we generally find that the moments to which μ is most sensitive tend to be AS treatment effects (e.g., the two most locally influential moments are the 75th QTE for the S + AS + MH group and the 95th QTE for the S + AS group), and the moments to which ξ is most sensitive tend to be MH treatment effects (e.g., the two most locally influential moments are the mobile money orders for the MH group and the 75th QTE for the S + AS + MH group).

panies themselves introducing ratings and reviews onto their platforms. Moreover, there is likely considerable variation across in the extent to which these would reduce the frictions. Because the variation in the frictions likely varies considerably across the different policies and tools, as well as a cross contexts and technology design, we instead take a broader view and plot the returns to alleviating the frictions by varying extents, both individually and jointly.

We show the results in Figure 8, which plots average lifetime discounted profit as a function of μ for various values of $\tilde{\xi}$. The first takeaway is that the gains to alleviating the trust frictions are large. To take one numerical example: lowering μ by 21% to 0.6 increases lifetime discounted profits by 20%, and additionally lowering $\tilde{\xi}$ to 0.6 (a decrease of 26%) increases lifetime discounted profits by 33%. Thus, while we do not simulate any particular policy, any reasonable reduction in the frictions is likely to be highly valuable. The second main takeaway is that, as in the experiment, alleviating moral hazard alone has a relatively limited impact (lowering $\tilde{\xi}$ alone to 0.6 increases lifetime discounted profits by 5%), whereas doing so when adverse selection is also alleviated increases the return to 13% instead of 5% (reflecting the complementarity). In practice, many interventions have the potential to affect both—for example, pre-screening or verifying suppliers primarily targets adverse selection, but can also be made to target moral hazard if the screening or verification process takes time or is costly for the supplier (such that it becomes a valuable asset that the supplier does not want to risk losing). Our structural estimation quantifies the complementarity and suggests that it is worth ensuring that such policies can and do alleviate both where possible.

8 Conclusion

We study search and trust frictions in international sourcing, and whether the rapid growth of “social commerce” in lower-income countries can alleviate them. Finding and developing relationships with suppliers is a first-order issue, and thus understanding these frictions—as well as how they interact—is important. Moreover, the rapid recent growth of firms using social media to find and evaluate suppliers makes it especially timely to understand the role that these new digital tools play in alleviating information frictions.

Our results have two main implications. First, search and trust frictions are both substantial barriers for trade. Our mystery shopping exercise shows that lowering search costs by connecting firms to new suppliers abroad allows them to provide both a wider set of varieties and higher quality varieties to real customers. However, using both survey and mobile payments data, we find that these connections only develop into meaningful and lasting relationships when we also alleviate the trust frictions. This, in turn, leads to higher profits for some firms. Beyond the experiment, our structural exercise suggests that adverse selection may be the more important trust friction, though both are important and there is scope for large complementarities to alleviating them jointly. While our experiment focuses on international trade, the mechanisms we study are not specific to crossing borders. The same frictions arise when quality is difficult to verify, supplier

behavior is hard to observe, and formal enforcement is limited. These conditions are pronounced in international trade, but they also characterize many domestic markets where buyers and sellers are spatially distant, goods are differentiated, and alternative ways to mitigate the frictions are unavailable.

Second, the fact that we find these effects through interventions that deliberately mimic how firms use social media in the real world suggests that the rapid growth of social commerce should be at least partially understood as a new way for firms to mitigate search and trust frictions. These technologies are changing the structure of supply chains in ways that may complement or substitute for existing modes of sourcing: retailers can develop new international connections, and wholesalers may be able to reduce costly travel or use the increased information to gain more value from it. Moreover, these changes are here to stay, as smartphone access and mobile connectivity continue to grow, and as social media companies themselves introduce e-commerce features to their products.

Yet, despite substantial demand for online commerce, firms in our setting appear to rarely use formal B2B platforms like Alibaba. This does not appear to be simply a lack of knowledge. Why does social commerce succeed where formal platforms fail? Our evidence suggests that social commerce succeeds by capturing a mechanism with deep historical roots: embedding trust in social networks rather than in formal institutions. Supplier WhatsApp groups leverage existing relationships and communication norms; buyer discussion groups aggregate quality information through people who share social ties and can hold each other accountable; and collective enforcement operates through community reputation rather than platform-mediated dispute resolution. These are precisely the features that formal platforms lack in settings where institutional trust is low. As smartphone access and mobile connectivity continue to grow, understanding how to design platforms that complement—rather than replace—these relational features is an important direction for both research and policy.

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Tables

Table 1: Access to Foreign Goods

	Horizontal	Vertical				Price
	Has Product ≥ 3 Criteria (1)	High Qual Dummy (2)	Qual Score (/50) (3)	Made in Turkey (4)	Index (5)	Price (USD) (6)
<i>Panel A: Pooled</i>						
Treatment	0.071*** (0.027)	0.131* (0.067)	-0.391 (0.562)	0.162** (0.067)	0.412*** (0.135)	0.710 (0.699)
<i>Panel B: Individual Treatments</i>						
Search Only	0.107*** (0.035) [0.010]	0.175** (0.083) [0.126]	0.259 (0.675) [0.906]	0.138 (0.082) [0.183]	0.426** (0.166) [0.048]	1.360 (0.896) [0.369]
Search + Adverse Selection	0.041 (0.034) [0.244]	0.159* (0.087) [0.170]	-0.027 (0.858) [0.970]	0.130 (0.086) [0.183]	0.398** (0.169) [0.048]	0.445 (0.833) [0.813]
Search + Moral Hazard	0.083** (0.034) [0.045]	0.099 (0.082) [0.379]	-1.179 (0.789) [0.385]	0.183** (0.081) [0.068]	0.399** (0.162) [0.048]	1.064 (0.866) [0.465]
Search + AS + MH	0.050 (0.034) [0.244]	0.083 (0.084) [0.379]	-0.717 (0.814) [0.719]	0.211** (0.081) [0.037]	0.424** (0.171) [0.048]	-0.025 (0.818) [0.973]
Control Mean	0.307	0.431	43.064	0.477	0.000	19.990
% Increase (Pooled)	23.1%	30.4%	-0.9%	34.0%	N/A	3.6%
All Coefs Equal p -val	0.198	0.619	0.321	0.668	0.996	0.317
Adjusted R^2	0.06	0.04	0.34	0.09	0.02	0.40
N	1859	359	359	361	359	642

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is an indicator that is one if the firm has a good that matches at least 3 horizontal criteria. Column 2 is an indicator for whether the good's quality score is above the median product-group quality score. Column 3 is the raw quality score. Column 4 is an indicator for whether the good is made in Turkey, primarily inferred based on the label. See the text for full details of how this outcome is constructed. Column 5 is the [Anderson \(2008\)](#) index combining the vertical outcomes to account for multiple hypothesis testing across outcomes. Column 6 is the price in USD, which is only measured conditional on the firm finding a good matching at least three horizontal criteria.

Table 2: Supplier Relationships (Followup Survey)

	Regular Suppliers in Türkiye		Previous Suppliers		
	Any Sup in Türkiye (1)	Num Sup in Türkiye (2)	Num Sup Total (3)	Num Sup in Senegal (4)	Ended with Sup (5)
<i>Panel A: Pooled</i>					
Treatment	0.037* (0.021)	0.083** (0.035)	-0.102 (0.164)	-0.130 (0.170)	0.062*** (0.021)
<i>Panel B: Individual Treatments</i>					
Search Only	0.024 (0.027) [0.576]	0.081 (0.051) [0.263]	-0.101 (0.216) [0.885]	-0.111 (0.222) [0.924]	0.050* (0.028) [0.103]
Search + Adverse Selection	0.049* (0.029) [0.205]	0.063 (0.047) [0.321]	-0.171 (0.218) [0.836]	-0.245 (0.226) [0.636]	0.066** (0.028) [0.049]
Search + Moral Hazard	0.003 (0.026) [0.916]	0.003 (0.041) [0.936]	-0.118 (0.201) [0.885]	-0.064 (0.212) [0.924]	0.076** (0.029) [0.038]
Search + AS + MH	0.075** (0.028) [0.035]	0.188*** (0.057) [0.003]	-0.014 (0.222) [0.951]	-0.096 (0.225) [0.924]	0.053* (0.028) [0.103]
Control Mean	0.167	0.222	3.700	3.213	0.135
% Increase (Pooled)	22.2%	37.4%	-2.8%	-4.0%	45.9%
All Coefs Equal p -val	0.075	0.013	0.919	0.861	0.826
Adjusted R^2	0.14	0.12	0.32	0.26	0.05
N	1680	1680	1681	1681	1671

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm says that they have a regular supplier in Türkiye. Column 2 is the number of regular suppliers in Türkiye. Column 3 is the total number of regular suppliers. Column 4 is the number of regular suppliers in Senegal. Column 5 is 1 if the firm has ended a relationship with a regular supplier in the past 3 months. A regular supplier is defined as a supplier from whom the firm has made two or more orders with the intention to continue the relationship.

Table 3: Order Value (Mobile Money Data)

	Entire Period			Post Mystery Shopping		
	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
Search + Adverse Selection	4.18 (5.28) [0.910]	-0.03 (0.11) [0.993]	23.33** (10.68) [0.092]	3.12 (4.44) [0.784]	-0.06 (0.10) [0.920]	5.62 (22.31) [0.891]
Search + Moral Hazard	4.79 (6.31) [0.910]	0.01 (0.12) [0.993]	17.18* (8.57) [0.099]	5.60 (5.60) [0.718]	0.03 (0.11) [0.958]	11.16 (10.59) [0.790]
Search + AS + MH	12.94 (19.15) [0.910]	0.00 (0.15) [0.993]	44.50 (28.92) [0.209]	16.10 (19.10) [0.784]	0.03 (0.14) [0.958]	40.45 (20.44) [0.457]
Search Only Mean	14.31	0.36	39.24	7.92	0.23	33.95
% Increase (AS)	29.2%	-7.8%	59.4%	39.4%	-24.4%	16.6%
% Increase (MH)	33.5%	3.0%	43.8%	70.7%	13.8%	32.9%
% Increase (AS+MH)	90.4%	0.4%	113.4%	203.2%	12.6%	119.1%
All Coefs Zero p -val	0.879	0.995	0.088	0.737	0.878	0.516
Adjusted R^2	0.01	0.04	0.03	0.00	0.03	0.09
N	1500	1500	538	1500	1500	351

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. This table shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments, where Search Only is the omitted category. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is the total value of transfers to study suppliers. Column 2 is the total number of weeks in which a firm made a transfer to a study supplier. Column 3 is the mean value transferred to study suppliers in weeks with a positive transaction value. Column 4 is the total value of transfers to study suppliers, excluding the first four weeks. Column 5 is the total number of weeks in which a firm made a transfer to a study supplier, excluding the first four weeks. Column 6 is the mean value transferred to study suppliers in weeks with a positive transaction value, excluding the first four weeks. Mystery shopping generally took place during within 2-3 weeks of the firm entering the study. The data covers up to 22 months after being added to the study. Columns 3 and 6 are at the firm-week level, with standard errors clustered by firm. All values are in USD.

Table 4: Profit and Sales

	Raw			Winsorized (1%)		
	Profit (USD) (1)	Sales (USD) (2)	Index (3)	Profit (USD) (4)	Sales (USD) (5)	Index (6)
<i>Panel A: Pooled</i>						
Treatment	82.4** (31.6)	245.2* (108.4)	0.314** (0.107)	45.5** (21.2)	121.5 (78.6)	0.174** (0.076)
<i>Panel B: Individual Treatments</i>						
Search Only	31.7 (28.8) [0.561]	311* (177) [0.183]	.197* (.114) [0.187]	22.1 (25.4) [0.650]	178 (112) [0.270]	.21* (.108) [0.124]
Search + Adverse Selection	43.5 (37.7) [0.561]	80.8 (114) [0.729]	.193 (.129) [0.235]	30.8 (30.7) [0.650]	73.1 (100) [0.695]	.124 (.102) [0.387]
Search + Moral Hazard	15.8 (25.1) [0.562]	-24 (106) [0.829]	.036 (.085) [0.695]	8.21 (23.2) [0.741]	-25.3 (87) [0.786]	6.2e-03 (.078) [0.943]
Search + AS + MH	254*** (89.3) [0.004]	636** (267) [0.031]	.88*** (.315) [0.005]	128*** (39.2) [0.002]	269** (124) [0.084]	.371*** (.137) [0.018]
Control Mean	188.3	609.5	0.000	188.3	609.5	0.000
% Increase (Pooled)	43.8%	40.2%	N/A	24.2%	19.9%	N/A
All Coefs Equal p -val	0.051	0.041	0.041	0.024	0.054	0.027
Adjusted R^2	0.13	0.12	-0.01	0.25	0.29	-0.02
N	1351	1378	1298	1351	1378	1298

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is total profit from the past 30 days in USD. Column 2 is total sales from the past 30 days in USD. Column 3 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Column 4 is total profit from the past 30 days in USD, winsorizing the top 1%. Column 5 is total sales from the past 30 days in USD, winsorizing the top 1%. Column 6 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Profit is measured using the survey question from [De Mel, McKenzie, and Woodruff \(2009\)](#). Sales is measured using a similar survey question.

Table 5: Effect of Alibaba Training

	Heard of Alibaba (1)	Searched on Alibaba (2)	Compared Prices with Supplier (3)	Bought on Alibaba (4)
e-Commerce Treatment	0.065** (0.025)	0.113** (0.052)	0.087* (0.049)	0.014 (0.035)
Control Mean	0.908	0.423	0.319	0.135
% Increase	7.2%	26.8%	27.4%	10.7%
Adjusted R^2	0.07	0.09	0.14	0.22
N	340	340	340	340

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. Conventional robust standard errors are reported in parentheses. This table shows the effect of the Alibaba training treatment on Alibaba usage.

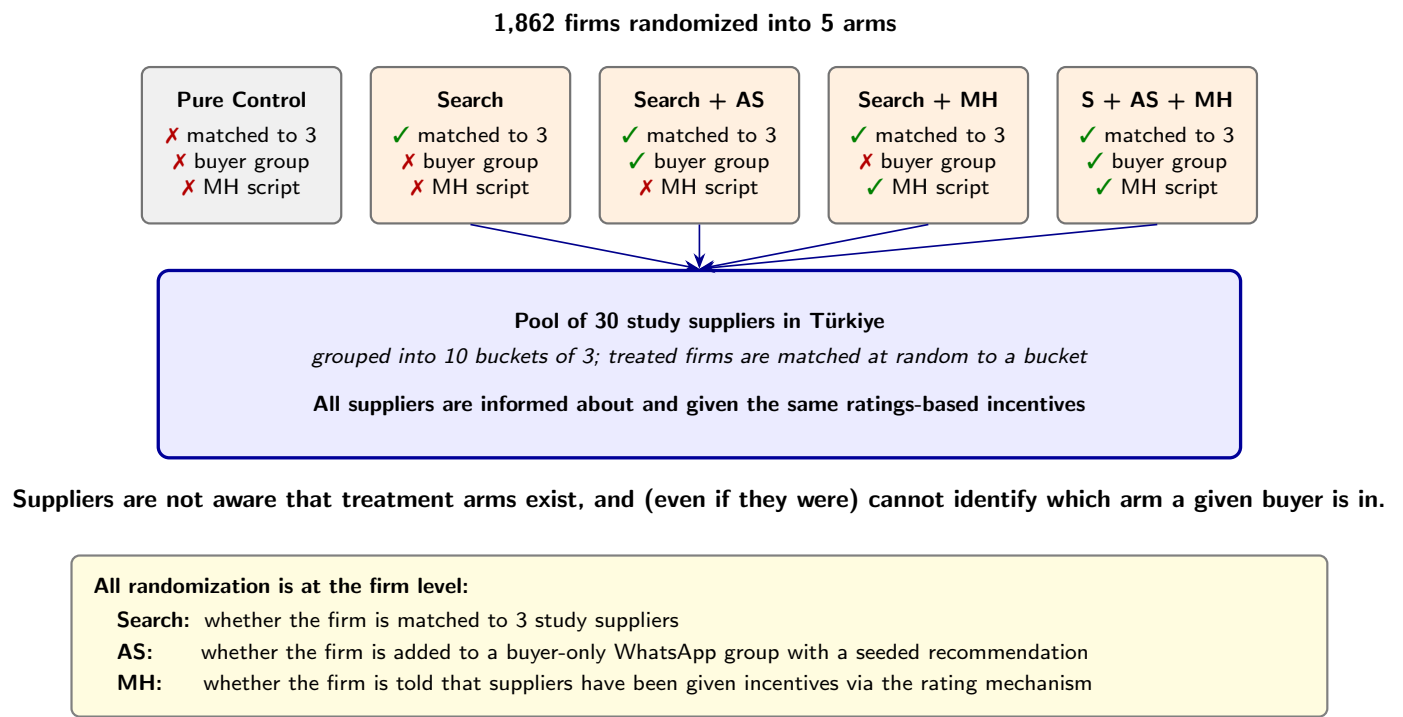
Table 6: Estimated Structural Parameters

Description	Parameter	Estimate
Share of bad types	μ	0.76 (0.08)
Moral hazard composite	$\tilde{\xi}$	0.81 (0.19)
Sequential search cost (USD)	s	73 (96)
Match productivity (mean)	ψ_μ	-3.10 (0.64)
Match productivity (std dev)	ψ_σ	0.50 (0.27)
Bad type quality probability	λ	0.025 (0.17)

Notes: This table shows the estimated structural parameters. We estimate the parameters using Simulated Method of Moments, where we simulate the treatment in the model and match the model-implied treatment effects to the experimental treatment effects on profit, whether the firm has a supplier in Turkey, and orders from mobile money. We weight the moments by the inverse of the empirical variance-covariance matrix. We present bootstrapped standard errors in parentheses, which we calculate as the standard deviation of the results of the estimation procedure run on 200 sets of treatment effects obtained by bootstrapping the experimental data. The parameters for the Lognormal match-specific productivity distribution are for the underlying Normal. Please see Appendix E for more detail on the procedure to solve and estimate the model.

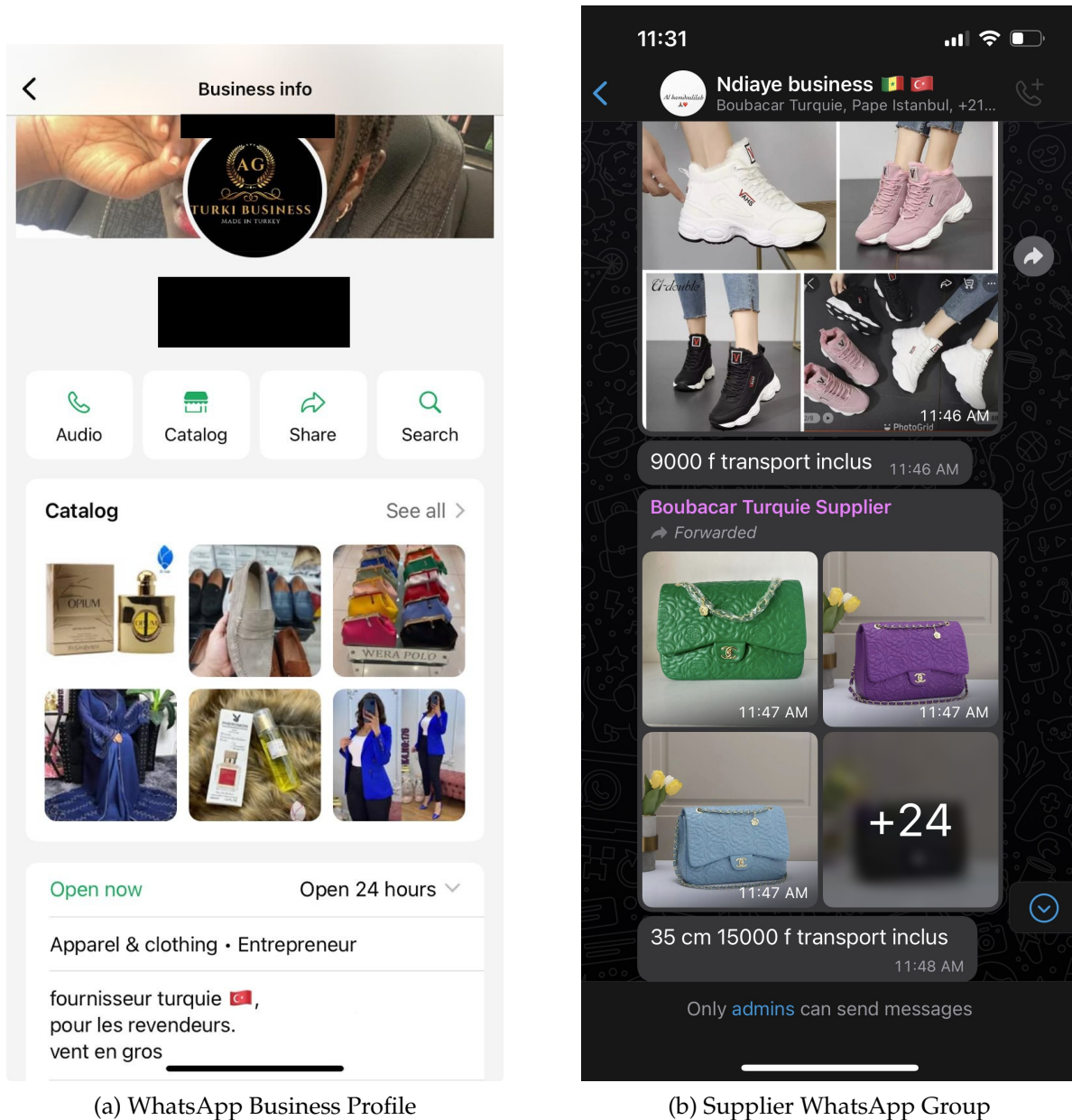
Figures

Figure 1: Experimental Design



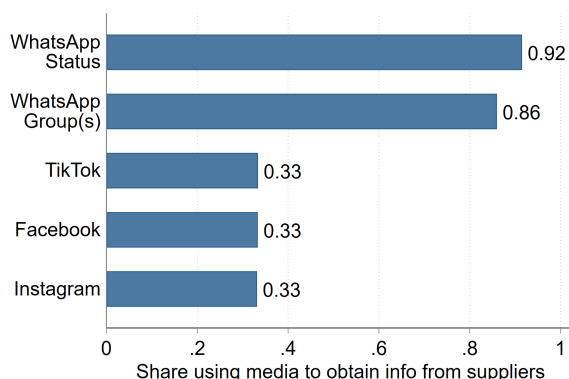
Note: This figure presents the design of the experiment, which generates variation designed to target three frictions—search, adverse selection, and moral hazard—by introducing randomized variation in matches with suppliers abroad and information about those suppliers. Treatments are described in detail in Section 4.1. Among recruited firms, 80% of them are randomly assigned to the Search treatment, in which they are matched via WhatsApp groups to three suppliers of Turkish-made goods. Within the Search group, 50% are cross-randomized into the Adverse Selection treatment, where they are placed in peer groups to share supplier experiences and receive a seeded recommendation from another (non-study) firm. Separately, 50% of firms in the Search group are cross-randomized into the Moral Hazard treatment, in which firms are told (truthfully) that suppliers have been given strong incentives to do a good job with study firms. Two additional sub-treatments (not shown)—Alibaba training and placebo peer groups—are cross-randomized within the pure control group to test secondary hypotheses. The first provides a brief Alibaba training to assess whether limited platform knowledge constrains B2B e-commerce adoption. The second creates peer WhatsApp groups without supplier matches to test whether connecting firms alone drives outcomes, serving as a placebo for the Adverse Selection treatment. Neither sub-treatment yields meaningful effects.

Figure 2: Supplier WhatsApp Profile

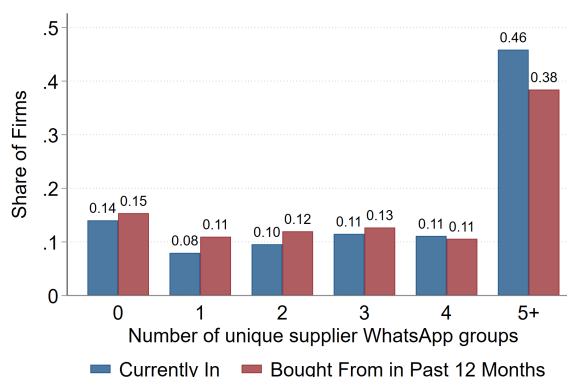


Note: These pictures show two common ways that suppliers use WhatsApp for social commerce, as described in Section 2.3. Panel (a) shows a typical WhatsApp Business profile, which allows suppliers to present key business information, including a catalog of goods. WhatsApp Business includes several features to facilitate commerce, such as the ability to run targeted advertisements on Facebook or Instagram that click through to this profile, and tools to automate and manage responses to customer inquiries. Panel (b) shows a typical supplier WhatsApp group. These groups function as virtual storefronts: the supplier regularly posts photos or videos showcasing products and can use the group to advertise promotions or new varieties. Clients—usually 50 to 100 per group—observe the posts but can rarely interact publicly; instead, they inquire or negotiate privately with the supplier via direct messages. Such groups could help reduce search costs by giving buyers access to a steady stream of updated product information and facilitating seamless communication. They might also reduce trust frictions to a lesser extent, as visible group membership and repeated engagement raise the reputational costs of cheating. Many suppliers operate both a supplier WhatsApp group and a WhatsApp Business profile, using the WhatsApp Business profile as the public-facing storefront and the supplier WhatsApp group as a way to engage with regular customers.

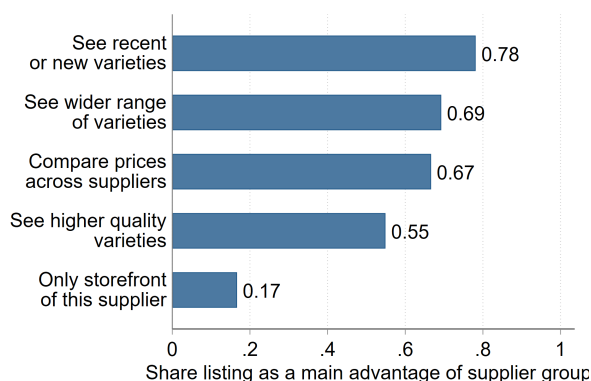
Figure 3: Firm Social Media Usage to obtain Information about Suppliers



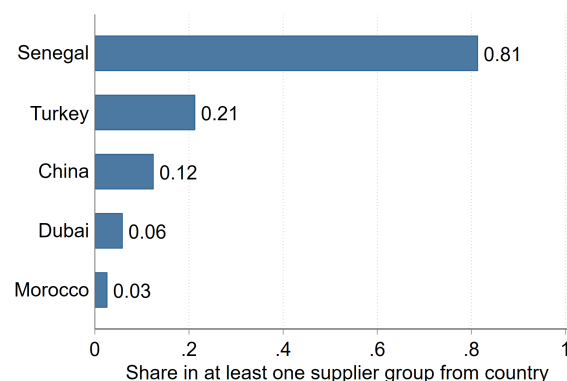
(a) Social Media Usage



(b) Supplier WhatsApp Groups



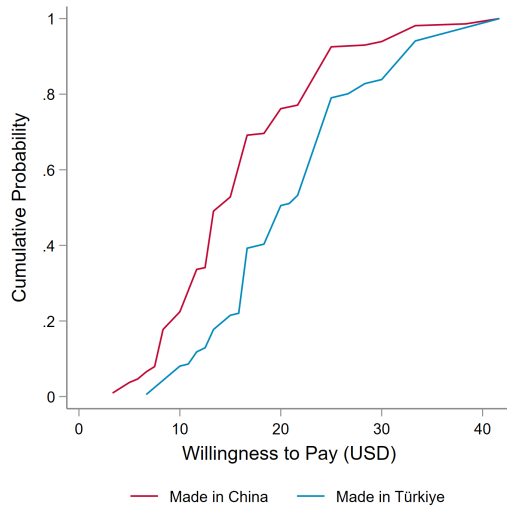
(c) Advantages of Supplier Groups



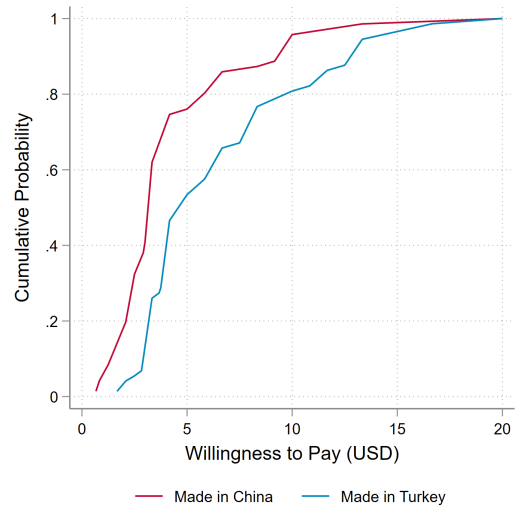
(d) Supplier Groups, by Supplier Country

Note: This figure shows a number of statistics about how firms in our sample use social media to obtain information about suppliers. All data is from our baseline survey with 1,862 firms. Panel (a) shows the results of a question asking firms to select all social media that they use to obtain information about suppliers for their business. Panel (b) shows the distribution of the number of supplier WhatsApp groups a firm is in at the time of the baseline survey, as well as the distribution of the number of such groups that the firm has directly made at least one purchase from in the past 12 months. Supplier WhatsApp groups are defined as WhatsApp groups in which the primary purpose is for suppliers to advertise their wares to downstream clients. Panel (c) shows the results of a question asking firms that use supplier WhatsApp groups to select all reasons why they find these groups useful. Panel (d) shows the share of firms who are in at least one supplier WhatsApp group where the supplier located and based in the country listed.

Figure 4: Willingness to Pay by Product Origin



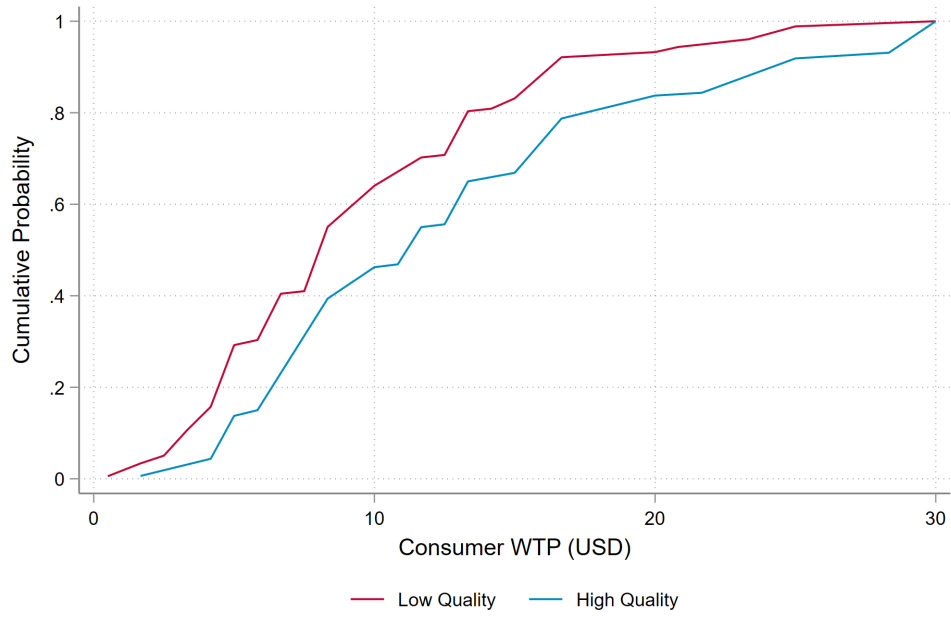
(a) Consumer Survey ($N = 400$)



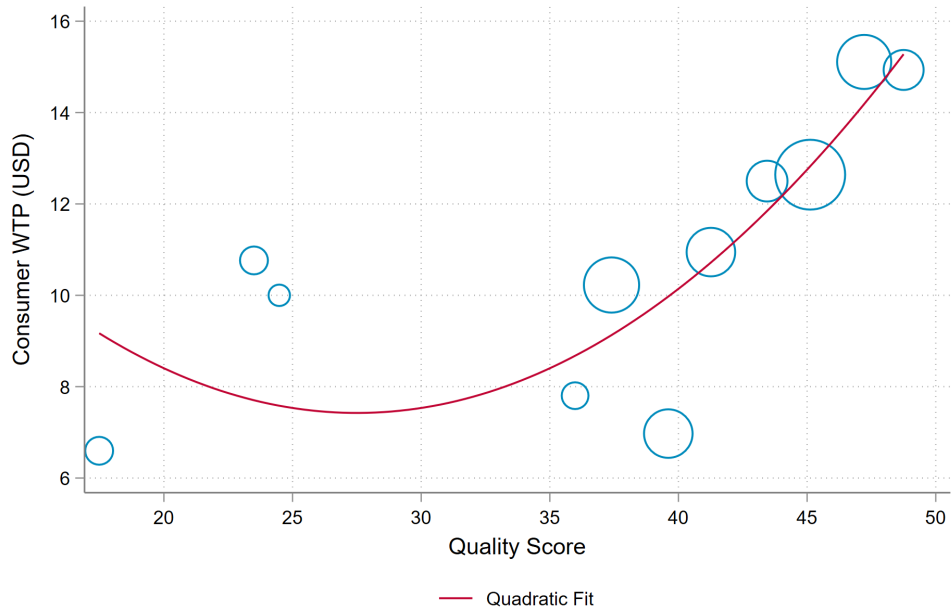
(b) Firm Survey ($N = 144$)

Note: This figure shows the results of two exercises in which surveyors showed various images of garments to respondents, randomising whether they stated that the good was made in Türkiye or made in China, and elicited willingness to pay for the garments. Panel (a) shows the CDF of WTP in the consumer survey, separately by whether the surveyor stated the good was made in Turkey or China. The distribution is truncated at 40 USD for ease of readability. Panel (b) shows the CDF of WTP for a small, separate survey of firms (for a different set of goods), with distribution truncated at 20 USD for ease of readability.

Figure 5: Consumer Willingness to Pay for Quality



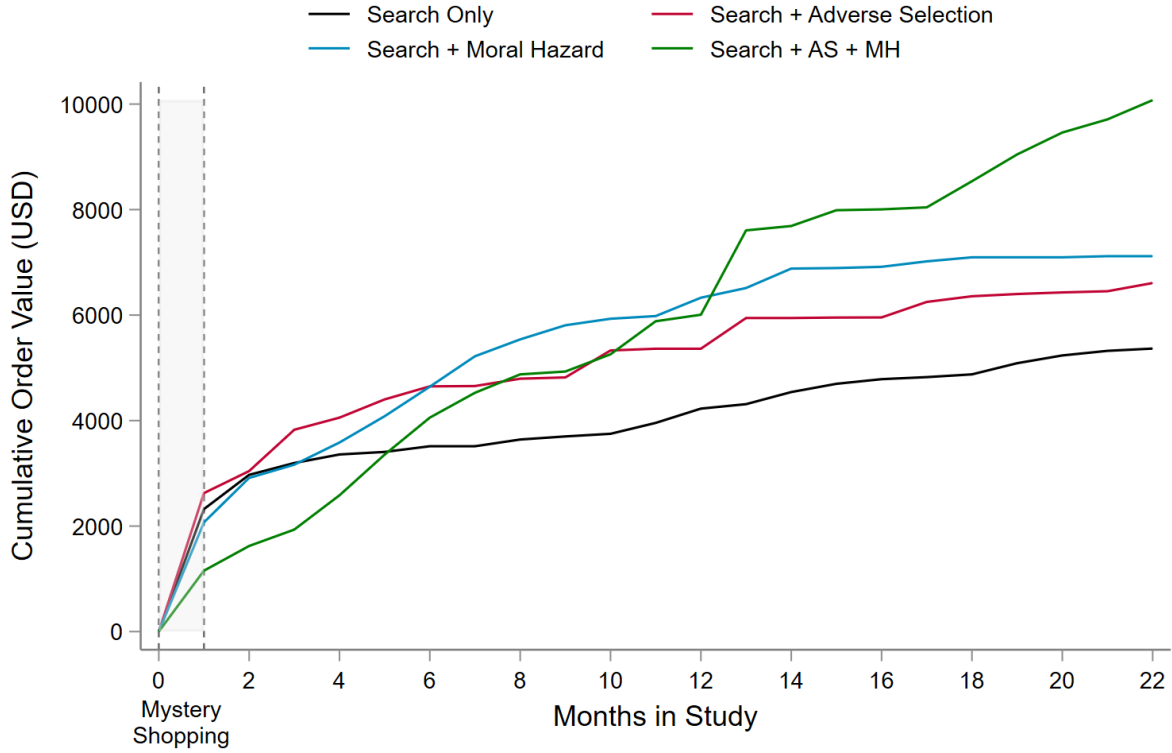
(a) CDF of Willingness to Pay by High or Low Quality



(b) Raw Quality Score

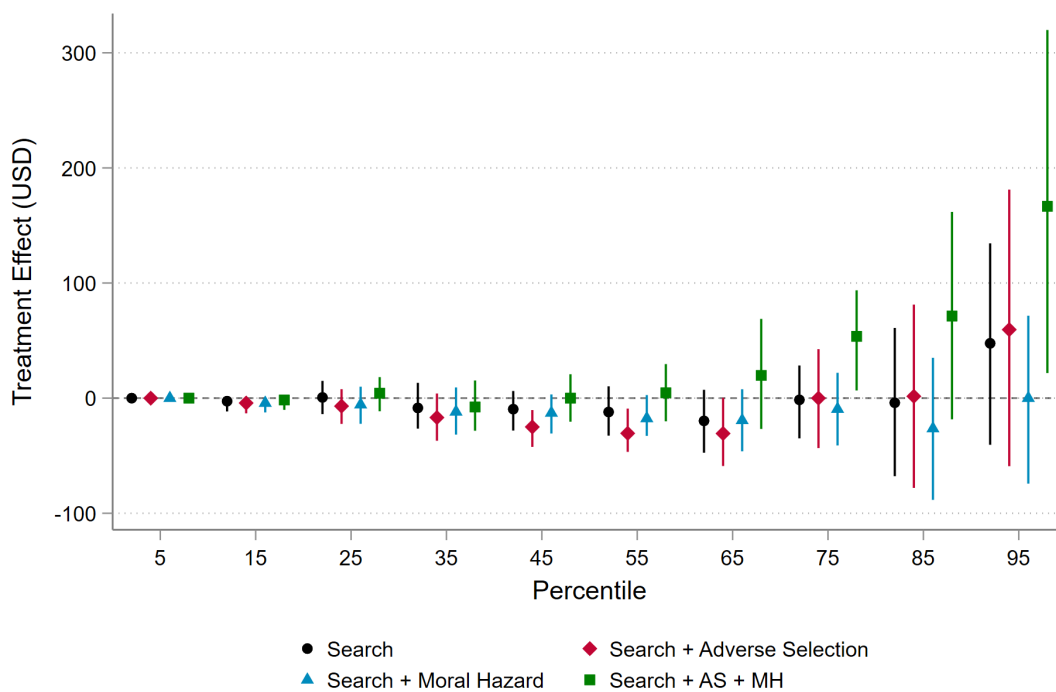
Note: Panel (a) shows the CDF of consumer willingness to pay separately based on whether the garment met our definition of high quality. We truncate willingness to pay at 30 USD to avoid unnecessarily stretching the x-axis. See the main text for full details on the consumer survey and variable construction. Panel (b) shows a binscatter of consumer willingness to pay for garments (as measured by the consumer survey) against the quality score of the garments. The size of each bubble is proportional to the number of observations.

Figure 6: Cumulative Order Value (Mobile Money Data)

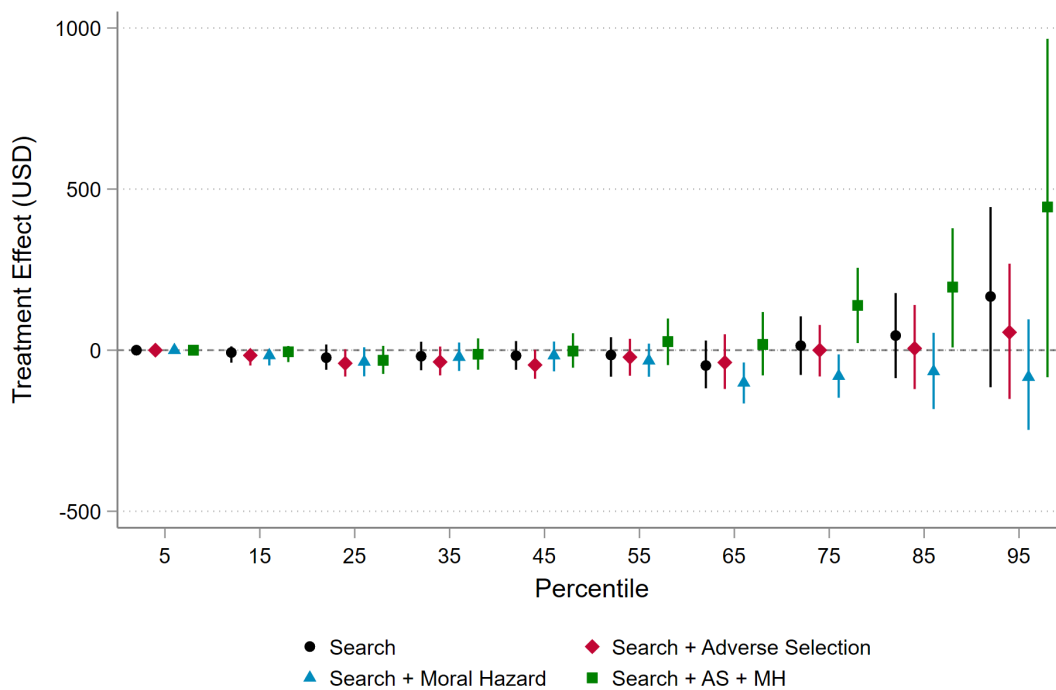


Note: This figure shows the total value flowing from study firms to study suppliers in the mobile money data, in each treatment group, as a function of number of months since the firm entered the study. As the group sizes are slightly different due to integer indivisibility issues, we normalize group-level totals by N_j/N_{avg} , where N_j is the number of units in treatment group j and N_{avg} is the average number of units across all four treatment groups. Pure control is omitted as they were not connected to any study suppliers.

Figure 7: Quantile Treatment Effects



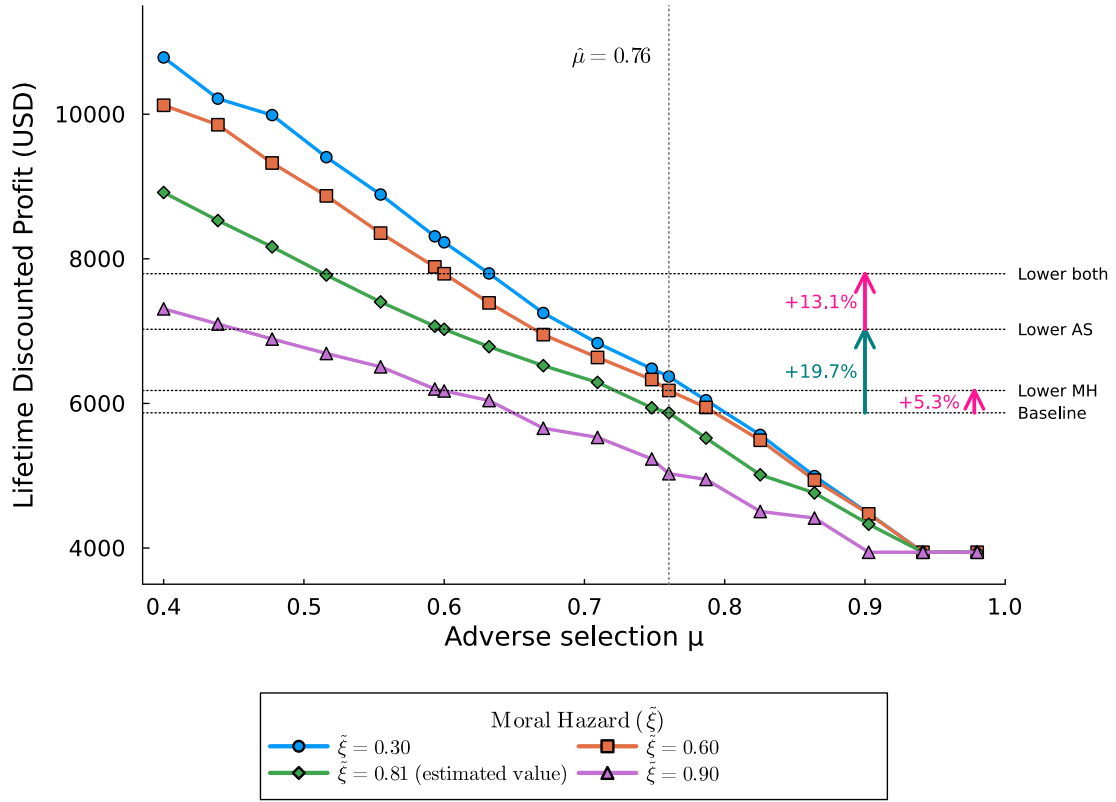
(a) Profit



(b) Sales

Note: This figure shows the coefficients from quantile regressions of profit and sales on the four treatment groups. All quantile regressions include the outcome measured at baseline (if available), but otherwise do not include any covariates. We plot 95% confidence intervals constructed using randomisation inference, defined as the set of sharp nulls that do not reject at the 5% level.

Figure 8: Impact of lowering the frictions market-wide



Note: This figure plots lifetime discounted profits as we counterfactually lower the trust friction parameters from the estimated values. Each line shows lifetime discounted profits for a range of μ holding $\tilde{\xi}$ fixed, with different lines representing different values of $\tilde{\xi}$. The estimated values are $(\hat{\mu}, \hat{\tilde{\xi}}) = (0.76, 0.81)$. All other parameters, including the distribution of match-specific productivity, are held fixed at the estimated values. The dotted lines represent the welfare gains from lowering each of the parameters to 0.6, which represents roughly a 20%-25% decrease.

Appendix A - Additional Tables and Figures

Tables

Table A1: Balance Table

	Control	Search	Search AS	Search MH	Search AS MH	p(2)	p(3)	p(4)	p(5)	Joint
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Female	0.49	0.51	0.49	0.53	0.49	0.71	0.9	0.29	0.84	0.68
Business Age (Years)	4.85	4.74	5.2	4.72	5.21	0.67	0.32	0.87	0.3	0.47
# Supp Groups	5.43	5.96	5.22	5.38	5.78	0.31	0.59	0.88	0.51	0.52
In Supp Group Turkey	0.22	0.21	0.22	0.21	0.21	0.77	0.85	0.83	0.87	0.99
Uses FB for Suppliers	0.32	0.31	0.35	0.31	0.37	0.8	0.34	0.63	0.13	0.21
Share Cust Turkey	0.43	0.45	0.45	0.44	0.45	0.48	0.45	0.67	0.53	0.96
Travelled Business (5y)	0.09	0.08	0.11	0.09	0.11	0.47	0.31	0.96	0.42	0.42
Profit USD (30d)	221	221	262	195	236	0.92	0.25	0.47	0.59	0.47
Bought Alibaba Ever	0.16	0.13	0.15	0.18	0.13	0.15	0.48	0.64	0.17	0.22
# Customers (3d)	9.46	8.93	9.09	8.61	10.07	0.43	0.47	0.28	0.44	0.39
# Suppliers (12m)	5.52	5.82	5.41	5.18	5.56	0.39	0.75	0.27	0.91	0.49
# Suppliers Turkey (12m)	0.42	0.33	0.4	0.32	0.39	0.22	0.84	0.2	0.71	0.51
Avg Order Size (USD)	463	468	499	421	527	0.99	0.76	0.65	0.51	0.85
Modal Good Price (USD)	17.67	17.03	17.66	17.61	17.41	0.5	1	0.9	0.78	0.95
Pays Supp Mobile Money	0.87	0.84	0.87	0.86	0.88	0.24	0.96	0.53	0.8	0.6
<i>N</i>	362	373	379	381	367					

Note: Columns 1–5 report means by treatment cell. Columns 6–9 report randomization-inference p -values for each arm relative to control (2vs1, 3vs1, 4vs1, 5vs1), computed following the regression specification in (2). Column 10 reports the RI p -value from a joint test all four coefficients are zero. Finally, we run a multinomial logit of the four treatment groups on all variables in the table; a joint test that all coefficients are zero has RI p -value of 0.958.

Table A2: One Week Survey

	Indicator for Answers 1 or 2 (out of 5)	
	Find Products From Turkey (1)	Trust Supplier (2)
<i>Panel A: Pooled</i>		
Treatment	0.110*** (0.029)	
<i>Panel B: Individual Treatments</i>		
Search Only	0.078** (0.038) [0.047]	
Search + Adverse Selection	0.140*** (0.037) [0.000]	0.009 (0.036) [0.791]
Search + Moral Hazard	0.087** (0.037) [0.044]	-0.025 (0.036) [0.694]
Search + AS + MH	0.136*** (0.038) [0.000]	0.075** (0.038) [0.105]
Control Mean	0.453	0.303
% Increase (Pooled)	24.3%	6.6%
All Coefs Equal p -val	0.241	N/A
All Coefs Zero p -val	N/A	0.054
Adjusted R^2	0.06	0.06
N	1636	1201

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is an indicator for whether they can find Turkish products. Column 2 is 1 if they reported a trust level 1 or 2, 0 if they reported a trust level of 3 or 4 or 5, and missing otherwise. Because control firms were not connected to any suppliers, they were not asked to provide ratings. Accordingly, Panel B, Column 2, omits the control group. Panel A, Column 2, reports estimates comparing the pooled trust treatments (AS, MH, and AS+MH) to the Search Only group.

Table A3: Horizontal Outcomes (Detailed)

	Extensive vs Intensive Margin		Number of Criteria	
	Agree Search (1)	Find Product Conditional (2)	Num Criteria Unconditional (3)	Num Criteria Conditional (4)
<i>Panel A: Pooled</i>				
Treatment	0.015 (0.027)	0.103*** (0.034)	0.281** (0.112)	0.368*** (0.139)
<i>Panel B: Individual Treatments</i>				
Search Only	0.020 (0.034) [0.888]	0.143*** (0.043) [0.004]	0.531*** (0.148) [0.002]	0.701*** (0.178) [0.001]
Search + Adverse Selection	0.004 (0.034) [0.993]	0.069 (0.043) [0.116]	0.143 (0.144) [0.503]	0.191 (0.182) [0.301]
Search + Moral Hazard	0.037 (0.033) [0.626]	0.103** (0.043) [0.040]	0.318** (0.143) [0.071]	0.326* (0.174) [0.149]
Search + AS + MH	-0.003 (0.034) [0.993]	0.096** (0.044) [0.047]	0.132 (0.142) [0.503]	0.249 (0.179) [0.283]
Control Mean	0.673	0.453	1.421	2.094
% Increase (Pooled)	2.2%	22.7%	19.8%	17.6%
All Coefs Equal p -val	0.597	0.369	0.025	0.020
Adjusted R^2	0.06	0.04	0.04	0.03
N	1859	1274	1859	1274

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm agrees to sell or search for the product. Column 2 is 1 if the firm has a suitable product, conditional on agreeing to sell or search for the product. Column 3 is the number of horizontal criteria of the product, and is 0 if the firm either did not agree to sell or search for a product or agreed but never sent any product. Column 4 is the number of horizontal criteria of the product, conditional on agreeing to sell or search for a product, and is 0 if the firm agreed but never sent any product.

Table A4: Vertical Outcomes (Detailed)

	From Turkey		
	Made in Turkey (Label) (1)	Made in Turkey (Tailor Judgement) (2)	Made in Turkey (Label + Tailors) (3)
<i>Panel A: Pooled</i>			
Treatment	0.210** (0.081)	0.099 (0.081)	0.159** (0.072)
<i>Panel B: Individual Treatments</i>			
Search Only	0.169* (0.096) [0.141]	0.134 (0.095) [0.437]	0.138 (0.086) [0.195]
Search + Adverse Selection	0.172* (0.100) [0.141]	0.010 (0.102) [0.927]	0.107 (0.089) [0.221]
Search + Moral Hazard	0.241** (0.094) [0.038]	0.132 (0.095) [0.437]	0.192** (0.084) [0.087]
Search + AS + MH	0.271*** (0.097) [0.028]	0.096 (0.097) [0.515]	0.189** (0.084) [0.087]
Control Mean	0.489	0.581	0.544
% Increase (Pooled)	42.9%	17.0%	29.2%
All Coefs Equal p -val	0.538	0.489	0.625
Adjusted R^2	0.10	-0.00	0.07
N	255	287	330

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the label says Made in Turkey, 0 if the label says for X other than Turkey, and missing otherwise. Column 2 is 1 if both tailors independently determined that the product was made in Turkey, and is 0 if both tailors independently determined that the product was not made in Turkey. It is missing if the tailors disagreed. For shoes, as there was only one expert shoemaker, we take their opinion directly. Column 3 is an indicator that combines the label and tailor measures of whether the good was made in Turkey. It is equal to the label measure where available, and the tailor measure otherwise.

Table A5: Access to Foreign Goods (Selection and Robustness Analysis)

	Combined		Horiz	Vertical				Price
	Sold HQ (1)	Sold Turkish (2)	Has Product (3)	HQ Dummy (4)	Q Score (/50) (5)	Made in Turkey (6)	Index (7)	Price (USD) (8)
<i>Panel A: Pooled</i>								
Treatment	0.037* (0.021)	0.064*** (0.021)	0.065** (0.027)	0.111 (0.069)	-0.767 (0.633)	0.123* (0.063)	0.308** (0.137)	0.590 (0.703)
<i>Panel B: Individual Treatments</i>								
Search Only	0.060** (0.028) [0.111]	0.070** (0.029) [0.046]	0.106*** (0.035) [0.011]	0.150* (0.085) [0.250]	-0.378 (0.706) [0.819]	0.104 (0.076) [0.303]	0.398** (0.159) [0.056]	1.054 (0.867) [0.541]
Search + Adverse Selection	0.029 (0.027) [0.543]	0.040 (0.028) [0.142]	0.029 (0.034) [0.396]	0.153* (0.088) [0.250]	-0.088 (0.948) [0.930]	0.081 (0.082) [0.316]	0.358** (0.169) [0.096]	0.829 (0.816) [0.605]
Search + Moral Hazard	0.030 (0.027) [0.543]	0.068** (0.028) [0.046]	0.077** (0.035) [0.071]	0.092 (0.084) [0.443]	-1.314 (0.868) [0.381]	0.146** (0.073) [0.156]	0.266 (0.167) [0.209]	0.655 (0.874) [0.660]
Search + AS + MH	0.029 (0.026) [0.543]	0.077*** (0.029) [0.031]	0.047 (0.035) [0.301]	0.062 (0.085) [0.447]	-1.109 (0.846) [0.424]	0.148** (0.076) [0.156]	0.210 (0.181) [0.261]	-0.214 (0.824) [0.778]
Control Mean	0.097	0.105	0.307	0.431	43.064	0.477	0.000	19.990
% Increase (Pooled)	38.1%	61.0%	21.2%	25.8%	-1.8%	25.8%	N/A	3.0%
All Coefs Equal p -val	0.673	0.627	0.114	0.654	0.502	0.772	0.659	0.356
Adjusted R^2	0.02	0.04	0.07	0.07	0.35	0.25	0.01	0.45
N	1438	1438	1859	359	359	361	356	642

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm both sold a suitable good and it was high quality (following the definition in Column 2), and 0 otherwise (regression excludes cases where firm was randomly selected for non-purchase). Column 2 is 1 if the firm both sold a suitable good and it was from Turkey (following the definition in Column 4), and 0 otherwise (regression excludes cases where firm was randomly selected for non-purchase). Column 3 is an indicator that is one if the firm has a good that matches at least 3 horizontal criteria, and is missing if the firm never replied to the mystery shopper or was otherwise unreachable. Column 4 is an indicator for whether the good's quality score is above the median product-group quality score. Column 5 is the raw quality score. Column 6 is an indicator for whether the good is made in Turkey, primarily inferred based on the label. See the text for full details of how this outcome is constructed. Column 7 is the [Anderson \(2008\)](#) index combining the vertical outcomes to account for multiple hypothesis testing across outcomes. Column 8 is the price in USD, which is only measured conditional on the firm finding a good matching at least three horizontal criteria. All columns from 3 onwards include fixed effects for each of the 28 products.

Table A6: Profit and Sales (Poisson Regression)

	Profit		Sales	
	Profit 30 Days (USD) (1)	Profit 30 Days Winsorized 1% (USD) (2)	Sales 30 Days (USD) (3)	Sales 30 Days Winsorized 1% (USD) (4)
<i>Panel A: Pooled</i>				
Treatment	0.307* (0.132)	0.188 (0.112)	0.217 (0.142)	0.118 (0.126)
<i>Panel B: Individual Treatments</i>				
Search Only	0.088 (0.147) [0.916]	0.074 (0.133) [0.826]	0.315 (0.210) [0.431]	0.198 (0.158) [0.511]
Search + Adverse Selection	0.097 (0.189) [0.916]	0.119 (0.162) [0.826]	-0.041 (0.167) [0.830]	-0.026 (0.159) [0.876]
Search + Moral Hazard	-0.047 (0.138) [0.916]	-0.009 (0.143) [0.950]	-0.151 (0.174) [0.683]	-0.147 (0.143) [0.546]
Search + AS + MH	0.865*** (0.214) [0.001]	0.480*** (0.142) [0.007]	0.475** (0.212) [0.159]	0.260 (0.162) [0.396]
Control Mean	188.3	188.3	609.5	609.5
% Increase (Pooled)	35.9%	20.7%	24.2%	12.5%
All Coefs Equal p -val	0.009	0.020	0.067	0.057
Adjusted R^2	0.34	0.38	0.46	0.48
N	1351	1351	1378	1378

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is total profit from the past 30 days in USD. Column 2 is total profit from the past 30 days in USD, winsorizing the top 1%. Column 3 is total sales from the past 30 days in USD. Column 4 is total sales from the past 30 days in USD, winsorizing the top 1%. Profit is measured using the survey question from [De Mel, McKenzie, and Woodruff \(2009\)](#). Sales is measured using a similar survey question.

Table A7: Travel

	Wholesalers			Retailers		
	Any Travel (1)	Travel China (2)	Travel Turkey (3)	Any Travel (4)	Travel China (5)	Travel Turkey (6)
<i>Panel A: Pooled</i>						
Treatment	-0.033 (0.036)	-0.035 (0.030)	-0.020 (0.019)	-0.004 (0.014)	-0.021* (0.011)	0.003 (0.005)
<i>Panel B: Individual Treatments</i>						
Search Only	-0.001 (0.049) [0.980]	-0.009 (0.039) [0.936]	-0.020 (0.026) [0.856]	-0.005 (0.017) [0.984]	-0.022 (0.013) [0.272]	-0.004 (0.004) [0.692]
Search + Adverse Selection	0.007 (0.041) [0.980]	-0.011 (0.034) [0.936]	-0.012 (0.022) [0.856]	-0.001 (0.018) [0.997]	-0.025** (0.012) [0.126]	0.004 (0.008) [0.813]
Search + Moral Hazard	-0.051 (0.038) [0.425]	-0.050* (0.030) [0.272]	-0.014 (0.020) [0.856]	-0.008 (0.017) [0.977]	-0.015 (0.014) [0.298]	0.010 (0.008) [0.586]
Search + AS + MH	-0.052 (0.037) [0.425]	-0.043 (0.032) [0.399]	-0.016 (0.020) [0.856]	-0.001 (0.020) [0.997]	-0.022 (0.014) [0.272]	0.001 (0.006) [0.865]
Control Mean	0.130	0.090	0.040	0.041	0.033	0.004
% Increase (Pooled)	-25.4%	-38.9%	-50.0%	-9.8%	-63.6%	75.0%
All Coefs Equal p -val	0.168	0.248	0.987	0.986	0.866	0.151
Adjusted R^2	0.18	0.28	0.09	0.04	0.07	0.01
N	546	546	546	1125	1125	1125

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available).

Column 1 is 1 if the firm travelled internationally for business in the past 3 months. Column 2 is 1 if the firm travelled for business to China in the past 3 months. Column 3 is 1 if the firm travelled for business to Turkey in the past 3 months. Column 4 is 1 if the firm travelled internationally for business in the past 3 months. Column 5 is 1 if the firm travelled for business to China in the past 3 months. Column 6 is 1 if the firm travelled for business to Turkey in the past 3 months. Travel is 1 if either the firm owner or someone closely involved with the firm travelled internationally for firm-specific business in the past 3 months.

Table A8: Lee Bounds on Main Survey Outcomes

Outcome	Treatment	Coef	SE	Lee LB	LB SE	Lee UB	UB SE
Any Sup Turkey	Pooled	0.039*	0.021	0.030	0.029	0.089***	0.032
Any Sup Turkey	S Only	0.024	0.027	0.013	0.035	0.062*	0.037
Any Sup Turkey	S+AS	0.051*	0.029	0.044	0.036	0.093**	0.040
Any Sup Turkey	S+MH	0.006	0.027	-0.005	0.034	0.061	0.038
Any Sup Turkey	S+AS+MH	0.075***	0.029	0.068*	0.035	0.143***	0.042
Num Sup Turkey	Pooled	0.085**	0.035	0.073	0.046	0.171***	0.047
Num Sup Turkey	S Only	0.080	0.051	0.064	0.062	0.164***	0.059
Num Sup Turkey	S+AS	0.069	0.046	0.059	0.056	0.138**	0.062
Num Sup Turkey	S+MH	0.002	0.040	-0.012	0.051	0.098**	0.049
Num Sup Turkey	S+AS+MH	0.193***	0.058	0.185***	0.066	0.278***	0.078
Profit (W. 1%)	Pooled	42.346**	21.414	21.431	22.939	69.399*	40.765
Profit (W. 1%)	S Only	19.276	26.129	-16.096	112.056	10.220	30.194
Profit (W. 1%)	S+AS	30.947	31.180	14.094	33.439	57.659	54.219
Profit (W. 1%)	S+MH	-4.981	24.780	-31.444	27.323	26.762	49.715
Profit (W. 1%)	S+AS+MH	131.630***	44.353	103.253**	46.973	189.377***	46.504
Sales (W. 1%)	Pooled	133.128	83.904	60.980	88.451	300.616***	75.961
Sales (W. 1%)	S Only	185.777	123.823	113.197	127.067	328.926**	143.754
Sales (W. 1%)	S+AS	70.475	109.672	16.607	116.684	250.794**	111.430
Sales (W. 1%)	S+MH	-59.735	92.478	-141.587	99.540	108.860	132.227
Sales (W. 1%)	S+AS+MH	349.621**	148.693	268.666*	158.297	558.849***	146.320

Note: This table shows [Lee \(2009\)](#) bounds for the main outcome variables from the survey. Columns 1 and 2 present the main coefficient and standard error. Column 3 presents the Lee lower bound and Column 4 presents its standard error. Column 5 presents the Lee upper bound and Column 6 presents its standard error. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. As Lee bounds are based on bivariate regressions and do not easily permit covariates, we only include the baseline value of the outcome: we use it directly to tighten the bounds if it is binary, and otherwise we convert it to binary by creating an indicator for above or below the median.

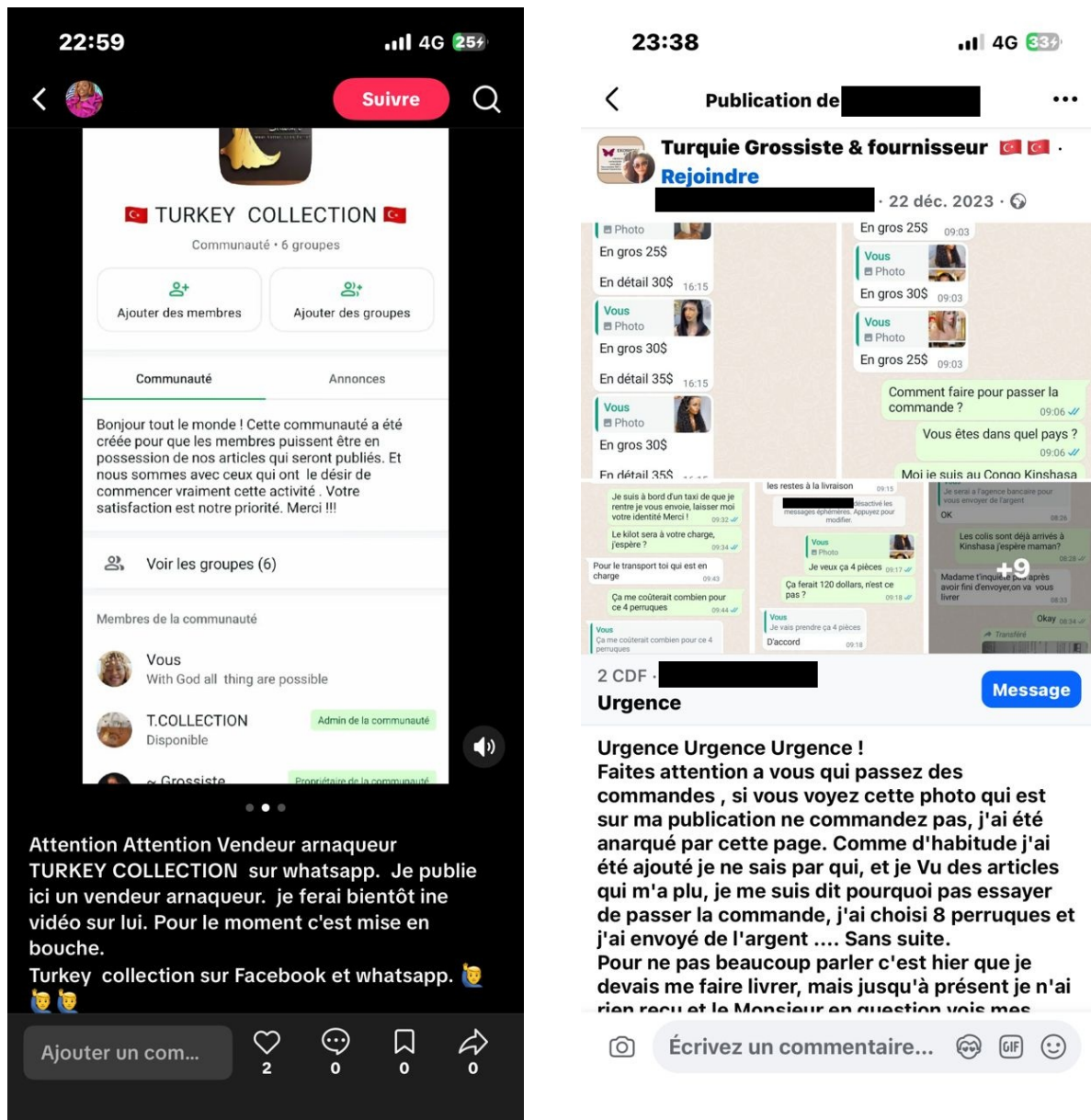
Table A9: Effect of Placebo Firm Groups

	Horizontal Dummy (1)	Vertical Dummy (2)	Num Supp In Turkey (3)	Profit Sales Index (1%) (4)
Placebo Group Treatment	-0.034 (0.054)	-0.013 (0.125)	0.013 (0.063)	0.012 (0.123)
Control Mean	0.374	0.438	0.216	-0.006
% Increase	-9.2%	-3.0%	6.0%	N/A
Adjusted R^2	-0.00	-0.02	-0.00	-0.00
N	311	65	342	266

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. Conventional robust standard errors are reported in parentheses. This table shows the effect of the placebo firm discussion groups on the main outcomes (see Section 4.1 for details). The omitted category is the half of the pure control group that did not receive as this treatment. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 2 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

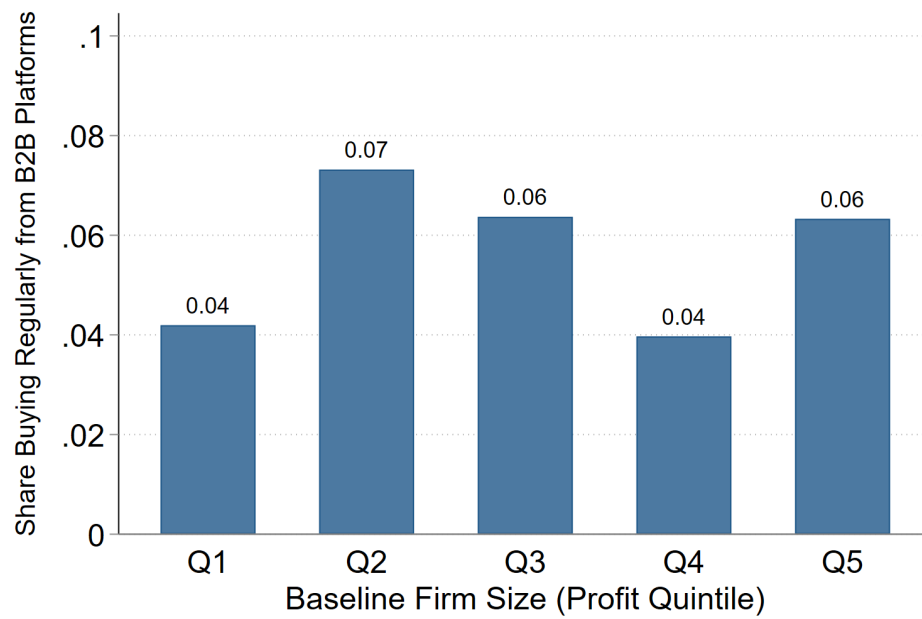
Figures

Figure A1: Examples of Group Punishment / Warning about Suppliers

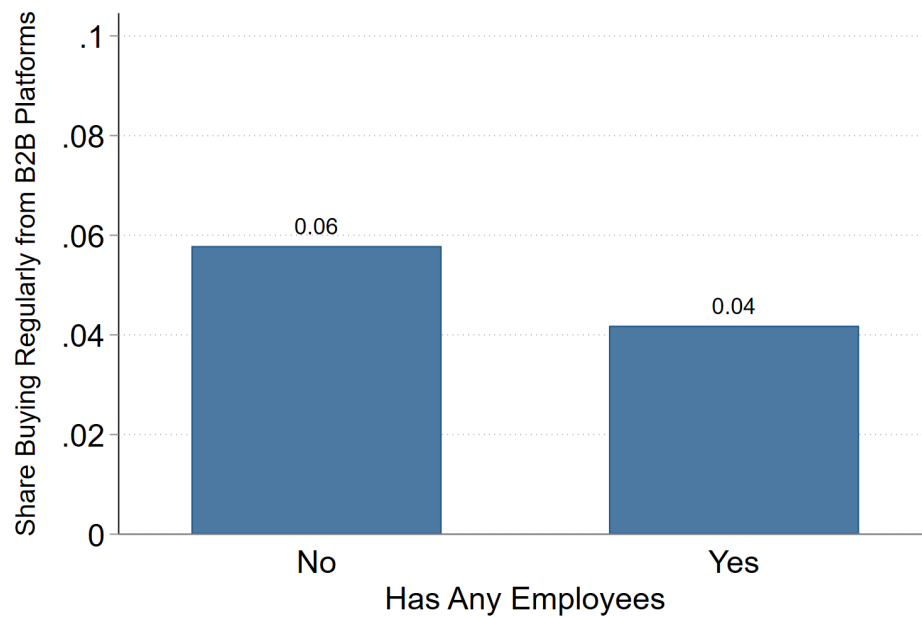


Note: This figure shows two examples of firms broadcasting on social media to alert other firms about bad experiences with particular suppliers in Turkey. In the left hand side image, the text states that this supplier is a scammer and that the firm will make and post a video with details, but is posting in the meantime to alert others about the supplier. In the right hand side image, the post explains that this firm made an order of 8 wigs (worth around USD 240), but, after paying, nothing has happened, and the supplier has repeatedly obfuscated.

Figure A2: B2B Platform Use by Firm Size



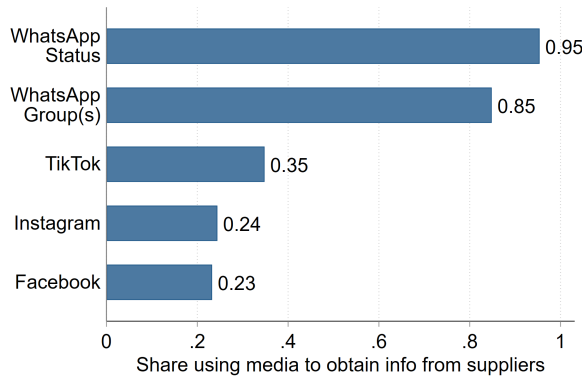
(a) By Profit Quintiles



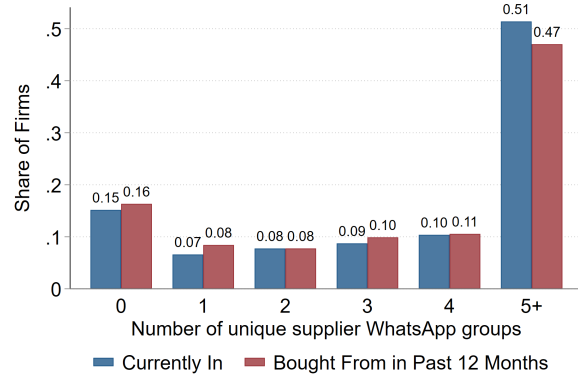
(b) By Any Employees

Note: This figure shows the share of firms that reported in the baseline survey that they made purchases from dedicated B2B e-commerce platforms at least once per month, as a function of baseline firm size. Panel (a) divides firms into quintiles based on baseline profit. Panel (b) divides firms into two groups based on whether they have at least one employee or not. In both panels, we cannot reject the null that the shares across all groups are equal.

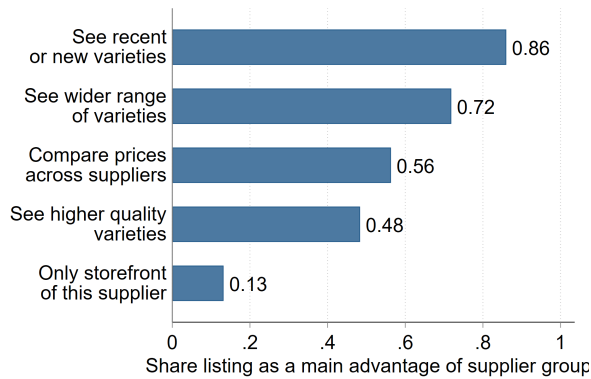
Figure A3: Social Media Usage (Physical Store Only)



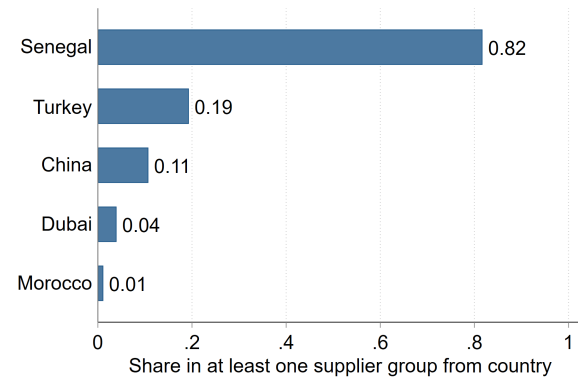
(a) Social Media Usage



(b) Supplier WhatsApp Groups



(c) Advantages of Supplier Groups



(d) Supplier Groups, by Supplier Origin

Note: This figure shows a number of statistics about how firms in our sample use social media to obtain information about suppliers. It is the same as Figure 3, but instead calculates statistics only for the 607 firms that have physical stores. Panel (a) shows the results of a question asking firms to select all social media that they use to obtain information about suppliers for their business. Panel (b) shows the distribution of the number of supplier WhatsApp groups a firm is in at the time of the baseline survey, as well as the distribution of the number of such groups that the firm has directly made at least one purchase from in the past 12 months. Supplier WhatsApp groups are defined as WhatsApp groups in which the primary purpose is for suppliers to advertise their wares to downstream clients. Panel (c) shows the results of a question asking firms that use supplier WhatsApp groups to select all reasons why they find these groups useful. Panel (d) shows the share of firms who are in at least one supplier WhatsApp group where the supplier located and based in the country listed.

Figure A4: Business Cards for Moral Hazard Treatment



Note: This figure shows the business cards used in the Moral Hazard treatment. In the treatment version, the card prominently displays a phone number for firms to rate suppliers or report problems, reinforcing the message that supplier performance will be monitored and can have consequences. In the control version, the card contains the same phone number but refers only to general questions about the study, with no mention of ratings or suppliers.

Figure A5: Mystery Shopping Goods (Examples)



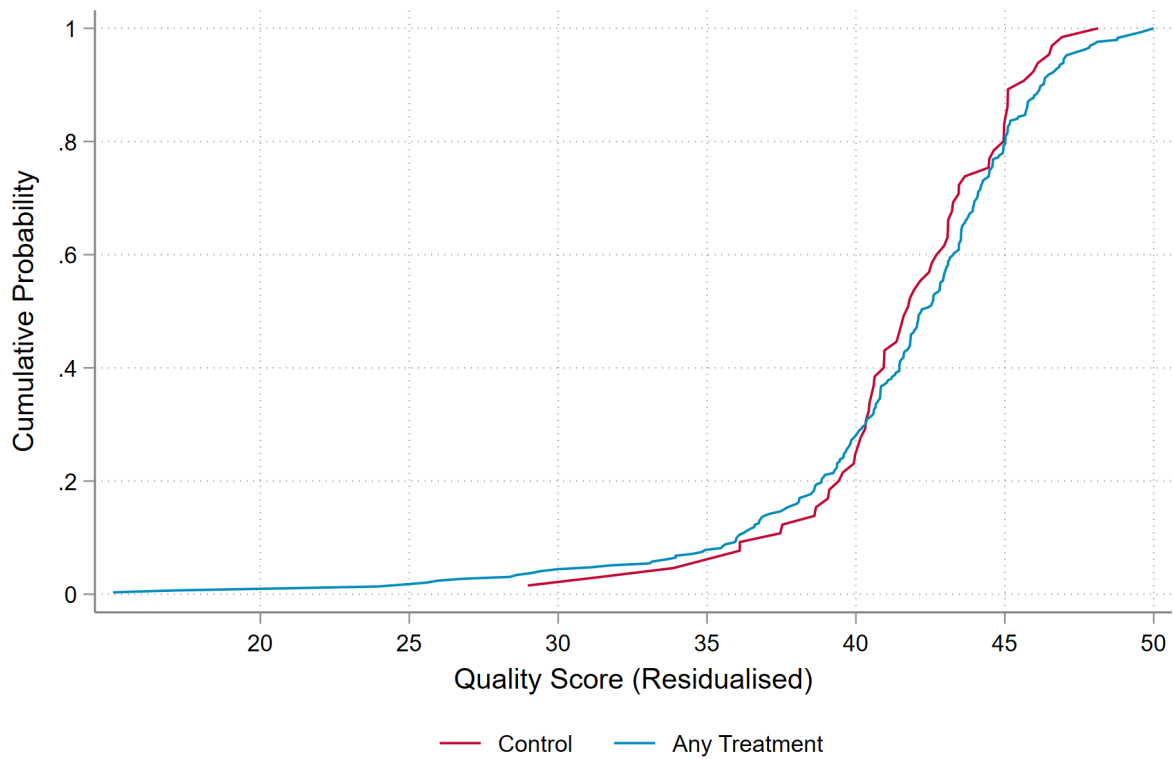
1. Red
2. One colour
3. No large logo/print
4. 3 black buttons
5. Size L



1. Black
2. Long sleeves
3. Knee length
4. Turtleneck
5. Ribbed Cotton

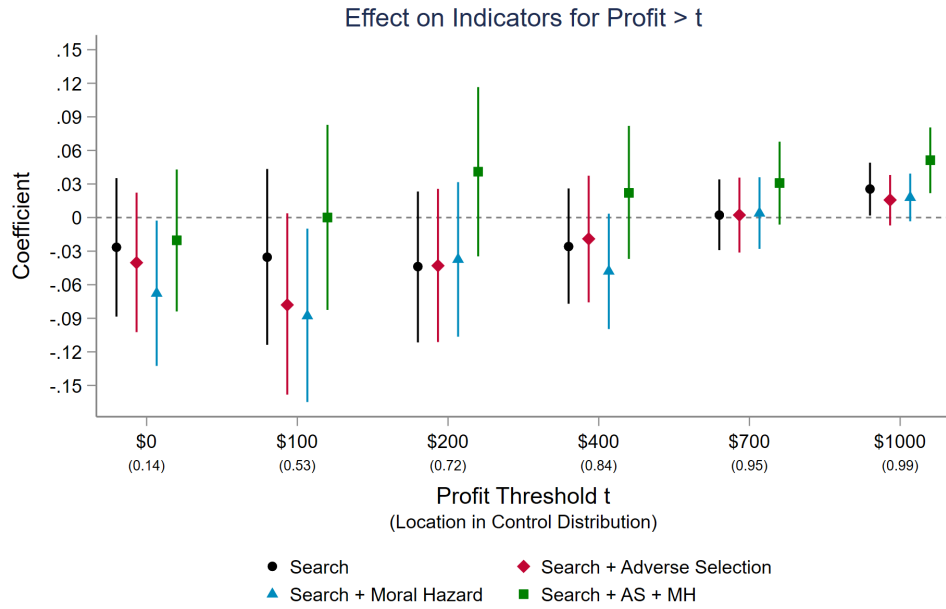
Note: Examples of goods requested in the mystery shopping exercise. In total, there were 28 different goods.

Figure A6: Quality Score Distribution

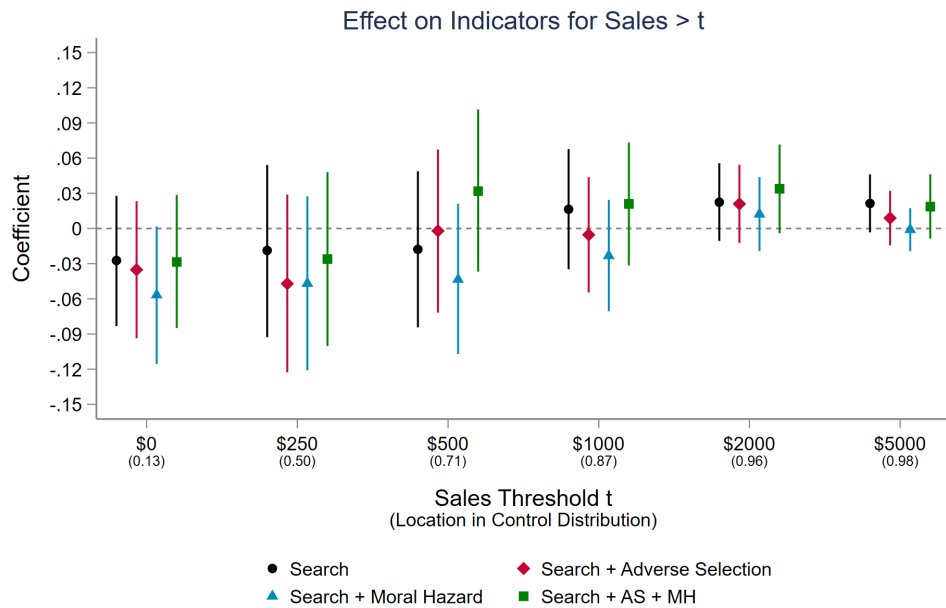


Note: This figure shows CDF of the quality score separately by treatment status, with all four treatment groups (Search Only, Search + Adverse Selection, Search + Moral Hazard, Search + AS + MH) pooled for visual ease. To be consistent with the regression in the table, we first residualise quality using stratum fixed effects and any covariates selected by PDS Lasso in the regression.

Figure A7: Threshold Regressions for Profit and Sales



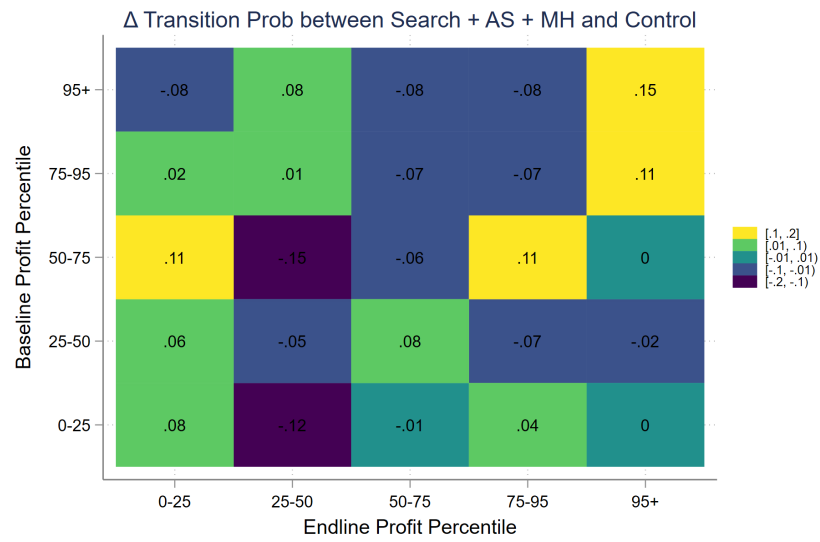
(a) Profit



(b) Sales

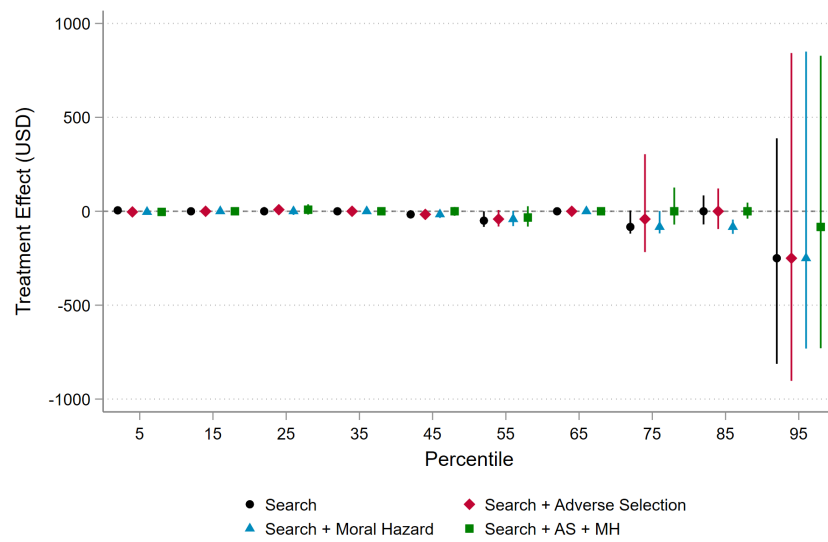
Note: This figure shows the coefficients from regressions of indicators for whether profit and sales are above some threshold t , for a range of t . All regressions include covariates selected by Double Lasso (Belloni, Chernozhukov, and Hansen, 2014), as well as the outcome measured at baseline (if available). The numbers in parentheses show the percentiles at which t is located in the distribution of the pure control group. We plot 95% confidence intervals constructed using randomisation inference, defined as the set of sharp nulls that do not reject at the 5% level, using the procedure in Young (2024).

Figure A8: Treatment Effects on Transition Matrix



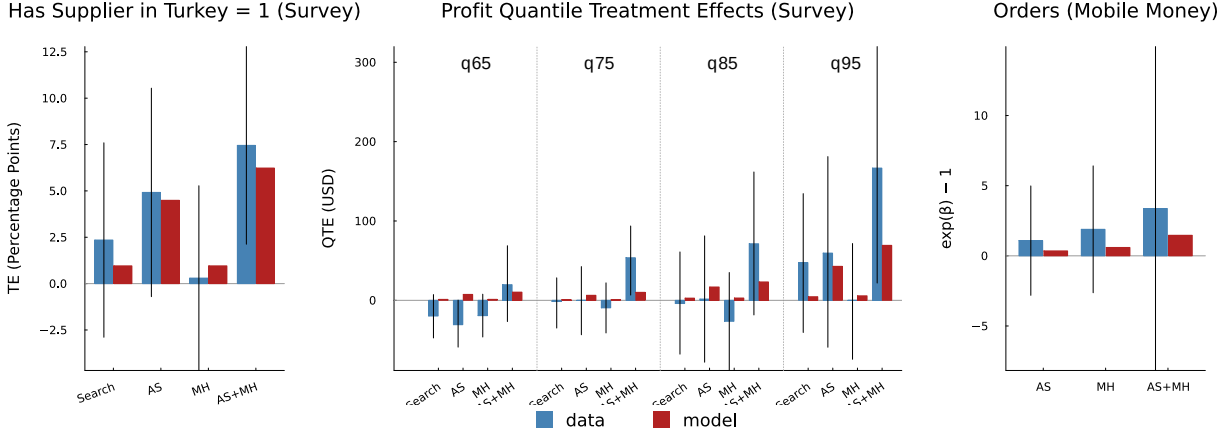
Note: This figure shows the treatment effects for the Search + Adverse Selection + Moral Hazard treatment on indicators for whether a firm transitioned between different percentile profit buckets between baseline and endline.

Figure A9: Placebo Check: Quantile Treatment Effects on Baseline Profit

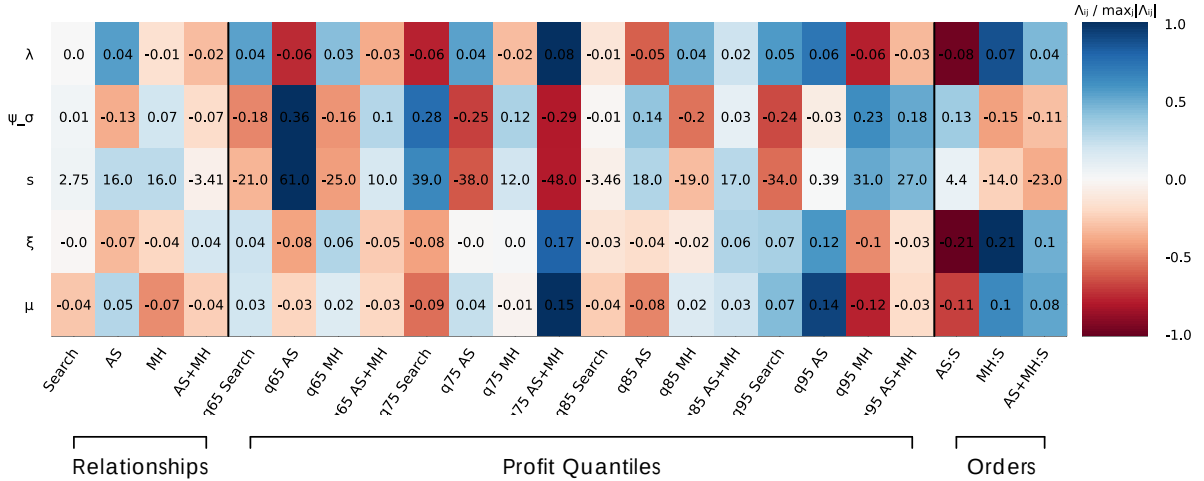


Note: This figure shows the coefficients from quantile regressions of baseline profit on the four treatment groups, intended as a placebo test. As the only covariate included in the main quantile regressions is the outcome measured at baseline, which is itself the outcome here, we do not include any covariates. We plot 95% confidence intervals constructed using randomisation inference, defined as the set of sharp nulls that do not reject at the 5% level.

Figure A10: Model Fit



(a) Fit on Targeted Moments



(b) Parameter Sensitivity (Andrews et al., 2017)

Note: Panel A shows the moments targeted in the structural estimation, with confidence intervals on the empirical moments shown to highlight the relative precisions. We weight moments by the inverse of the variance-covariance matrix of the treatment effects. Panel B shows the Andrews et al. (2017) measure of parameter sensitivity to estimation moments, $\Lambda = -(G'WG)^{-1}G'WK$, where K is such that Λ_{ij} can be interpreted as the local sensitivity of parameter i to a one standard deviation increase in empirical moment j . Reading within a row provides information about which moments most influence a given parameter. Thus, within each row (i.e., for a given parameter), we shade cells according to their relative influence—specifically, the shading of value Λ_{ij} corresponds to $\Lambda_{ij} / \max_j |\Lambda_{ij}| \in [-1, 1]$, such that moments with dark blue cells (shading near 1) have the largest positive influence on that parameter and the dark red cells (shading near -1) have the largest negative influence on that parameter. We omit ψ_μ as it is solved for internally to each guess of the other 5 parameters.

Appendix B - Mathematical Appendix

In this Appendix, we provide further detail on the model. In the main text, we defined the contract as a sequence of quantities, q_t , and prices, p_t . For the purposes of stating and characterizing the solution, it is easier to define the contract as a sequence of quantities, q_t , and transfers, τ_t , such that the firm orders quantity q_t and pays transfer τ_t . This is purely notational and does not affect the solution, as the price can always be backed out as $p_t = \tau_t/q_t$.

Further Details on the Optimal Contract

Assumption 1. (No trade with bad types)

$$\max_q \lambda r(q) - (1 - \xi)cq < \max_q r(q) - p_l q$$

This assumption states that if a firm knew that the supplier was a bad type, they would prefer to order from the local supplier.

Proposition 1. *It is not optimal to offer a menu of contracts that separates good and bad suppliers.*

Proof. Suppose the contrary, and consider the state of the world where the supplier is a bad type. Since the menu fully separates the types, the firm's posterior is then that the supplier is a bad type with probability 1. Because the supplier has limited liability, the maximum that the firm can earn under any such contract is the full surplus, i.e., $(\max_q \lambda r(q) - (1 - \xi)cq)/(1 - \delta)$. Assumption 1 implies that the firm can always do better than this, because at the very least they can buy from a local supplier in every period. Since the contract is relational, the firm would thus renege before sending the first transfer. Thus, the expected payoff to the bad type from accepting the revealing contract is 0. So long as the contract recommended to the good type involves positive quantity, the bad type can always earn a positive expected payoff by accepting the good type's contract, because limited liability ensures that $\tau_t \geq cq_t > (1 - \xi)cq_t$ for all t . Therefore, the bad type would not accept the contract that reveals their type, which is a contradiction. $\square$

The original program is as follows:

$$\begin{aligned} L = & \min_{\{\rho_t\}, \{\eta_t\}, \{\gamma_t\}} \max_{\{q_t\}, \{\tau_t\}} \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) ((1 - \mu_t(1 - \lambda)r(q_t) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}) \\ & + \sum_{t=0}^{\infty} \delta^t \rho_t \left[\sum_{\tau=t+1}^{\infty} \delta^{\tau-t} (R_{\tau} - cq_{\tau}) - \xi cq_t \right] \\ & + \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \eta_t \left[\sum_{\tau=t}^{\infty} \delta^{\tau-t} (1 - \mu_{\tau}(1 - \lambda^{\tau-t})) ((1 - \mu_{\tau}(1 - \lambda)r(q_t) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}) - \bar{U}) \right] \\ & + \sum_{t=0}^{\infty} \delta^t \gamma_t [\tau_t - cq_t] \end{aligned}$$

The modified program is as follows.

$$\begin{aligned}
W_t(U_t, V_t, \mu_t) = & y(q_t, \tau_t, \mu_t) + \rho_t (\delta(1 - \lambda)V_{t+1} - \xi c q_{t+1}) \\
& + \eta_t (U_{t+1} - \bar{U}) \\
& + \gamma_t (\tau_t - c q_t) \\
& + \nu_t^b (y(q_t, \tau_t, \mu_t) + \delta(1 - \mu_t(1 - \lambda))U_t + \delta\mu_t(1 - \lambda)\bar{U} - U_t) \\
& + \nu_t^s (\tau_t - c q_t + \delta V_{t+1} - V_t) \\
& + \delta(1 - \mu_t(1 - \lambda))W_{t+1}(U_{t+1}, V_{t+1}, \mu_{t+1})
\end{aligned}$$

The FOCs are as follows

$$(1 - \mu_t(1 - \lambda))r'(q_t)(1 + \nu_t^b) = \rho_t \xi c + (\gamma_t + \nu_t^s) c \quad (q_t)$$

$$1 + \nu_t^b = \gamma_t + \nu_t^s \quad (\tau_t)$$

$$\rho_t(1 - \lambda) + \nu_t^s = -(1 - \mu_t(1 - \lambda)) \frac{\partial W_{t+1}}{\partial V_{t+1}} \quad (V_{t+1})$$

$$\eta_{t+1} + \delta(1 - \mu_t(1 - \lambda))\nu_t^b = -\delta(1 - \mu_t(1 - \lambda)) \frac{\partial W_{t+1}}{\partial U_{t+1}} \quad (U_{t+1})$$

Substituting the FOC for τ_t into the FOC for q_t and re-arranging gives

$$r'(q_t) = \frac{1}{1 - \mu_t(1 - \lambda)} \left(1 + \xi \frac{\rho_t}{1 + \nu_t^b} \right) c.$$

The Envelope Condition implies that $\frac{\partial W_{t+1}}{\partial V_{t+1}} = -\nu_{t+1}^s$ and $\frac{\partial W_{t+1}}{\partial U_{t+1}} = -\nu_{t+1}^b$. Combining FOCs 2-4 then gives the following equation relating γ_t and γ_{t+1} that we will make extensive use of in the following proofs.

$$\rho_t(1 - \lambda) + 1 - \gamma_t + \mu_t(1 - \lambda)\nu_t^b = (1 - \gamma_{t+1})(1 - \mu_t(1 - \lambda)) + \frac{\eta_{t+1}}{\delta}. \quad (4)$$

Remark 1. $\nu_{t+1}^b > \nu_t^b \iff \eta_{t+1} > 0$

Proof. This follows immediately from the FOC for U_{t+1} after substituting in the Envelope Condition. $\square$

The optimal contract is generally not available in closed form, but we state and prove some properties in the following proposition.

Proposition 2. *There exists finite T^* such that:*

1. *The agent earns zero stage profits for all $t < T^*$ (i.e., LL binds for all $t < T^*$).*
2. *The principal earns $(1 - \delta)\bar{U}$, that is, zero stage profits net of their outside option, for all $t > T^*$ (i.e., DEC binds for all $t > T^*$).*
3. *q_t is strictly increasing for $t < T^*$.*

4. ICC binds for at least some $t < T^*$.

Proof. We prove this through a series of Lemmas, which we state and prove below. For parts 1 and 2, see Lemma 3. For part 3, see Lemma 1. For part 4, see Lemma 5. $\square$

Before formally stating the Lemmas, we first provide an intuitive sketch of the approach. First, we show that q_t must be strictly increasing whenever LL binds. Intuitively, a (weakly) decreasing q_t despite beliefs improving would imply that ICC strongly becomes “more binding” over time. But when LL is binding, the supplier is earning zero stage profits, so the ICC must be getting less binding over time.

Second, we show that if LL in t binds, then DEC in t cannot bind. Intuitively, both parties cannot be earning their outside option at the same time, as belief improving and q_t growing would imply that in other periods one of them must be making a loss.

Third, we show that the problem can be divided into two phases: LL will bind for all early periods and be slack for all late periods. “Backloading” results of this kind are standard in the dynamic moral hazard literature. Backloading happens for two reasons. The first reason is that incentives must be given to the agent at some point, and backloading incentives is efficient because it improves both early and late ICCs (whereas frontloading or even-loading still improves early ICCs but improves late ICCs less). The second reason is that adverse selection means that the “good type” agent is more patient than the principal, as the good type knows their own type. This means that it is always cheaper for the principal to backload payments. As a corollary, the DEC must bind for all late periods, as the principal wants to backload as much as possible, and will continue to do so until the DEC binds.

Finally, we show that ICC must bind for some t in the early section. Intuitively, if it didn’t, then the principal would just extend the early period—where they earn all the surplus—for longer. The only reason to ever end this early phase is precisely because an ICC eventually binds (the principal has to pay the agent eventually, and in absence of the ICC would always prefer not to).

Lemma 1. *If LL binds in $t + 1$, then $q_{t+1} > q_t$.*

Proof. Suppose that LL binds in $t + 1$ and $q_{t+1} \leq q_t$. The FOC for q_t is

$$r'(q_t) = \frac{1}{1 - \mu_t(1 - \lambda)} \left(1 + \xi \frac{\rho_t}{1 + \nu_t^b} \right) c.$$

We already know that $\nu_{t+1}^b \geq \nu_t^b$, and that $\mu_{t+1} \leq \mu_t$. Then, the hypothesis $q_{t+1} \leq q_t$ implies that $\rho_{t+1} > \rho_t$. Then, we can write

$$\delta(1 - \lambda)V_{t+1} < (1 - \lambda)V_{t+1} = (1 - \lambda)(R_{t+1} - cq_{t+1} + \delta V_{t+2}) = \delta(1 - \lambda)V_{t+2} = \xi cq_{t+1} \leq \xi cq_t.$$

The first equality is a definition, the second equality follows from the fact that LL binds in $t + 1$, the third equality follows from the fact that $\rho_{t+1} > \rho_t$ implies that $\rho_{t+1} > 0$, which implies that ICC binds in $t + 1$. The final inequality follows from the hypothesis that $q_{t+1} \leq q_t$. Thus, we have shown that

$$\delta(1 - \lambda)V_{t+1} < \xi cq_t.$$

But this implies that ICC in t is violated, which is a contradiction. $\square$

Lemma 2. *If LL in $t + 1$ binds, then DEC in $t + 1$ is slack.*

Proof. Suppose that DEC in $t + 1$ binds. Then, we can write

$$\begin{aligned}
U_t &= (1 - \mu_t(1 - \lambda))r(q_t) - \tau_t + \delta(1 - \mu_t(1 - \lambda))U_{t+1} + \delta\mu_t(1 - \lambda)\bar{U} \\
&= (1 - \mu_t(1 - \lambda))r(q_t) - \tau_t + \delta\bar{U} \\
&\leq (1 - \mu_t(1 - \lambda))r(q_t) - cq_t + \delta\bar{U} \\
&< (1 - \mu_{t+1}(1 - \lambda))r(q_t) - cq_t + \delta\bar{U} \\
&\leq (1 - \mu_{t+1}(1 - \lambda))r(q_{t+1}) - cq_{t+1} + \delta\bar{U} \\
&\leq (1 - \mu_{t+1}(1 - \lambda))r(q_{t+1}) - cq_{t+1} + \delta(1 - \mu_{t+1}(1 - \lambda))U_{t+2} + \delta\mu_{t+1}(1 - \lambda)\bar{U} \\
&= (1 - \mu_{t+1}(1 - \lambda))r(q_{t+1}) - R_{t+1} + \delta(1 - \mu_{t+1}(1 - \lambda))U_{t+2} + \delta\mu_{t+1}(1 - \lambda)\bar{U} \\
&= U_{t+1} \\
&= \bar{U}.
\end{aligned}$$

The first line is the definition of U_t . The second line follows from DEC binding in $t + 1$. The third line follows from LL in t . The fourth line follows from $\mu_{t+1} < \mu_t$. The fifth line follows from Lemma 1. The sixth line follows from DEC in period $t + 2$, i.e., $U_{t+2} \geq \bar{U}$. The seventh line follows from the hypothesis that LL binds in $t + 1$, i.e., $\tau_t = cq_t$. The eighth line is the definition of U_{t+1} . The ninth line follows from DEC binding in $t + 1$.

The above thus establishes that $U_t < \bar{U}$. But this is a violation of DEC in period t , which is a contradiction. $\square$

Corollary 1. *If LL is slack in t , then it is also slack in $t + 1$.*

Proof. LL slack in t means $\gamma_t = 0$. Suppose it binds in $t + 1$, which means $\gamma_{t+1} > 0$. Lemma 2 then implies that $\eta_{t+1} = 0$. But, Equation (4) gives

$$\gamma_{t+1}(1 - \mu_t(1 - \lambda)) = \frac{\eta_{t+1}}{\delta} - \rho_t(1 - \lambda) - \mu_t(1 - \lambda)(1 + \nu_t^b).$$

It must then be that $\eta_{t+1} > 0$, which is a contradiction. $\square$

Corollary 2. *If LL is slack in t , then DEC binds in $t + 1$.*

Proof. Corollary 1 implies that $\gamma_t = \gamma_{t+1} = 0$. Then, the FOC implies

$$\frac{\eta_{t+1}}{\delta} = \rho_t(1 - \lambda) + \mu_t(1 - \lambda)(1 + \nu_t^b) \geq \mu_t(1 - \lambda)(1 + \nu_t^b).$$

The final term is strictly positive, which implies that $\eta_{t+1} > 0$. $\square$

Lemma 3. *If any trade occurs, then there exists finite $T^* \geq 1$ such that (i) LL binds for all $t < T^*$ and is slack for all $t \geq T^*$, and (ii) DEC is slack for all $t < T^*$ and binds for all $t > T^*$.*

Proof. For (i): Corollary 1 shows that $\gamma_t = 0 \implies \gamma_{t+1} = 0$. Thus, if there exists T^* such that $\gamma_t = 0$, then $\gamma_s = 0$ for all $s \geq T^*$. We already know that $\gamma_0 = 1$, so this is not the case for $t = 0$. Suppose that $\gamma_t > 0$ for all t . Then, the supplier earns zero profit, which implies that all ICCs will fail unless $q_t = 0$ for all t , which violates the supposition that trade occurs at some point. Thus, there must be at least one $t \geq 1$ such that LL is slack in t . If there are multiple, define T^* as the earliest such t .

For (ii): Since LL is slack for all $t \geq T^*$, Corollary 2 implies that DEC binds for all $t > T^*$. $\square$

Corollary 3. $\nu_t^b = 0$ for all $t \leq T^*$, and $\nu_{t+1}^b > 0$ for all $t > T^*$.

Proof. This follows from Remark 1 and Lemma 3. $\square$

Lemma 4. $\gamma_{t+1} \leq \gamma_t$, with inequality strict if $\gamma_t \in (0, 1)$.

Proof. If $\gamma_{t+1} = 0$, then this holds trivially. We thus need to establish the claim for $\gamma_{t+1} > 0$. Suppose then that $\gamma_{t+1} > \gamma_t$ with $\gamma_{t+1} > 0$. Then, Equation (4) implies

$$0 > \rho_t(1 - \lambda) - \frac{\eta_{t+1}}{\delta} + \mu_t(1 - \lambda)(1 + \nu_t^b - \gamma_{t+1}).$$

Since $\gamma_{t+1} > 0$, Lemma 2 implies that $\eta_{t+1} = 0$. We are thus left with

$$\begin{aligned} 0 &> \rho_t(1 - \lambda) + \mu_t(1 - \lambda)(1 + \nu_t^b - \gamma_{t+1}) \\ &= \rho_t(1 - \lambda) + \mu_t(1 - \lambda)(1 + \nu_{t+1}^b - \gamma_{t+1}) \\ &= \rho_t(1 - \lambda) + \mu_t(1 - \lambda)\nu_{t+1}^s \end{aligned}$$

where the first equality follows from the fact that the FOC for U_{t+1} implies that $\eta_{t+1} = 0 \implies \nu_t^b = \nu_{t+1}^b$, and the second equality follows from the FOC for R_{t+1} . But the RHS is weakly positive, so this is a contradiction, which establishes that $\gamma_{t+1} \leq \gamma_t$.

Then, to establish the claim about strict inequality, suppose instead that $\gamma_{t+1} = \gamma_t$. We instead have

$$0 = \rho_t(1 - \lambda) + \mu_t(1 - \lambda)\nu_{t+1}^s,$$

which is only possible if $\rho_t = \nu_{t+1}^s = 0$. But this implies that $\gamma_{t+1} = 1 + \nu_t^b \geq 1$. If $\gamma_t \in (0, 1)$, this implies $\gamma_{t+1} > \gamma_t$, which is a contradiction. $\square$

Lemma 5. ICC binds for some $t < T^*$.

Proof. We prove this by iterating forward Equation (4). Since $\eta_t = \nu_t^b = 0$ for all $t < T^*$, the equation can be written

$$\gamma_t = \mu_t(1 - \lambda) + \rho_t(1 - \lambda) + \gamma_{t+1}(1 - \mu_t(1 - \lambda)).$$

Starting with $\gamma_0 = 1$ and iterating this until $T^* - 1$, for which $\gamma_{t+1} = 0$, we get

$$1 + (1 - \mu_0(1 - \lambda^{T^*})) \frac{\eta T^*}{\delta} = \sum_{t=0}^{T^*-1} (1 - \mu_0(1 - \lambda^t)) [\mu_t(1 - \lambda) + \rho_t(1 - \lambda)].$$

Note that $LHS \geq 1$. The first term on the RHS simplifies to

$$\sum_{t=0}^{T^*-1} (1 - \mu_0(1 - \lambda^t)) \mu_t(1 - \lambda) = (1 - \lambda) \mu_0 \sum_{t=0}^{T^*-1} \lambda^t = \mu_0(1 - \lambda^{T^*}) < 1.$$

Thus, it cannot be that $\rho_t = 0$ for all $t \leq T^* - 1$, as otherwise $LHS > RHS$. So there must be some $t \leq T^*$ for which the ICC binds. $\square$

Proposition 3. *The period-0 value of the relationship is decreasing in μ_0 and ξ .*

Proof. We first show the case for ξ . Using the Envelope Theorem to differentiate the original infinite horizon Lagrangian with respect to ξ gives

$$\frac{dL}{d\xi} = - \sum_{t=0}^{\infty} \delta^t \rho_t c q_t.$$

Since Proposition 2 established that the DICC binds for at least some t , we know that $\rho_t > 0$ for at least some t . So this is strictly negative except for the trivial case where $q_t = 0$ for all t , in which case it is zero.

Second, for μ_0 , it is easier to work with the recursive formulation. Differentiating the period-0 program gives

$$\frac{dW_0}{d\mu_0} = \underbrace{-(1 - \lambda)r(q_0)(1 + \nu_0^b) - \delta(1 - \lambda)\nu_0^b(U_1 - \bar{U}) - \delta(1 - \lambda)W_1}_{\equiv \phi_0} + \delta(1 - \mu_0(1 - \lambda)) \frac{d\mu_1}{d\mu_0} \frac{\partial W_1}{\partial \mu_1}.$$

The first three terms are all negative. The only term that we cannot immediately sign is $\partial W_1 / \partial \mu_1$. However, since the problem is recursive, we can simply repeatedly lead ϕ_t to obtain

$$\frac{dW_0}{d\mu_0} = \sum_{t=0}^{\infty} \delta^t \prod_{s=0}^t (1 - \mu_s(1 - \lambda)) \frac{d\mu_{s+1}}{d\mu_s} \phi_t.$$

This is strictly negative, except in the trivial case where the firm never purchases anything with the supplier abroad. $\square$

Proposition 4. *Under a technical condition described in Equation (5), the buyer's period-0 value function satisfies $\partial^2 U_0(\mu, \xi) / \partial \mu \partial \xi > 0$.*

Proof. We know that

$$dU_0/d\xi = -\sum_{t=0}^{\infty} \delta^t \rho_t c q_t = -\sum_{t=0}^{T-2} \delta^t \rho_t c q_t - \sum_{t=T-1}^{\infty} \delta^t \rho_t c q_t,$$

where T is the first period of the mature phase of the relationship (and $T - 1$ is the “handoff” phase). We want to differentiate this with respect to μ . Start with the second term, as the buildup and mature structure of the solution means that it will be relatively unimportant (as the frictions have essentially no interaction in the mature phase). For $t \geq T$, the supplier earns all of the surplus, with the solution converging to the stationary solution where $\mu_s = 0$ for all s . To facilitate intuition, we can approximate this term with the stationary solution (Ray (2002)). The stationary solution is the solution to

$$\max_{\tau, q} \tau - cq$$

subject to $\delta(1 - \lambda) \frac{\tau - cq}{1 - \delta} \geq \xi cq$ (the DICC) and $r(q) - \tau \geq (1 - \delta)\bar{U}$ (the DEC). We can see that the DEC always binds and $\tau = r(q) - (1 - \delta)\bar{U}$. We can then plug this into the objective function and DICC. If the DICC is slack at the first-best value of q , then the solution is q^{FB} ; otherwise, it is obtained by solving the DICC at equality. Either way, the solution to this program yields $V^* = (r(q^*) - (1 - \delta)\bar{U} - cq^*)/(1 - \delta)$ and a value for the multiplier $\rho^*(q^*)$.

Returning to $\sum_{t=T-1}^{\infty} \delta^t \rho_t c q_t$, and continuing with the stationary approximation, if the stationary DICC is slack then we simply have $\rho_t = 0$ and this term is zero. If the DICC does bind, then we have $cq_t = \delta(1 - \lambda)V^*/\xi$ and $\rho_t = (1 - \mu(1 - \lambda^t))\rho^*$, so plugging in gives

$$\begin{aligned} -\sum_{t=T-1}^{\infty} \delta^t \rho_t c q_t &= -\frac{\delta(1 - \lambda)}{\xi} \sum_{t=T-1}^{\infty} \delta^t (1 - \mu(1 - \lambda^t)) \rho_t V_t^* \\ &\approx -\frac{\delta(1 - \lambda)V^*\rho^*}{\xi} \sum_{t=T-1}^{\infty} \delta^t (1 - \mu(1 - \lambda^t)). \end{aligned}$$

The term in the summation is decreasing in μ , causing the overall object to be increasing in μ . The only concern is the approximation step, as ρ_t and V_t do have a dependence on μ_t as the surplus does expand as beliefs improve. However, this region only depends on μ to the extent that the surplus is still expanding due to beliefs improving, and since this is conditional on surviving deep into the relationship this is of order $(\delta\lambda)^t$ for $t \geq T$, and is thus negligible unless λ is very close to 1.

The first term, $-\sum_{t=0}^{T-2} \delta^t \rho_t c q_t$, is more important, as this is the region where the frictions more meaningfully interact. Since the supplier earns zero stage profits in periods 0 through $T - 2$, we can write $V_t = \delta^{T-1-t} V_{T-1}$. This means that we can simplify the DICC in this period as

$$\delta(1 - \lambda)\delta^{T-t} V_{T-1} \geq \xi c q_t.$$

Since $\rho_t = 0$ for any t for which this does not bind, we can substitute this directly into the earlier

derivative to obtain

$$-\sum_{t=0}^{T-2} \delta^t \rho_t c q_t = -\frac{\delta^{T-1}(1-\lambda)}{\xi} V_{T-1} \sum_{t=0}^{T-2} \rho_t,$$

Moreover, we already showed that $\sum_{t=0}^{T-2} \rho_t = 1 - \mu(1 - \lambda^{T-1})$, and we know that $V_0 = \delta^{T-1} V_{T-1}$. The expression thus simplifies to

$$-\sum_{t=0}^{T-2} \delta^t \rho_t c q_t = -\frac{(1-\lambda)}{\xi} (1 - \mu(1 - \lambda^{T-1})) V_0.$$

Thus, we have that

$$\begin{aligned} \frac{d^2 U_0(\mu, \xi)}{d\mu d\xi} &= -\frac{d}{d\mu} \left[\frac{(1-\lambda)}{\xi} (1 - \mu(1 - \lambda^{T-1})) V_0 \right] - \frac{d}{d\mu} \left[\frac{\delta(1-\lambda)}{\xi} \sum_{T-1}^{\infty} \delta^t (1 - \mu(1 - \lambda^t)) \rho_t^* V_t^* \right] \\ &\approx -\frac{d}{d\mu} \left[\frac{(1-\lambda)}{\xi} (1 - \mu(1 - \lambda^{T-1})) V_0 \right] - \frac{d}{d\mu} \left[\frac{\delta(1-\lambda)}{\xi} \rho^* V^* \sum_{T-1}^{\infty} \delta^t (1 - \mu(1 - \lambda^t)) \right] \end{aligned}$$

As discussed above, the second term is positive. The first term has two components. The direct component, $1 - \mu(1 - \lambda^{T-1})$, is positive. The only possible negative component comes from V_0 . That is, if improving the buyer's beliefs causes the supplier's period-0 value of the relationship to fall. This can in principle happen if improving beliefs causes the buyer to strongly increase the rent that they extract in period $T - 1$ and thereby *decrease* quantity in the buildup phase, which would occur if the revenue function is highly elastic (so they respond strongly to a change in the perceived intertemporal exchange rate) and if λ is very high (so the first-order desire to increase q early on due to the improved beliefs is muted). This decreases the total rents paid to the supplier in the buildup. Note that $dV_0/d\mu$ need not be positive; it can—and typically is—negative, as the supplier typically prefers the buyer to have better beliefs.

Thus, a necessary and sufficient condition for decreases in the frictions to be complementary is

$$\frac{dV_0}{d\mu} < \frac{V_0(1 - \lambda^T)}{1 - \mu(1 - \lambda^T)} - \frac{1}{1 - \mu(1 - \lambda^T)} \frac{d}{d\mu} \left[\sum_{T-1}^{\infty} \delta^{t+1} (1 - \mu(1 - \lambda^t)) \rho_t^* V_t^* \right]. \quad (5)$$

In words, any potential decrease in rent paid to the supplier must not be so large as to outweigh the direct positive effect coming from $(1 - \mu(1 - \lambda^{T-1}))$. Moreover, a sufficient condition (ignoring the near-negligible second term) is $\frac{dV_0}{d\mu} < 0$. $\square$

While V_0 is endogenous this condition amounts to requiring that the revenue function is not too elastic and that λ is not too large. We view this as mild for two reasons. First, Assumption (1) already requires that λ be small enough that bad types are not worth trading with. Second, in simulations with CES utility with elasticity σ , the necessary and sufficient condition holds for almost all but very extreme parameter constellations. Relatedly, this condition holds under the parameters estimated in the structural estimation section.

Further Details on the Search Process

In this sub-section, we prove (1) that the sequential search process behaves according to the standard results despite the fact that the outside option, $\bar{U}$, enters the utility function directly and (2) that if there is a firm-specific heterogeneity term z , then there exists a cutoff $\bar{z}$ such that only firms with $z > \bar{z}$ engage in search.

The search process works as follows. To match with a random supplier abroad, a firm can pay a one-time search cost $s > 0$. Upon matching, both parties immediately observe a realisation of a match-specific productivity term, $\psi \sim G$, that is fixed over time. We denote the period-0 value of a relationship with a supplier abroad with match-specific productivity ψ as $U_0(\psi, \bar{U})$, where we include $\bar{U}$ to make explicit that U_0 depends on the outside option directly. It does so through two channels. First, the adverse selection problem means that the firm will eventually earn $\bar{U}$ on path if the supplier is a bad type (because a low quality realization will eventually occur), which will cause U_0 to be increasing in $\bar{U}$. Second, the limited commitment problem means that the firm cannot make promises that would result in earning less than $\bar{U}$ (i.e., the DEC), which limits the enforceable surplus—this causes U_0 to be decreasing in $\bar{U}$. Since this is direct dependence on the outside option is non-standard in sequential search models, we prove the standard results here.

Proposition 5. *The optimal policy in the search process is a cutoff rule, where the cutoff value equals the expected utility from the search process.*

Proof. First, note that the search problem is recursive. That is, the entire history of searching is summarized by the highest value of ψ drawn to date, which, we denote with ψ_b . Second, U_0 is monotonic in ψ , strictly so in the region where the firm develops a relationship. As is standard, these together imply that the optimal policy is to search until the firm receives a draw above an endogenous cutoff $\bar{\psi}$.

For a firm whose best current draw is ψ_t , the expected benefit from obtaining one more draw is

$$\int_{\psi}^{\infty} (U_0(\psi, \bar{U}) - U_0(\psi_b, \bar{U})) dG(\psi),$$

where $\bar{U}$ is the expected utility of terminating the current relationship and re-searching. The expected cost is s . By definition, the cutoff value of $\bar{\psi}$ equates expected benefits with expected costs,

$$\int_{\bar{\psi}}^{\infty} (U_0(\psi, \bar{U}) - U_0(\bar{\psi}, \bar{U})) dG(\psi) = s.$$

As the stopping time is geometrically distributed, the expected number of draws until the firm obtains a realisation above $\bar{\psi}$ is $1/(1 - G(\bar{\psi}))$. This means that the expected utility from engaging in search, $\bar{U}$, is

$$\bar{U} = \frac{\int_{\bar{\psi}}^{\infty} U_0(\psi, \bar{U}) dG(\psi)}{1 - G(\bar{\psi})} - \frac{s}{1 - G(\bar{\psi})}.$$

Re-arranging, we can obtain

$$s = \int_{\bar{\psi}}^{\infty} U_0(\psi, \bar{U}) dG(\psi) - \bar{U} \int_{\bar{\psi}}^{\infty} dG(\psi) = \int_{\bar{\psi}}^{\infty} (U_0(\psi, \bar{U}) - \bar{U}) dG(\psi)$$

which, when compared with the definition of the cutoff, implies that $\bar{U} = U_0(\bar{\psi}, \bar{U})$. Thus, despite the fact that $\bar{U}$ enters the utility function directly, it does not change the fundamental structure of the search process. $\square$

Proposition 6. *The cutoff $\bar{\psi}$ is decreasing in s .*

Proof. It is easier to eliminate the $\bar{\psi}$ in the integral limit by rewriting the cutoff rule definition as follows

$$\int_{-\infty}^{\infty} \max\{U_0(\psi, \bar{U}) - \bar{U}, 0\} dG(\psi) = s,$$

where $\bar{U} = U_0(\bar{\psi}, \bar{U})$. Note that

$$\frac{ds}{d\bar{U}} = \int_{-\infty}^{\infty} \left(\frac{\partial U_0(\psi, \bar{U})}{\partial \bar{U}} - 1 \right) 1\{U_0(\psi, \bar{U}) - \bar{U} > 0\} dG(\psi).$$

In standard sequential search models, the first term in parentheses is zero, so this is clearly negative. Here, we will show that this new term is always strictly smaller than 1, which is sufficient to establish the claim.

We know that the U_0 is given by the following dynamic program (and we know that a solution exists for the particular draw of ψ under consideration as we only need to consider the cases where $U_0(\psi, \bar{U}) - \bar{U} > 0$),

$$\begin{aligned} L = & \min_{\{\rho_t\}, \{\eta_t\}, \{\gamma_t\}} \max_{\{q_t\}, \{\tau_t\}} \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) ((1 - \mu_t(1 - \lambda)r(\psi q_t) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}) \\ & + \sum_{t=0}^{\infty} \delta^t \rho_t \left[\sum_{\tau=t+1}^{\infty} \delta^{\tau-t} (R_{\tau} - cq_{\tau}) - \xi cq_t \right] \\ & + \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \eta_t \left[\sum_{\tau=t}^{\infty} \delta^{\tau-t} (1 - \mu_{\tau}(1 - \lambda^{\tau-t})) ((1 - \mu_{\tau}(1 - \lambda)r(\psi q_t) - \tau_t + \delta\mu_{\tau}(1 - \lambda)\bar{U}) - \bar{U}) \right] \\ & + \sum_{t=0}^{\infty} \delta^t \gamma_t [\tau_t - cq_t]. \end{aligned}$$

The Envelope Theorem implies that $\frac{dL}{d\bar{U}}$ is given by

$$\sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \delta\mu_t(1 - \lambda) + \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \eta_t \left[\sum_{\tau=t}^{\infty} \delta^{\tau-t} (1 - \mu_{\tau}(1 - \lambda^{\tau-t})) \delta\mu_{\tau}(1 - \lambda) - 1 \right].$$

Since Bayesian updating implies that we can write $\mu_t = \mu_0 \lambda^t / (1 - \mu_0(1 - \lambda^t))$, we can rewrite the first term as $\mu_0 \delta(1 - \lambda) \sum_{t=0}^{\infty} \delta^t \lambda^t$, or $\frac{\mu_0 \delta(1 - \lambda)}{1 - \delta\lambda}$, which is strictly less than 1. Intuitively, this is a (discounted) sum of probabilities and thus cannot exceed 1—indeed, it is easy to see that if we

set δ to 1 then this simplifies to just μ_0 . The first term is therefore strictly less than 1. The second term is negative, because the first term in square brackets is strictly less than 1 by the exact same argument. Since this argument did not rely on the value of ψ , we have that $dL/d\bar{U} < 1$ for all ψ for which $U_0(\psi, \bar{U}) - \bar{U} > 0$, which implies that

$$\frac{ds}{d\bar{U}} = \int_{-\infty}^{\infty} \left(\frac{\partial U_0(\psi, \bar{U})}{\partial \bar{U}} - 1 \right) 1\{U_0(\psi, \bar{U}) - \bar{U} > 0\} dG(\psi) < 0.$$

□

Note that a firm-specific multiplicative utility shifter that takes the form is equivalent to firms having heterogeneous search costs s/z as we simply divide through the cutoff definition equation by z . This proposition then implies that the cutoff utility is increasing in z , and thus that if there is a “global” outside option (from not searching at all) U_l , then there will exist some $\bar{z}$ such that the cutoff $\bar{U}(\bar{z})$ exactly equals U_l . In particular, the cutoff is defined such that,

$$\int_{-\infty}^{\infty} \max\{U_0(\psi, U_l) - U_l, 0\} dG(\psi) = s/\bar{z}.$$

Lemma 6. *The cutoff $\bar{z}$ is increasing in μ_0 and ξ .*

Proof. Since Proposition 3 establishes that U_0 is decreasing in μ_0 and ξ , the LHS of the equation defining $\bar{z}$ is decreasing in μ_0 and ξ , which means that $\bar{z}$ must be increasing in μ_0 and ξ to ensure that the RHS decreases. □

Appendix C - Heterogeneity Analysis

Table C1: Heterogeneity by Retailer vs Wholesaler

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.147*** (0.043)	0.158 (0.097)	0.021 (0.034)	0.004 (0.104)
Search + AS	0.031 (0.042)	0.221** (0.105)	0.053 (0.036)	0.013 (0.121)
Search + MH	0.097** (0.042)	0.119 (0.100)	-0.006 (0.033)	0.004 (0.091)
Search + AS + MH	0.005 (0.043)	0.058 (0.110)	0.054 (0.037)	0.024 (0.103)
S Only * Wholesaler	-0.105 (0.075)	-0.071 (0.194)	-0.011 (0.067)	0.469 (0.324)
S + AS * Wholesaler	0.061 (0.073)	-0.179 (0.193)	0.001 (0.067)	0.255 (0.323)
S + MH * Wholesaler	0.013 (0.075)	-0.139 (0.190)	0.016 (0.066)	-0.144 (0.231)
S + AS + MH * Wholesaler	0.142* (0.073)	0.005 (0.187)	0.023 (0.068)	1.209*** (0.437)
All Inter Zero p -val	0.023	0.783	0.989	0.034
Adjusted R^2	0.01	-0.01	0.00	0.05
N	1859	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm sells at least partially wholesale, defined as having a positive share of sales that are wholesale to other firms. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C2: Heterogeneity by Physical Store vs Online Only

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.176*** (0.044)	0.113 (0.097)	0.040 (0.038)	-0.058 (0.076)
Search + AS	0.091** (0.043)	0.139 (0.104)	0.065* (0.039)	-0.017 (0.098)
Search + MH	0.161*** (0.043)	0.074 (0.097)	0.005 (0.036)	-0.053 (0.060)
Search + AS + MH	0.098** (0.044)	0.033 (0.104)	0.081** (0.040)	-0.026 (0.066)
S Only * Physical Store	-0.181*** (0.073)	0.090 (0.200)	-0.066 (0.056)	0.653* (0.337)
S + AS * Physical Store	-0.135* (0.071)	0.048 (0.194)	-0.035 (0.060)	0.519 (0.376)
S + MH * Physical Store	-0.208*** (0.072)	-0.079 (0.209)	-0.036 (0.056)	0.165 (0.285)
S + AS + MH * Physical Store	-0.120* (0.072)	0.098 (0.185)	-0.031 (0.061)	1.582*** (0.497)
All Inter Zero p -val	0.042	0.924	0.836	0.028
Adjusted R^2	0.03	-0.01	0.02	0.12
N	1859	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm had a physical store at baseline. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C3: Heterogeneity by Direct Import Experience

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.126*** (0.044)	0.114 (0.104)	0.050 (0.030)	0.074 (0.121)
Search + AS	0.059 (0.043)	0.120 (0.106)	0.070** (0.031)	0.008 (0.102)
Search + MH	0.085** (0.043)	0.045 (0.106)	0.024 (0.028)	-0.135* (0.075)
Search + AS + MH	0.053 (0.043)	0.023 (0.108)	0.072** (0.031)	0.416** (0.192)
S Only * Experience	-0.033 (0.074)	0.076 (0.177)	-0.104 (0.068)	0.180 (0.278)
S + AS * Experience	-0.030 (0.073)	0.110 (0.190)	-0.053 (0.069)	0.246 (0.329)
S + MH * Experience	0.042 (0.074)	0.091 (0.178)	-0.080 (0.067)	0.285 (0.230)
S + AS + MH * Experience	0.006 (0.074)	0.114 (0.179)	-0.014 (0.072)	0.319 (0.430)
All Inter Zero p -val	0.840	0.968	0.515	0.762
Adjusted R^2	0.00	-0.01	0.05	0.02
N	1859	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm has direct online importing experience, defined as having bought goods from another country online directly at least once in the past 12 months. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C4: Heterogeneity by Baseline Turkish Share

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.059 (0.056)	0.180 (0.145)	0.024 (0.043)	0.038 (0.173)
Search + AS	-0.010 (0.054)	0.121 (0.155)	0.052 (0.044)	-0.033 (0.176)
Search + MH	0.026 (0.054)	0.200 (0.150)	-0.020 (0.038)	-0.168 (0.125)
Search + AS + MH	0.026 (0.056)	0.354** (0.141)	0.041 (0.045)	0.255 (0.261)
S Only * High Share	0.099 (0.073)	-0.053 (0.178)	-0.012 (0.059)	0.161 (0.236)
S + AS * High Share	0.098 (0.071)	0.046 (0.187)	0.015 (0.061)	0.204 (0.248)
S + MH * High Share	0.109 (0.072)	-0.177 (0.182)	0.023 (0.056)	0.214 (0.178)
S + AS + MH * High Share	0.045 (0.073)	-0.427** (0.176)	0.037 (0.062)	0.420 (0.359)
All Inter Zero p -val	0.471	0.073	0.940	0.692
Adjusted R^2	0.01	0.01	0.01	0.01
N	1814	353	1636	1261

Note: This table shows the main results with treatment interacted with an indicator for whether the share of the firm's customers in the past 30 days that buy Turkish-made products is above the median. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C5: Heterogeneity by Sex of Owner

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.020 (0.050)	0.111 (0.128)	-0.028 (0.037)	0.439** (0.216)
Search + AS	-0.045 (0.048)	0.148 (0.127)	-0.010 (0.038)	0.240 (0.206)
Search + MH	0.026 (0.050)	-0.033 (0.120)	-0.056 (0.036)	0.005 (0.152)
Search + AS + MH	-0.035 (0.048)	0.040 (0.122)	0.039 (0.041)	0.904*** (0.321)
S Only * Female	0.189*** (0.070)	0.052 (0.173)	0.090 (0.058)	-0.574** (0.232)
S + AS * Female	0.191*** (0.069)	0.017 (0.177)	0.134** (0.060)	-0.313 (0.242)
S + MH * Female	0.143** (0.070)	0.200 (0.170)	0.097* (0.056)	-0.047 (0.182)
S + AS + MH * Female	0.185*** (0.070)	0.046 (0.173)	0.060 (0.062)	-0.822** (0.343)
All Inter Zero p -val	0.021	0.768	0.222	0.029
Adjusted R^2	0.02	-0.01	0.02	0.05
N	1859	359	1680	1298

Note: This table shows the main results with treatment interacted with an indicator for whether the firm owner is female. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Table C6: Heterogeneity by Baseline Turkish Supplier Group

	Horizontal Dummy (1)	High Quality Dummy (2)	Any Supp in Türkiye (3)	Profit Sales Index (1%) (4)
Search Only	0.107*** (0.040)	0.184* (0.097)	0.053** (0.027)	0.177 (0.127)
Search + AS	0.033 (0.039)	0.245** (0.099)	0.054** (0.028)	0.196 (0.148)
Search + MH	0.092** (0.040)	0.133 (0.096)	0.028 (0.026)	-0.044 (0.093)
Search + AS + MH	0.043 (0.039)	0.125 (0.100)	0.095*** (0.030)	0.438** (0.182)
S Only * In Turkish Group	0.029 (0.088)	-0.173 (0.191)	-0.144 (0.089)	-0.221 (0.314)
S + AS * In Turkish Group	0.024 (0.086)	-0.295 (0.210)	-0.033 (0.088)	-0.547** (0.239)
S + MH * In Turkish Group	0.039 (0.088)	-0.221 (0.200)	-0.116 (0.088)	0.042 (0.267)
S + AS + MH * In Turkish Group	0.055 (0.087)	-0.237 (0.192)	-0.126 (0.090)	0.421 (0.564)
All Inter Zero p -val	0.983	0.656	0.415	0.111
Adjusted R^2	0.01	-0.00	0.10	0.01
N	1840	357	1662	1286

Note: This table shows the main results with treatment interacted with an indicator for whether the firm was in a supplier WhatsApp group with a supplier based in Türkiye at baseline. p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. The outcome in Column 1 is the same as Column 1 of Table 1. The outcome in Column 2 is the same as Column 2 of Table 1. The outcome in Column 3 is the same as Column 1 of Table 2. The outcome in Column 4 is the same as Column 6 of Table 4.

Appendix D - Robustness Checks

We pre-specified that we would use the regression specification specified in Equation (2). Nonetheless, in this section, we run analysis using additional specifications to verify that results are not driven by this (pre-specified) choice. We first replicate all of the main tables in the analysis using the following simpler regression specification that does not include any covariates:

$$y_i = \alpha + \sum_{j=1}^4 \beta_j T_{ji} + \varepsilon_i,$$

where, as before, Pure Control is the omitted category.

Then, we run a specification that is the same as Equation (2) but also includes supplier fixed effects. Specifically, suppliers were divided into 10 buckets of 3 suppliers each, and then firms were randomly allocated among the buckets that matched their sub-sector. We therefore include fixed effects for the bucket that a firm was assigned to. This is not defined for the Pure Control group as they were not matched with any suppliers, so we must exclude them from the regression (indeed, the indicator for Pure Control is perfectly collinear with bucket fixed effects). We thus present regressions for the main outcomes where Search Only is the omitted category (and Pure Control is dropped altogether), both with and without bucket fixed effects, in Table D5.

Table D1: Access to Foreign Goods (No Covariates)

	Horizontal	Vertical				Price
	Has Product ≥ 3 Criteria (1)	High Qual Dummy (2)	Qual Score (/50) (3)	Made in Turkey (4)	Index (5)	Price (USD) (6)
<i>Panel A: Pooled</i>						
Treatment	0.080*** (0.027)	0.107 (0.068)	-0.636 (0.591)	0.145** (0.068)	0.379*** (0.138)	1.656** (0.840)
<i>Panel B: Individual Treatments</i>						
Search Only	0.115*** (0.035) [0.005]	0.139* (0.084) [0.231]	-0.177 (0.683) [0.951]	0.112 (0.081) [0.315]	0.389** (0.169) [0.084]	2.444** (0.991) [0.051]
Search + Adverse Selection	0.049 (0.035) [0.199]	0.154* (0.087) [0.229]	-0.026 (0.845) [0.966]	0.108 (0.085) [0.315]	0.358** (0.173) [0.084]	1.409 (1.049) [0.287]
Search + Moral Hazard	0.100*** (0.035) [0.010]	0.076 (0.084) [0.577]	-1.284 (0.843) [0.357]	0.154* (0.080) [0.139]	0.360** (0.165) [0.084]	1.746* (0.976) [0.171]
Search + AS + MH	0.055 (0.035) [0.199]	0.062 (0.085) [0.577]	-0.934 (0.846) [0.549]	0.203** (0.081) [0.053]	0.405** (0.176) [0.084]	0.932 (0.994) [0.347]
Control Mean	0.307	0.431	43.064	0.477	0.000	19.990
% Increase (Pooled)	26.1%	24.8%	-1.5%	30.4%	N/A	8.3%
All Coefs Equal p -val	0.158	0.609	0.445	0.549	0.992	0.358
Adjusted R^2	0.01	0.00	0.32	0.09	0.00	0.23
N	1859	359	359	361	359	642

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms.

Column 1 is an indicator that is one if the firm has a good that matches at least 3 horizontal criteria. Column 2 is an indicator for whether the good's quality score is above the median product-group quality score. Column 3 is the raw quality score. Column 4 is an indicator for whether the good is made in Turkey, primarily inferred based on the label. See the text for full details of how this outcome is constructed. Column 5 is the [Anderson \(2008\)](#) index combining the vertical outcomes to account for multiple hypothesis testing across outcomes. Column 6 is the price in USD, which is only measured conditional on the firm finding a good matching at least three horizontal criteria.

Table D2: Supplier Relationships (Followup Survey) (No Covariates)

	Regular Suppliers in Türkiye			Previous Suppliers		
	Any Sup in Türkiye (1)	Num Sup in Türkiye (2)	Index (3)	Num Sup Total (4)	Num Sup in Senegal (5)	Ended with Sup (6)
<i>Panel A: Pooled</i>						
Treatment	0.035 (0.023)	0.079** (0.038)	0.121* (0.062)	-0.032 (0.193)	-0.092 (0.191)	0.070*** (0.022)
<i>Panel B: Individual Treatments</i>						
Search Only	0.018 (0.029) [0.761]	0.069 (0.055) [0.336]	0.085 (0.080) [0.477]	0.092 (0.269) [0.867]	0.088 (0.266) [0.918]	0.062** (0.029) [0.052]
Search + Adverse Selection	0.056* (0.030) [0.150]	0.077* (0.048) [0.245]	0.146* (0.084) [0.214]	-0.154 (0.253) [0.867]	-0.236 (0.249) [0.722]	0.067** (0.029) [0.050]
Search + Moral Hazard	-0.001 (0.029) [0.971]	-0.009 (0.043) [0.829]	0.009 (0.078) [0.910]	-0.197 (0.248) [0.832]	-0.168 (0.244) [0.825]	0.091*** (0.029) [0.006]
Search + AS + MH	0.069** (0.031) [0.085]	0.183*** (0.062) [0.007]	0.249*** (0.086) [0.016]	0.139 (0.258) [0.867]	-0.049 (0.251) [0.918]	0.057** (0.029) [0.052]
Control Mean	0.167	0.222	0.000	3.700	3.213	0.135
% Increase (Pooled)	21.0%	35.6%	N/A	-0.9%	-2.9%	51.9%
All Coefs Equal p -val	0.072	0.012	0.050	0.487	0.642	0.707
Adjusted R^2	0.00	0.01	0.01	-0.00	-0.00	0.00
N	1680	1680	1680	1681	1681	1671

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms.

Column 1 is 1 if the firm says that they have a regular supplier in Türkiye. Column 2 is the number of regular suppliers in Türkiye. Column 3 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Column 4 is the total number of regular suppliers. Column 5 is the number of regular suppliers in Senegal. Column 6 is 1 if the firm has ended a relationship with a regular supplier in the past 3 months. A regular supplier is defined as a supplier from whom the firm has made two or more orders with the intention to continue the relationship.

Table D3: Order Value (Mobile Money Data) (No Covariates)

	Entire Period			Post Mystery Shopping		
	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)	Total Order Value (USD)	Num Weeks Ordered	Avg Order Value (USD)
	(1)	(2)	(3)	(4)	(5)	(6)
Search + Adverse Selection	3.58 (5.20) [0.930]	-0.03 (0.11) [0.990]	15.05 (10.00) [0.454]	2.56 (4.13) [0.806]	-0.06 (0.10) [0.915]	26.22 (17.91) [0.450]
Search + Moral Hazard	4.67 (6.05) [0.930]	0.02 (0.12) [0.990]	9.95 (7.82) [0.549]	5.45 (5.24) [0.680]	0.04 (0.11) [0.929]	15.04 (9.74) [0.450]
Search + AS + MH	12.63 (19.04) [0.930]	-0.00 (0.15) [0.990]	35.66 (28.71) [0.549]	15.79 (18.92) [0.806]	0.02 (0.14) [0.929]	58.61 (32.68) [0.378]
Search Only Mean	14.31	0.36	39.24	7.92	0.23	33.95
% Increase (AS)	25.0%	-8.1%	38.4%	32.3%	-25.4%	77.2%
% Increase (MH)	32.6%	5.8%	25.3%	68.9%	17.0%	44.3%
% Increase (AS+MH)	88.3%	-1.4%	90.9%	199.3%	9.8%	172.6%
All Coefs Zero p -val	0.894	0.989	0.371	0.730	0.863	0.238
Adjusted R^2	-0.00	-0.00	0.02	-0.00	-0.00	0.06
N	1500	1500	538	1500	1500	351

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. This table shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments, where Search Only is the omitted category.

Column 1 is the total value of transfers to study suppliers. Column 2 is the total number of weeks in which a firm made a transfer to a study supplier. Column 3 is the mean value transferred to study suppliers in weeks with a positive transaction value. Column 4 is the total value of transfers to study suppliers, excluding the mystery shopping period. Column 5 is the total number of weeks in which a firm made a transfer to a study supplier, excluding the mystery shopping period. Column 6 is the mean value transferred to study suppliers in weeks with a positive transaction value, excluding the mystery shopping period. Mystery shopping generally took place during within 2-3 weeks of the firm entering the study. The data covers up to 22 months after being added to the study. Columns 3 and 6 are at the firm-week level, with standard errors clustered by firm. All values are in USD.

Table D4: Profit and Sales (No Covariates)

	Raw			Winsorized (1%)		
	Profit (USD) (1)	Sales (USD) (2)	Index (3)	Profit (USD) (4)	Sales (USD) (5)	Index (6)
<i>Panel A: Pooled</i>						
Treatment	57.3* (28.8)	204.0* (108.1)	0.311** (0.101)	24.1 (20.9)	83.0 (83.6)	0.172** (0.075)
<i>Panel B: Individual Treatments</i>						
Search Only	.566 (29.1) [0.984]	254 (187) [0.377]	.194* (.114) [0.203]	-5.26 (26.2) [0.847]	107 (125) [0.582]	.208* (.108) [0.132]
Search + Adverse Selection	30.6 (36.4) [0.644]	56 (119) [0.660]	.194 (.122) [0.204]	18.4 (31.5) [0.775]	38.1 (111) [0.725]	.124 (.099) [0.352]
Search + Moral Hazard	-27.4 (24.8) [0.565]	-103 (103) [0.516]	.037 (.082) [0.702]	-27.4 (24.8) [0.543]	-121 (93.6) [0.394]	6.7e-03 (.078) [0.935]
Search + AS + MH	244*** (93.2) [0.011]	640** (282) [0.050]	.873*** (.31) [0.004]	120*** (45.2) [0.019]	325** (152) [0.089]	.367*** (.135) [0.027]
Control Mean	188.3	609.5	0.000	188.3	609.5	0.000
% Increase (Pooled)	30.4%	33.5%	N/A	12.8%	13.6%	N/A
All Coefs Equal p -val	0.032	0.027	0.036	0.022	0.018	0.024
Adjusted R^2	0.01	0.01	0.01	0.01	0.01	0.01
N	1351	1378	1298	1351	1378	1298

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all treated groups. Panel B shows the coefficients corresponding to treatment indicators for each of the four treatment arms.

Column 1 is total profit from the past 30 days in USD. Column 2 is total sales from the past 30 days in USD. Column 3 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Column 4 is total profit from the past 30 days in USD, winsorizing the top 1%. Column 5 is total sales from the past 30 days in USD, winsorizing the top 1%. Column 6 is the [Anderson \(2008\)](#) index combining the previous two columns to account for multiple hypothesis testing across outcomes. Profit is measured using the survey question from [De Mel, McKenzie, and Woodruff \(2009\)](#). Sales is measured using a similar survey question.

Table D5: Robustness to Supplier Fixed Effects

	Horizontal Dummy		High Quality Dummy		Any Supp in Türkiye		Ord Value (MoMo)		Profit Sales Index	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Treatments vs Search Only</i>										
Search + Adverse Selection	-.068*	-.076**	.049	.052	.03	.029	4.18	2.43	-.078	-.07
	(.035)	(.035)	(.085)	(.089)	(.029)	(.03)	(5.28)	(5.52)	(.118)	(.118)
	[0.132]	[0.083]	[0.562]	[0.605]	[0.479]	[0.522]	[0.910]	[0.973]	[0.508]	[0.552]
Search + Moral Hazard	-.025	-.027	-.087	-.071	-.017	-.017	4.79	3.89	-.202**	-.196*
	(.035)	(.035)	(.086)	(.088)	(.027)	(.028)	(6.31)	(6.55)	(.102)	(.103)
	[0.464]	[0.435]	[0.485]	[0.605]	[0.528]	[0.552]	[0.910]	[0.973]	[0.124]	[0.147]
Search + AS + MH	-.061*	-.062*	-.107	-.107	.054*	.053*	12.9	12.1	.165	.154
	(.035)	(.035)	(.089)	(.091)	(.029)	(.03)	(19.1)	(18.5)	(.153)	(.152)
	[0.143]	[0.138]	[0.485]	[0.496]	[0.156]	[0.172]	[0.910]	[0.973]	[0.466]	[0.514]
Search Only Mean	0.422	0.422	0.564	0.564	0.185	0.185	14.31	14.31	0.208	0.208
Supplier Bucket FEs	N	Y	N	Y	N	Y	N	Y	N	Y
All Coefs Zero p -val	0.195	0.131	0.234	0.309	0.066	0.086	0.879	0.956	0.025	0.026
Adjusted R^2	0.05	0.05	0.01	0.03	0.14	0.14	0.01	0.01	-0.03	-0.02
N	1498	1498	292	291	1338	1338	1500	1500	1032	1032

Note: p -values are computed using randomisation inference. Specifically, we compute the randomised randomisation- t p -value from [Young \(2019\)](#) using 2000 reps. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$. We report conventional robust standard errors in parentheses, although we do not use these directly for inference. We also report Romano-Wolf multiple-testing adjusted p -values in square brackets ([Romano and Wolf, 2005](#)). To test the null that the trust treatments have no effect, we report at the bottom of the table the p -value for an F -test that all coefficients are equal, computed by permuting the F -statistic. Panel A shows the coefficient from a regression on an indicator that pools all trust treated groups, where Search Only is the omitted category. Panel B shows the coefficients corresponding to treatment indicators for each of the three treatment groups with trust treatments. All regressions include any covariates selected by Double Lasso ([Belloni, Chernozhukov, and Hansen, 2014](#)), as well as stratum fixed effects and the outcome measured at baseline (if available). Column 1 is the same as Column 1 of Table 1. Column 2 is the same as the previous column but includes supplier bucket fixed effects. Column 3 is the same as Column 2 of Table 1. Column 4 is the same as the previous column but includes supplier bucket fixed effects. Column 5 is the same as Column 1 of Table 2. Column 6 is the same as the previous column but includes supplier bucket fixed effects. Column 7 is the same as Column 4 of Table 3. Column 8 is the same as the previous column but with supplier bucket fixed effects. Column 9 is the same as Column 3 of Table 4. Column 10 is the same as the previous column but with supplier bucket fixed effects. The suppliers were divided into 10 buckets of 3 suppliers each, and firms were randomly allocated across buckets that matched their sub-sector. The regressions in this table exclude Pure Control as they were not matched with any suppliers and so supplier bucket fixed effects are not well-defined. All comparisons are therefore with reference to Search Only.

Appendix E - Further Details on Solving and Estimating the Model

In this Appendix, we provide more detail on our algorithm to numerically solve and estimate the model.

Solving the Model

As is often the case in dynamic optimisation, the original infinite horizon program is very difficult to work with directly. It is much more tractable to find a way to work with a recursive formulation. The challenge is that, unlike standard dynamic problems encountered in macro, we have two constraints (the DEC and DICC) that are forward looking, and, in particular, forward-looking to an infinite horizon.

We follow the approach of [Marcet and Marimon \(2019\)](#). The idea is that the original Lagrangean can be rewritten recursively as a pseudo planner's problem, where the Pareto weights are state variables that evolve endogenously to completely summarise historical binding constraints. Intuitively, if the DICC is binding in period 0, which implies that the agent must be delivered a certain amount of utils at some point in the future, the Pareto weight on the agent increases over time to ensure that the planner delivers precisely the required amount of utils. This allows the problem to be written recursively because the principal can trade off the benefit of making a constraint "more binding" today against the cost of increasing next period's Pareto weight on the agent. The recursive formulation delivers a Saddle Point Functional Equation, which is analogous to the familiar Bellman Equation but for saddle point problems, which satisfies a number of familiar properties that permit the use of dynamic programming techniques.

We first rewrite our problem as a recursive Lagrangean and thus derive the Saddle Point Functional Equation. To avoid cumbersome algebra and focus on the core intuition of the method, the derivations that follow assume that the firm can only purchase from one supplier at a time. We write the full version, allowing for purchases from both a new and existing supplier, at the end. Rewriting the objective function as an infinite sum and including the constraints with Lagrange multipliers, the program can be expressed as follows

$$\begin{aligned}
 L = & \min_{\{\rho_t\}, \{\eta_t\}, \{\gamma_t\}} \max_{\{q_t\}, \{\tau_t\}} \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) ((1 - \mu_t(1 - \lambda)r(q_t) + \mu_t(1 - \lambda)r(0) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}) \\
 & + \sum_{t=0}^{\infty} \delta^t \rho_t \left[\sum_{s=t+1}^{\infty} \delta^{s-t} (\tau_s - cq_s) - \xi cq_t \right] \\
 & + \sum_{t=0}^{\infty} \delta^t (1 - \mu_0(1 - \lambda^t)) \eta_t \left[\sum_{s=t}^{\infty} \delta^{s-t} (1 - \mu_s(1 - \lambda^{s-t})) ((1 - \mu_s(1 - \lambda)r(q_s) + \mu_s(1 - \lambda)r(0) - \tau_s + \delta\mu_s(1 - \lambda)\bar{U}) \right. \\
 & \left. + \sum_{t=0}^{\infty} \delta^t \gamma_t [\tau_t - cq_t] \right]
 \end{aligned}$$

Then, with some algebra, we can collect the Lagrange terms directly inside the first infinite sum to express the Lagrangean as a function of "Pareto weights", Lagrange multipliers, state variables,

and time-invariant functions.

$$L = \min_{\{\rho_t\}, \{\eta_t\}, \{\gamma_t\}} \max_{\{q_t\}, \{\tau_t\}} \sum_{t=0}^{\infty} \delta^t [(\zeta_t^b + \eta_t)h_0^b(q_t, \tau_t, \mu_t) + \eta_t h_1^b(q_t, \tau_t, \mu_t) + (\zeta_t^s + \gamma_t)h_0^s(q_t, \tau_t) + \rho_t h_1^s(q_t, \tau_t)]$$

where $\zeta_{t+1}^b \equiv (\zeta_t^b + \eta_t)(1 - \mu_t(1 - \lambda))$, $\zeta_{t+1}^s \equiv \zeta_t^s + \rho_t$, and

$$\begin{aligned} h_0^b(q_t, \tau_t, \mu_t) &\equiv [(1 - \mu_t(1 - \lambda))r(q_t) + \mu_t(1 - \lambda)r(0) - \tau_t + \delta\mu_t(1 - \lambda)\bar{U}] \\ h_1^b(q_t, \tau_t, \mu_t) &\equiv -\bar{U} \\ h_0^s(q_t, \tau_t) &\equiv \tau_t - cq_t \\ h_1^s(q_t, \tau_t) &\equiv -\xi cq_t \end{aligned}$$

ζ_t^b and ζ_t^s behave like Pareto weights and are equal to the sum of all prior Lagrange multipliers from the forward-looking constraints for the principal (the buyer) and the agent (the seller), respectively. This is the sense in which they fully summarise the shadow cost of constraints from earlier periods of the problem. For example, if DICC is “very binding” in early periods, meaning ρ_t is large for early t , then this is reflected in a large value of ζ_t^s for later t , which causes the “planner” to endogenously choose a high value of τ_t and thus give the agent utility. Note that there is a small difference relative to how the weights evolve in [Marcet and Marimon \(2019\)](#) as the buyer’s weight here includes the term $(1 - \mu_t(1 - \lambda))$ in the evolution, which appears because the Adverse Selection problem means that we only reach the next period in the relationship with probability $(1 - \mu_t(1 - \lambda))$.

Distinguishing between the foreign good, q_t , and the existing supplier q_{et} , we can then write this recursively as a Saddle Point Functional Equation (SPFE) as follows,

$$\begin{aligned} W(\zeta_t^b, \zeta_t^s, \mu_t) &= \min_{\eta_t, \rho_t, \gamma_t} \max_{q_t, q_{et}, \tau_t} (\zeta_t^b + \eta_t)h_0^b(q_t, q_{et}, \tau_t, \mu_t) + \eta_t h_1^b(q_t, \tau_t, \mu_t) \\ &\quad + (\zeta_t^s + \gamma_t)h_0^s(q_t, \tau_t) + \rho_t h_1^s(q_t, \tau_t) + \delta W(\zeta_{t+1}^b, \zeta_{t+1}^s, \mu_{t+1}), \end{aligned}$$

with

$$\begin{aligned} h_0^b(q_t, \tau_t, \mu_t) &\equiv [(1 - \mu_t(1 - \lambda))r(q_t, q_{et}) + \mu_t(1 - \lambda)r(0, q_{et}) - \tau_t - p_{et}q_{et} + \delta\mu_t(1 - \lambda)\bar{U}] \\ h_1^b(q_t, \tau_t, \mu_t) &\equiv -\bar{U} \\ h_0^s(q_t, \tau_t) &\equiv \tau_t - cq_t \\ h_1^s(q_t, \tau_t) &\equiv -\xi cq_t \end{aligned}$$

subject to

$$\begin{aligned} \zeta_{t+1}^b &= (\zeta_t^b + \eta_t)(1 - \mu_t(1 - \lambda)) \\ \zeta_{t+1}^s &= \zeta_t^s + \rho_t, \end{aligned}$$

and the evolution of μ_t . For the evolution of μ_t , to prevent a discontinuity in learning when going from $q_t = 0$ to any positive value, while also ensuring the principal's choice set remains convex, we impose a technical assumption that for q_t below some small $\underline{q}$, posteriors after observing a high quality realisation are given by $\mu_{t+1} = \mu_{t+1}^{\text{Bayes}} + (\mu_t - \mu_{t+1}^{\text{Bayes}}) \cdot (1 - q_t/\underline{q})^p$, where μ_{t+1}^{Bayes} is the posterior implied by Bayes' Rule and $p \in (1, 2)$. We set $p = 1.4$ as numbers around this value tended to perform well in numerical experiments. Making learning a continuous function of q_t ensures the problem satisfies appropriate regularity conditions as it makes the buyer's choice set convex. Intuitively, it captures the idea of a minimum order size (which is the case in our empirical setting), and we set $\underline{q}$ to 6 as packets of 6 are a very common minimum order size in our empirical setting. This learning function makes little difference in practice as it only applies in rare cases where the optimal q is below $\underline{q}$, but it is important to ensure the problem satisfies the appropriate regularity conditions.

The Saddle Point Functional Equation is analogous to the Bellman Equation for saddle point problems, and [Marcet and Marimon \(2019\)](#) prove that—under some regularity conditions—it has the usual desirable properties associated with dynamic programming problems. In particular, this means that we can obtain solutions to the original problem by using dynamic programming techniques to solve for the value function and policy functions associated with the SPFE.

We solve the model by iterating on the value function, W . Since W is homogeneous of degree one in the Pareto weights, we can write $W(\zeta_t^b, \zeta_t^s, \mu_t) = \zeta_t^b W(1, \frac{\zeta_t^s}{\zeta_t^b}, \mu_t)$. This means that we can eliminate one state variable. We define a discrete grid over the two state variables and linearly interpolate over the grid.⁴¹ Within each iteration, we solve the min max as follows. First, the FOC with respect to τ_t gives implies $\zeta_t^b + \eta_t = \zeta_t^s + \gamma_t$. Since we know from [Appendix B](#) that we will never have $\eta_t > 0$ and $\gamma_t > 0$ simultaneously, it must then be that $\eta_t = \max(\zeta_t^s - \zeta_t^b, 0)$ and $\gamma_t = \max(\zeta_t^b - \zeta_t^s, 0)$. We can then obtain τ_t as either cq_t if $\gamma_t > 0$ or $(1 - \mu_t(1 - \lambda))r(q_t, q_e) + \mu_t(1 - \lambda)r(0, q_e) + \delta\mu_t(1 - \lambda)\bar{U} - p_e q_e + \delta(1 - \mu_t(1 - \lambda))\frac{\partial W}{\partial \zeta_{t+1}^b} - \bar{U}$ if $\eta_t > 0$. (We discuss the edge-case where $\eta_t = \gamma_t = 0$ below). To obtain q_t, q_{et} , for a given ρ_t we can solve for q_t and q_{et} analytically using the FOCs if $q_t > \underline{q}$; otherwise we have to solve for them numerically as a term $\frac{\partial W}{\partial \mu_{t+1}}$ appears in this region as the choice of q affects μ_{t+1} through the smoothed learning function. The FOC with respect to ρ_t includes $\frac{\partial W}{\partial \zeta_{t+1}^s}$, which must be evaluated numerically and thus we cannot solve for ρ_t analytically. We thus solve for it numerically in an outer loop, with the solver for q_t, q_{et} in the inner loop.

Once we have obtained the value function, we iterate forward from the initial conditions at $t = 0$ to get the solution path. Since the agent's utility is linear in τ_t , the value function has a kink in the neighbourhood of $\zeta_t^b = \zeta_t^s$, which [Marimon and Werner \(2021\)](#) show results in non-unique policy functions and thus inconsistent utility promises. This kink is also the point at which the surplus shifts from the buyer to the seller, and the period where it occurs is exactly $T^* - 1$ (see [Appendix B](#)), where τ_t is interior. We resolve this by doing the following. First, since $\pi_t = 0$ for all $t < T^* - 1$, we know that $V_0 = \delta^t V_t$ for any $t < T^* - 1$. Starting with $t = 1$, we can calculate $V_0 - \delta^t V_t$ for each t , and find $T^* - 1$ by checking when this object switches sign. Second, once we know $T^* - 1$, we iterate forward normally until $T^* - 2$, at which point we set ρ_{T^*-2} to be such that $\zeta_{T^*-1}^b = \zeta_{T^*-1}^s$ and we calculate τ_{T^*-1} as the value that ensures that the Envelope Selection Condition in [Marimon and Werner \(2021\)](#) holds.

⁴¹We experimented with both linear interpolation and cubic splines, and generally found that linear interpolation performed better.

Estimation

The above sub-section describes how we solve the model for a given guess of the parameters. In order to estimate the parameters, we need to find the parameters that best match the empirical moments, and thus solve the model for many combinations of parameters. We do this in two steps. First, we solve the model for a grid of $(\mu_0, \tilde{\xi}, c, \lambda, \bar{U})$, where c is the cost of supplier abroad (varying c and ψ are isomorphic and one can simply map between them by setting $\psi = 1/c$), and linearly interpolate over the grid. This gives *functions* for all of the relevant theoretical objects, such as $y_t(\mu_0, \tilde{\xi}, c, \delta, \lambda, \bar{U})$, which means that we do not need to further solve the model as we can simply evaluate these functions at a given $(\mu_0, \tilde{\xi}, c, \delta, \lambda)$. The model converges for 99.6% of the parameter combinations, and we impute the values for the combinations that do not converge using a simple nearest-neighbour means algorithm.

Second, we use the above functions for estimation. We define the theoretical moments described in the text, which are all treatment effects. These treatment effects provide exogenous variation to identify the parameters governing the frictions. We calculate the variance-covariance matrix of the empirical moments by drawing 1,000 bootstrap sub-samples of the experimental data and use the inverse of this as the weighting matrix.

As the moments only identify differences, we ensure that the model matches the control means through the choice of the z distribution and the ψ_μ parameter. For z , since our assumption on the baseline equilibrium is that each firm can be represented as having one supplier, the model implies that profit for a firm i with one input is given by $\pi_i = z_i^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p_i^{1-\sigma}$, where p_i is their input price. The input price is a function of z : firms with $z < \bar{z}$ have p_l ; other firms have an endogenous price determined by their reservation utility from the search game (which is a function of z). We thus have a many-to-one function that maps z onto p , which implies that there is a one-to-one mapping from z to π given by $\pi(z) = z^\sigma \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} p(z)^{1-\sigma}$. We thus draw at random from the empirical π_i distribution of the control group and invert this mapping to obtain an empirical distribution of z that rationalizes the baseline control distribution (and thus control mean). As firms often report profit in round numbers, there are several mass points in this distribution, so to improve numerical stability in the optimization routine, whenever we draw z we jitter it by adding $u \sim U[-1, 1]$. For ψ_μ , note that the share of firms importing in the control group is determined by the share of firms with $z > \bar{z}$. Thus, for a given guess of the parameters, we can solve for ψ_μ that sets this share to match the control mean. Intuitively, for any guess of the parameters, we can always shift the location of the distribution of match-specific productivities of the suppliers until exactly a share d falls above the cutoff.

The two dimensions of heterogeneity are the firm-specific productivities, z , and the match-specific productivities, ψ . For distribution of match-specific productivity, we directly estimate the parameters of the LogNormal, (ψ_μ, ψ_σ) . We winsorize the upper tail such that $c = 1/\psi$ never falls below 2. Note that the absolute level is not meaningful—the model is homogeneous of degree $1 - \sigma$ in c and p_l , so all that matters is the ratio c/p_l , and since we calibrate $p_l = 12.29$, a relative price of less than one sixth is already very low. We do this both because heavy tails make estimation sensitive to implementation details of the SMM and because the value function becomes very large as c becomes small (causing numerical issues with convergence). We actually compute the moments by Monte Carlo integration using $\Omega = 20,000$ draws of ψ .

As this is a non-stationary dynamic model, the questions of what is baseline and how changes in the environment affect existing relationships are non-trivial. We assume that the baseline equilibrium is the long-run steady state of the model. That means that all firms with $z \geq \bar{z}$, where

$\bar{z}$ is the endogenous cutoff above which a firm will engage in search for a supplier, have already searched and are in the “terminal” phase of the relationship. Recall from Section 3 that the optimal contract features a unique T^* such that the firm earns exactly their outside option (which is the maximum of re-searching or buying locally forever) for all $t > T^*$, which we refer to as the terminal phase. Thus, all firms with $z \leq \bar{z}$ earn $U_l = \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} z^\sigma p_l^{1-\sigma} / (1-\delta)$, while all firms with $z > \bar{z}$ earn $\bar{U}(z) > U_l$, where $\bar{U}(z)$ is the expected value of participating in the search game, defined implicitly by $\int_\psi \max\{U(\psi, z) - \bar{U}(z), 0\} dF(\psi) = s$. For simplicity, we simply assume that these firms earning $\bar{U}(z)$ also have access to a technology that allows them to frictionlessly purchase at a constant price $p_e(z)$ defined such that $\bar{U}(z) = \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} z^\sigma p_e(z)^{1-\sigma} / (1-\delta)$, i.e., the constant price that gives the same utility as they are currently earning. This simplifies the model as they can then combine orders from this technology with orders from the new supplier without us having to take a stand on how or when they might renegotiate their existing dynamic contract. Thus, the baseline is defined by firms with $z < \bar{z}$ having existing price $p_e = p_l$, while firms with $z > \bar{z}$ have existing price $p_e(z) < p_l$. Using the fact that the model is homogeneous of degree $1 - \sigma$ in p_e and c , we can then evaluate the interpolated objects from the grid of solved model runs.

For the theoretical moments, we calculate the theoretical profit in each treatment group, with the treatments implemented as described in the main text. For profit, we use y_3 as the time period, and construct the quantile treatment effect. For the binary outcome relationships, we set this to 1 if they already have an existing supplier abroad (i.e., firms with $z > \bar{z}$) and 0 if they have $z < \bar{z}$ and they choose $q_{f0}(c_{\min}) = 0$, where $c_{\min}$ is the minimum of the three free cost draws from the search treatment (or, equivalently, the maximum of the three productivity draws, $\psi = 1/c$). For firms with $q_{f0} > 0$ and $z < \bar{z}$, we set the moment equal to the probability that the relationship reaches the third period (as a function of μ_0). Finally, for mobile money, we calculate $\sum_{t=3}^{18} \tau_t$, reflecting the total value ordered after mystery shopping finished (and, as discussed above, we divide the treatment means by the control—i.e., search only—mean). The final moments are shown in Figure A10.

As the objective function may have local minima, and the quantile treatment effects can cause derivatives to be challenging to calculate reliably at a speed suitable for estimation, we implement the estimation using six independent starts of a Particle Swarm algorithm. We experimented with a number of global and local minimizers and found Particle Swarm to perform well for our problem, with most initializations arriving at similar estimates. To compute standard errors, we repeat the optimization procedure using 200 draws from the bootstrap distribution of the treatment effects, and calculate the standard deviation.

Appendix F - Survey and Protocol Details

F.1 Firm Eligibility Criteria

To ensure the relevance of the study, we administered a short screening survey with the following eligibility criteria. Firm owners were included if they: (i) managed their own orders with suppliers; (ii) averaged at least five ready-to-wear sales per week over the past 30 days; (iii) had access to WhatsApp; and (iv) had used WhatsApp to sell to clients in the past 30 days, including through posts or status updates. Respondents who did not meet these criteria were thanked for their interest but deemed ineligible to participate.

Among those who agreed to participate, 85% of online sellers and 89% of physical-store sellers met the eligibility criteria. This indicates that our final sample represents a substantial share of these two key populations of interest, while being appropriately filtered to focus on active garment retailers engaged in a minimum level of digital commerce.

F.2 Further Detail on Supplier Recruitment, Eligibility Criteria, and Matching

Suppliers were recruited in three stages. First, we hired two surveyors in Istanbul to conduct a listing of Senegalese suppliers in Aksaray and Laleli, two important quarters with a large presence of exporters and export infrastructure to Francophone West Africa. The listing was a short survey that asked questions necessary to assess inclusion criteria for the study. The inclusion criteria were the following: a supplier must (1) sell ready-to-wear garments manufactured in Turkey, (2) live in Istanbul, (3) visit Senegal in person no more than once per year on average, (4) report making at least 5 sales per week via WhatsApp on average, (5) be willing to make a WhatsApp group to advertise their products for the purpose of the study. The listing was a short survey that asked the questions necessary to assess the inclusion criteria. Suppliers with physical stores were listed by a census, and suppliers without physical stores were listed by being approached in cargo stores (businesses that manage transportation logistics for exporting).

Second, among listed suppliers who met the inclusion criteria, we conducted a mystery shopping exercise 3-4 months later. We hired and trained a team of 6 firms in Senegal to make a total of 2 separate orders at each supplier (one order of 2 dresses, and another order of 2 polo shirts). The firms did not mention the study, and, if asked, said that they received the supplier's number from a friend. As our target sample size was 30 suppliers, we rolled out this mystery shopping among the 89 suppliers that met the inclusion criteria in waves (with order randomized), inviting suppliers that were sufficiently active in responding to the mystery shoppers to join the study, until closing recruitment once the target sample size was reached. The purpose of this exercise was to screen out suppliers that would not engage with new customers or were otherwise unresponsive, as such suppliers would not be relevant for the study. If any supplier had scammed us in this exercise then we would have excluded them from the study, too, but this did not happen in practice (perhaps because the reputational cost of doing so would greatly exceed the gain for orders of this size).

Third, we required participating suppliers to agree to respect a few benign conditions governing appropriate use of the supplier WhatsApp group (posting regularly in the group, posting items of varying quality levels, posting only items within their chosen sector, including prices with their posts), which almost all suppliers agreed to.

We divided the 30 suppliers into 10 buckets of 3 suppliers each. Three buckets corresponded to men's clothing, five buckets corresponded to women's clothing, and two buckets corresponded

to shoes and accessories. In the Search treatment, which matched treated firms to suppliers, we randomly allocated treated firms to a bucket that matched their sub-sector. That is, for example, a firm whose main sub-sector is women's clothing was randomly allocated among the five women's clothing buckets. This means that suppliers should be balanced across the four treatment arms, and, in line with this, we show in Table D5 that controlling for supplier fixed effects makes no difference to the results.

F.3 Moral Hazard Treatment Details

The Moral Hazard treatment, as the model highlights, requires either shifting the suppliers' incentives or shifting firms' perceptions of suppliers' incentives. To do this, we read the information to treated firms at the end of the baseline survey to emphasize that study suppliers have strong incentives:

I have one last piece of information to give you. As you know, you have been added to WhatsApp groups of Senegalese suppliers in Turkey.

We are working with many suppliers as part of our study. We want to assure you that we have given them strong incentives to do a good job.

We would like to collect feedback on these suppliers so that we can recommend the best ones in the future. To do this, we will ask merchants in the study to rate their experience with the suppliers we introduced to them on a scale from 1 to 5, based on product arrival and quality. These ratings allow us to identify the best suppliers, which benefits them and allows us to continue recommending them to others. They therefore have an incentive to do a good job.

If a supplier receives bad ratings, we will investigate and, if we find they did not do a good job, we will remove them from the study. They would therefore lose access to around 150 traders.

I will give you a phone number that you can use to provide a rating or report a problem.

I want to emphasize that the suppliers are aware that they are being rated and that they risk being removed from the study if they receive bad ratings. We can thus assure you that they have incentives to do a good job with firms in the study.

F.4 Mystery Shopping Protocol

To evaluate firms' access to high-quality and differentiated foreign goods, we implemented a structured mystery shopping protocol. Surveyors contacted firms over WhatsApp posing as genuine customers interested in purchasing garments for an upcoming event. The goal was to simulate realistic sourcing and negotiation behavior while standardizing the interaction across sellers.

Initial contact. Two weeks after the baseline survey, trained surveyors messaged each firm over WhatsApp, referencing a product image and asking whether the firm had something similar. Firms had previously been told that they might be contacted by potential customers who were seeking high-quality goods.

Horizontal match criteria. The referenced product images were designed to include five horizontal characteristics that are orthogonal to quality (e.g., color, presence of graphic, sleeve style, neckline). If the firm proposed a good that met at least three of the five criteria, this was coded as a horizontal match. This is the primary outcome for the horizontal dimension.

Negotiation protocol. If a horizontal match was found, then the mystery shopper proceeded to negotiating the price and delivery details. All price interactions followed a standardized script:

- Surveyors first asked the seller for a quote (including delivery).
- They then proposed a counteroffer equal to 75% of the quoted price (rounded down to the nearest 500 FCFA), capped at 20,000 FCFA.
- The seller's final offer was recorded, along with whether they accepted payment on delivery or required an advance (and how much).

Purchase and fallback procedure. If a match was found, the seller agreed to a price at or below the 20,000 FCFA cap, and the seller accepted either 50% or full payment on delivery, the mystery shopper proceeded to buy the good in a randomly selected 80% of cases (due to our budget constraints). In the 20% of cases where the random assignment indicated to not buy the good, they explained that their plans had changed, and offered a nominal payment of about 2 USD (1,000 FCFA) as a gesture of appreciation. This fallback protocol was piloted to ensure natural exit and avoid upsetting sellers.

Rapid-order test. In a second stage, surveyors sent an image from a randomized set of “priority products” and asked if the seller could find the item within 24 hours. This assessed the firm's ability to respond quickly to buyer demand or to test new suppliers.

Secondary Outcomes. Surveyors recorded whether the merchant added them to WhatsApp status, if delivery date was confirmed, and if the merchant mentioned quality or reputation. They also tracked total time elapsed between contact and order.

Exit. Depending on the scenario (purchase/no purchase), shoppers used an exit script, including a fallback offer of 1,000 FCFA to compensate for the merchant's time in cases where no purchase occurred because of the 20% random draw.

Vertical quality scoring. Upon receipt, each good was evaluated by two professional tailors (and a shoemaker for footwear) using a structured 50-point scorecard developed for the study.

Each criterion was scored on a 0–2 or 0–3 scale (with a few exceptions), and we computed both the raw score and a binary high-quality indicator (equal to 1 if the item's score was above the median for its product type). As each product was independently assessed by both experts, we average the two assessments.

Origin recording. In addition to the quality scores, surveyors recorded whether the item was labeled as “Made in Türkiye,” a strong signal of quality in this setting. When labels were absent, tailors independently evaluated origin based on design and construction cues. The resulting variable is a binary indicator equal to 1 if the label—or both experts—indicated Turkish origin.

F.5 Vertical Quality Scoring — Scorecard

Upon receipt, each good was evaluated by two professional tailors (and a shoemaker for footwear) using a structured 50-point scorecard developed for the study. We designed this scorecard with our hired experts. We also drew inspiration from [Vitali \(2024\)](#), who uses a similar approach to assess garment quality in Kampala.

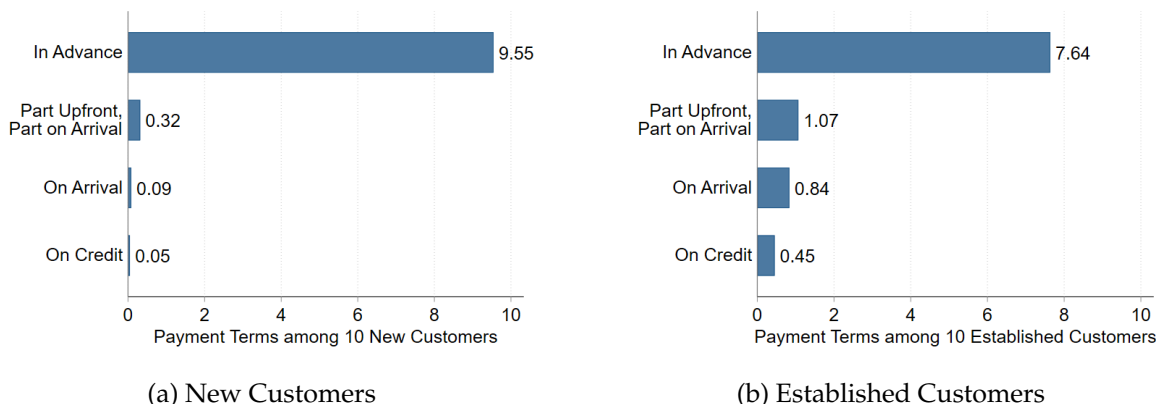
Table F1: Garment Quality Scorecard (50-point scale; higher = better)

Category	Question	Polo	Dress
Fabric	Is the fabric soft?	3	3
	Are there pills?	2	2
	Is it transparent under light?	4	4
	Are the threads consistent?	3	3
Sewing	Are the seams straight and neat?	3	3
	Do the seams come apart upon gentle pulling?	3	3
	Are there stray threads or loose stitches?	2	2
	Do the seams lie flat without puckering?	2	2
	Is the hem seam reinforced?	2	2
Shoulders	Do the shoulders fall correctly?	3	3
Collar and Buttons	Is the collar well-attached and flat?	3	–
	Are the buttons securely attached?	3	–
Darts	Are there darts on the bust and waist?	–	3
	Are the darts neatly sewn?	–	3
Colourfastness	Does it lose colour upon soaking?	7	7
Pattern	Does the pattern line up with the seams?	2	2
Hem	Is the hem well done?	4	4
Sleeves	Are the sleeves the same length and width?	4	4
Absence of Other Defects	Are there any other defects not covered above?	-10	-10
Total Score		50	50

Appendix G - Contract Terms and Relationship Dynamics

In this Appendix, we discuss descriptive evidence pertaining to the contract structure and relationship dynamics in the model and relate it to empirical findings in other settings. We discuss three features. First, for the contract structure itself, the model assumes that the buyer is required to pay for orders upfront. We make this assumption as this how the vast majority of contracts operate in our setting. As evidence for this, in a baseline survey that we conducted with 141 suppliers in Türkiye, we asked them, out of 10 customers, how many they would require to pay in full upfront, allow to pay on arrival, allow to pay on credit (i.e., at a later date than arrival), or pay some portion upfront and the remainder on arrival. We asked this separately for new and established customers. We plot the results in Figure G1. New customers are almost always required to pay in full upfront. This is relaxed slightly among established customers, although it is still the case that the majority of transactions involve upfront payment. [Startz \(2025\)](#) documents a similar payment structure among Nigerian importers, who prepay 97% of costs.

Figure G1: Contract Payment Terms out of 10 Customers

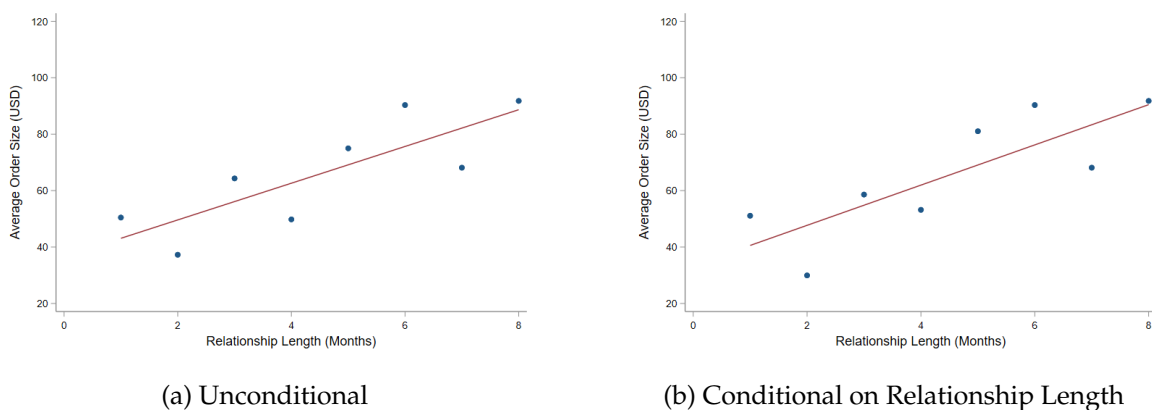


Note: This figure shows the mean response to two survey questions about payment terms in a baseline survey with 141 suppliers in Turkey. The first question, shown in Panel (a), asks, out of 10 new customers, how many the supplier would require to pay upfront, allow them to pay on arrival, allow them to pay on credit (i.e., at a later date than arrival), or pay some portion upfront and the remainder on arrival. The second question, shown in Panel (b), is the same except for established customers.

Second, on relationship dynamics, the optimal contract in the model is non-stationary and, in particular, is backloaded in value, with both quantity and price rising over time. Quantity and/or value rising over time occurs under models of both adverse selection and moral hazard and is well-established in several empirical settings. For example, [Monarch and Schmidt-Eisenlohr \(2023\)](#) find that value traded increases with relationship length using US import data matched with foreign supplier identifiers, and [Fitzgerald et al. \(2024\)](#) shows that Irish exporters grow quantities in new markets over time. We find the same pattern in our relationship data. In Figure G2, we plot the mean value traded between study firms and study suppliers—i.e., new matches created in the study—as a function of relationship length. Panel (a) shows this unconditionally, and Panel (b) restricts to only relationships that lasted at least six months, to account for potential selection into relationship length. In both cases, average value traded approximately doubles over the course of the relationship.

The dynamic price path is significantly more complicated to evaluate empirically as the theoretical prediction is about *surplus* increasing—i.e., markups rather than prices *per se*. This means that one needs to control for changes in cost, which may change over the course of a relationship due to, for example, economies of scale (since quantities grow over time), learning by doing, or initial fixed costs. [Monarch and Schmidt-Eisenlohr \(2023\)](#) study prices in relationships in US imports, and find that prices decrease with relationship length, while [Fitzgerald et al. \(2024\)](#), who study markups directly, find that markups are flat over the course of an export spell for Irish exporters (though the data is at the market level rather than relationship level). Another empirical challenge is that, in general, both the buyer and seller have contracting problems (whose relative magnitude varies with the context) and there are two instruments available to provide incentives: the timing of the payment, and the allocation of future surplus. Whether markups grow or shrink thus depends on which party is primarily being incentivized by future surplus. This means that there is likely to be important heterogeneity across settings. In our setting, since we see in Figure G1 that the buyer’s contracting problem is largely resolved in this setting through upfront payment, this leaves dynamic incentives for the seller side, which suggests that markups should at least be unlikely to decline. Unfortunately, our mobile payments data only contains value so ultimately we are not able to test this directly in our setting. That said, anecdotally, we heard often from suppliers that they provide new customer discounts and/or sell at cost in early phases of a relationship in order to allow new customers to observe quality and develop trust, i.e., investing in a reputation that they will later extract surplus from.

Figure G2: Average Value by Relationship Length



Note: This figure shows the mean value transferred from study firms to study suppliers—aggregated to the month level—in the mobile money data as a function of the number of months in which a given buyer-supplier pair has transacted. In Panel (b), to account for potential selection issues driven by the possibility that more optimistic firms are more likely to make both larger and repeat orders, we restrict the analysis to buyer-seller pairs with relationships lasting at least six months.